MILK AND MILK PRODUCTS TECHNOLOGY

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CHAPTER 1

Development of Milk Industry in India

During the early 1920's Military dairy farms were established for adequate supply of milk to the army stations. These farms were well maintained and, even in their early stages, were raising improved milch animals. Elsewhere, in urban areas, dairying was largely left in the hands of traditional producers, middlemen, and small producers, dealers of milk and milk products and vendors. To some extent, World War II gave impetus to private dairies with modernized processing facilities. The metropolitan cities such as Bombay, Calcutta and Delhi and some other large townships could claim of making available processed milk, table-butter and ice-cream, though not on a large scale.

In India the market milk industry may be said to have started in 1950-51 when the central dairy of Aarey Milk Colony was commissioning and went in to stream.

Milk Production in India

Year	Production (Million Tonnes)
1951-52	17.5
1961-62	20.4
1978-79	28.6

Sources; Dairying in India, IDSA publication, 1976.

Year	Production (Million Tonnes)
1991-92	55.7
2000-01	80.6
2004-05	92.5
2009-10	116.4
2010-11	121.8

Source: Department of Animal Husbandry, Dairying & Fisheries

Market Milk Industry in India

Although a beginning to organized milk handling was made in India with the establishment of Military Dairy Farms (oldest; Allahabad, 1889), the salient features of the market milk industry so far have been detailed below.

- Handling of milk in co-operative Milk Unions (oldest : Allahabad, 1913) established all over the country on a small scale in the early stages
- Long distance refrigerated rail-transport of milk from Anand to Bombay since 1945.
- Pasteurization and bottling of milk on a large scale for organized distribution was started at Aarey (1950), Worli (1961), Calcutta (Haringhata, 1959), Delhi (1959), Madras (1963), etc.
- Establishment of milk plants under the five year plans for Dairy Development all over India. These were taken up with the dual object of increasing the national level of milk consumption and ensuring better returns to the primary milk producer, their main aim was to produce more, better and cheaper milk.
- Dairy Cooperatives account for the major share of processed liquid milk marketed in the India. Milk is processed and marketed by 170 Milk Producers' Cooperative Unions, which federate into 15 State Cooperative Milk Marketing Federations. Over the years, several brands have been created by cooperatives like Amul (GCMMF), Vijaya (AP), Verka (Punjab), Saras (Rajasthan). Nandini (Karnataka), Milma (Kerala) and Gokul (Kolhapur).
- India has emerged as the largest milk producing country in the world with present level of annual milk production estimated as 121.8 million tones and almost reached 17 percent of world milk production. We expect a production level of 135 million tonnes by the year 2015. India has a large livestock population base constituting 278 million livestock including 180.5 million cattle, 82.8 million buffaloes, 4 million sheep and 9.2 million goats. The livestock population is projected to increase to 322 million by the year 2015.
- Per capita availability of milk was 281 grams per day in 2010-11 which increased from 241grams per day in 2005-06, up from 112 grams per day in 1968-69.

World Milk Production (Million Tonnes)

2009	2010 (estim)	2011 (f' cast)
701.4	713.6	727.6

Sources: FAO Food Outlook Report 2011

Per Capita Availability of Milk by States (gms/day)

State	1991-92	1995-96	2001-02	2005-06	2009-10	2010-11
All India	178	197	225	241	273	281
Andhra Pradesh	120	162	209	260	342	364
Arunachal Pradesh	22	119	105	113	59	63
Assam	77	95	70	72	69	71
Bihar	100	101	88	154	175	184
Goa	65	73	91	105	96	93
Gujarat	236	229	317	349	418	435
Haryana	586	618	645	628	662	679
Himachal Pradesh	313	329	341	373	397	446
Jammu & Kashmir	204	276	367	353	379	378
Karnataka	149	185	249	197	226	237
Kerala	167	198	234	171	201	210
Madhya Pradesh	197	199	240	262	278	287
Maharashtra	**136**	**163**	**172**	**178**	**190**	**197**
Meghalaya	76	83	78	82	83	83
Mizoram	31	31	43	43	29	31
Nagaland	96	88	78	96	96	93
Orissa	43	49	69	95	112	113
Punjab	722	847	892	943	944	937
Rajasthan	275	294	376	387	509	538
Sikkim	194	204	187	232	200	194
Tamilnadu	164	180	219	231	278	278

Tripura	31	35	77	70	77	80
Uttar Pradesh	199	216	241	262	283	289
West Bengal	120	130	120	126	133	137
A&N Islands	227	173	177	135	137	142
Chandigarh	141	143	131	116	95	87
Dadra & Nagar Haveli	59	87	100	53	86	83
Daman & Diu	0	25	17	11	15	14
Delhi	65	69	58	54	72	72
Lakshadweep	55	54	90	64	84	71
Pondicherry	90	90	104	108	96	94
Chhattisgarh	_	_	105	103	110	117
Uttaranchal	_	_	344	361	387	383
Jharkhand	_	_	96	126	130	136

Source: Various issues of BAHS, Department of Animal Husbandry, Dairying & Fisheries, Ministry of Agriculture, GoI.

Top Ten States of India in Milk Production (000 Tonnes)

State	2008-09	2009-10	2010-11
All India	112183	116425	121848
Uttar Pradesh	19537	20203	21031
Rajasthan	11931	12330	13234
Andhra Pradesh	9570	10429	11203
Punjab	9387	9389	9423
Gujarat	8386	8844	9321
Maharashtra	7455	7679	8044
Madhya Pradesh	6855	7167	7514
Tamil Nadu	6651	6787	6831
Bihar	5934	6124	6517
Haryana	5745	6006	6267

Source: Department of Animal Husbandry, Dairying & Fisheries, Ministry of Agriculture, GoI

Milk Utilisation Pattern in India

Year	1943	1956	2004
Milk Production (million tones)	23.5	17.8	91
Milk Utilisation (Percentage)	100	100	100
Liquid Milk	28.0%	39.2%	46.0%
Traditional Products	72.0%	60.8%	50.0%
Ghee/Makhan (clarified butter)	58.7%	46.0%	33.0%
Dahi (Yogurt-like)	5.2%	8.8%	7.0%
Khoya (Partially desiccated Milk	5.0%	4.4%	7.0%
Chhana and Paneer (unprocessed cottage cheese)	3.1%	1.6%	3.0%
Western Products: Milk Powder, etc	Neg	Neg	4.0%

Source: Handbook on Technology of Indian Milk Products

Production of Milk, Projections for 2011-2020

Countries	Production (000 tones)		Rate of growth (percent)	
	Average 2008-10 est.	2020	2001-10	2011-20
Developed countries	318980	349769	0.4	0.8
Developing countries	373667	503128	3.6	2.8
North Africa	11110	14533	3.9	2.6
Sub-Saharan Africa	25928	35878	2.5	2.9
Latin America and the Caribbean	75361	93848	2.2	2.0
Brazil	28674	35188	2.4	1.7
Asia and the Pacific	231824	314186	3.9	2.9
China	40332	59064	12.8	3.4
India	111147	153019	3.6	3.0
World	692647	852898	2.0	2.0

Source: OECD-FAO Agricultural Outlook 2011-2020

Species Contribution to Total Milk Production (Average 2006-2009)

Countries	Sheep milk (share in %)	Goat Milk (share in %)	Cow Milk (share in %)	Buffalo Milk (share in %)
India	0.0	3.3	48.7	48.0
Developed countries	0.9	0.8	98.1	0.1

Source: FAO, STAT (2011).

Operation Flood

The Government of India launched "Operation Flood" project in 1970. It was aimed at creating a 'flood' of rurally produced milk in the urban consuming centers. The first phase of the programme had 5-year duration from 1970 to 1975. The programme conducted in March 1981. It was intended to cover 10 states in this phase having 18 milk sheds. Subsequently, on the foundation laid in Operation Flood first phase, the Government of India, decided to launch its second phase which is almost 5 times in magnitude than the first phase.

Operation Flood I (1970-81)

Operation flood phase 1 was launched on July 01, 1970 and completed on 31st March, 1981. The project had an initial outlay of Rs. 95.4 crores which was later increased to Rs. 116.40 crores. It was mainly aimed at developing the milk marketing system in the country. As such major demand centers like Delhi, Calcutta, Bombay & Madras were linked with the rural milk producing pockets in the country.

Objectives

The programme had the following major objectives:

1. Increase in capacity of milk processing facilities.
2. Change in urban markets from traditional raw milk supplies to modern dairies milk supplies.
3. Resettlement of city based cattle in rural areas.
4. Development of long distance milk transport and storage facilities.
5. Development of milk procurement system on Anand Pattern.
6. Improvement in Dairy Farming Standards.

Line of Action

1. India received 1, 26,000 Metric Tones of Skim Milk Powder (SMP) and 42,000 Metric Tones of Butter Oil (B.O.). The donations were received over a period of 11 years.

2. Through sale of these commodities, funds to the extent of Rs. 114.68 crores were generated.
3. Thus generated funds were again ploughed back for dairy development and regeneration of funds to the same extent was done.

Achievements

The above actions resulted in the following major achievements:

1	Generation of funds	Rs. 114.68 crores
2	Anand Pattern Dairy Cooperative Societies	13270
3	Animals covered under production enhancement Programme	32 lakhs (Est.)
4	Bull Mother Farms	12
5	Number of farmer families covered	17.47 lakhs
6	Dairy plants built	17
7	Chilling Plants Built	25
8	National Milk Grid	Foundation laid
9	Share of milk markets commanded: Bombay	66%
	Calcutta	36%
	Delhi	60%
	Madras	43%
10	Imports of dairy equipment reduced to	10
11	Imports of Cattle Feed Plant equipment Reduced to	Zero(100% indigenous Manufacture)
12	Annual milk production increased by about 50%	2.07 to 3.02 crore Tones
13	Personnel trained	1779
14	Farmers inducted	18992

In Brief: Operation flood I was lunched in 1970 following an agreement with the food programme (WFP) which undertook to provide an aid 1,26,000 tones of skim milk powder (SMP) and 42,000 tones of butter oil (BO) for financing the programme. The main thrust was to setup dairy cooperative in the milk sheds, so as to link them to the four metro cities of Bombay, Calcutta, Delhi and Madras in which a commanding share of the milk markets or cities of Bombay, Calcutta, Delhi and Madras in which a commanding share of the milk markets were to be captured. Funds for operation flood I were generated by the sale of SMP and

BO. A total of 116.54 crore was invested in the implementation of the programme. The Delhi Government setup the Indian Dairy Corporation (IDC) in 1970.The IDC received the skimmed milk powder and butter oil, which it markets along with domestically produced milk powder.

Operation Flood II (1981-85)

This was launched on 2nd October 1979. The programme has an outlay of Rs. 485.5 crores. In this phase still more attention will be paid to increasing the milk production. The programme is designed to cover the whole country over a period of 7 years. Some 25 Cluster Federations covering 155 cooperative Unions will be formed. The phase – II was implemented during the 6th Five Year Plan (completed on 31st March 1985)

Objectives

The major objectives are as follows:

1. Coverage of 10 million rural milk producer families.
2. Creation of National Milch Herd of 14 million cross-bred cows and upgraded buffaloes
3. Strengthening of National Milk Grid (NMG) by linking milk supply and demand centers.
4. Base-structure for National Dairy Industry
5. Increase in per capita consumption of milk and milk products (114 gms/day)

Line of Action

1. India will receive aid in kind and soft-term loan.
2. Two-tier cooperative foundation will have one additional tier at state level.
3. Implementation to be done through farmers own federations.
4. All milk sheds will be linked to NMG and so also consuming cant.
5. Viable farmers' organizations will be established.

Achievements

Sr. No.	Particulars	Target	Achievement as on March, 1985
1	Anand pattern village co-operatives (No.)	29 000	34 523
2	Anand pattern village co-operatives under A.I., (No.)	8 000	7 543
3	Milk producers families covered, (lakhs)	34.8	36.31

4	Milch animals uner co-operative ambit, (lakh)	52.3	54.5
5	Annual milk production (1984-85) lakh (tones)	--	380
6	Per capita milk availability (1984-85), (g/day)	--	142
7	Average milk procurement, (lakh / pd) peak milk procurement	55.30	57.84
8	Urban milk marketing, (lakh /pd) capacity, peak throughout	43.00 --	35.00 50.11
9	Rural dairy plants, (lakh/ pd) Capacity, peak throughout	76.00 --	87.75 78.85
10	Milk powder manufacturing capacity(t/d)	--	507.50

In Brief: Operation flood II was started in April 1981 and ended in March 1985 with the expressed intention of creating a viable dairy industry to meet India's needs in milk and milk products. The European Community undertook to provide some 1,86,000 tonnes of skimmed milk powder and 76,000 tones of butter oil.

Operation Flood Phase III

Operation flood was a continuous programme. The OF Phase – III during the 7th Five year Plan period is for a period from April 1986 to March 1990. the key physical targets of Phase-III (7th Plan targets) are as follows:

Sr. No.	Particulars	Target
1	No. of Anand pattern dairy co-operative societies	55 000
2	No. of Anand pattern DCS under A.I.	25 000
3	Milch animals under co-operative ambit, in lakh	152.9
4	Milk procurement, peak, lakh / pd	183.3
5	Urban milk marketing, lakh / pd	124.2
6	Build up of dairy capacity, lakh / pd	200.0
7	DCS membership, lakh	100.2
8	Urban centres to be covered for marketing	200

Thus, India's dairy industry is progressing well. The tremendous strides taken towards modernization of dairying provided effective boost to indigenous manufacture of diary equipment, veterinary equipment including that required for semen banks, veterinary medicine and vaccines and long distance transport vehicles for milk. Development of trained manpower suitable to function at all levels provided backbone

for sustained and self reliant growth of Indian dairy industry. Its achievement during the past two decades is an example which can be followed by other developing countries of the world.

In Brief: The EC has been asked to continue its support by supplying a further 75,000 tones of skimmed milk powder and 25,000 tones of butter oil. India's white revolution has not only received support from the European Community and the World Bank, but also from a number of Western Governments, the United Nations food and Agriculture Organization (FAO), the United Nations Children Education Fund (UNICEF) and European NGOs such as the British relief agency OXFAM.

Summary of Operation Flood

The scheme sought to establish milk producers' cooperatives in the villages and make modern technology available to them. The broad objectives are to increase milk production ("a flood of milk"), augment rural incomes and transfer to milk producers the profits of milk marketing which are hitherto enjoyed by well-to-do-middlemen.

Phase I: Operation Flood was financed by the sale within India of skimmed milk powder and butter oil gifted by the EC countries via the World Food Program. As founder-chairman of the National Dairy Development Board (NDDB) of India, Dr Kurien finalized the plans and negotiated the details of EEC assistance. He looked after the administration of the scheme as founder-chairman of the erstwhile Indian Dairy Corporation, the project authority for Operation Flood. During its first phase, the project aimed at linking India's 18 best milksheds with the milk markets of the four metropolitan cities of Delhi, Mumbai, Calcutta and Madras.

Phase II: The project, implemented during 1981-85 raised this to some 136 milksheds linked to over 290 urban markets. The seed capital raised from the sale of WFP/EEC gift products and World Bank loan had created, by end 1985, a self-sustaining system of 43,000 village cooperatives covering 4.25 million milk producers. Milk powder production went up from 22,000 tonnes in the pre project year to 1,40,000 tonnes in 1989, thinks to dairies set up under Operation Flood. The EEC gifts thus helped to promote self-reliance. Direct marketing of milk by producers' cooperatives resulting in the transfer of profits from milk contracts increased by several million liters per day.

Phase III: Operation Flood (1985-1996) enabled dairy cooperatives to rapidly build up the basic infrastructure required to procure and market more and more milk daily. Facilities were created by the cooperatives to provide better veterinary first-aid health care services to their producer members.

Dairy Co-Operatives सहकार्य

(Role of Co-Operation In Dairy Industries)

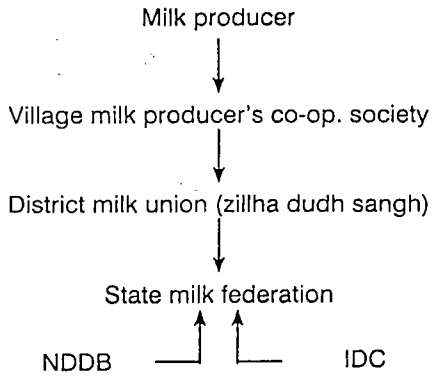
According to Co-Operative Planning Committee 1946, Co-Operation has been defined as a form origination in which persons voluntarily associate together on basis of equality for the promotion of their economic interests. The self help made effective through a mutual relation between individuals is known as co-operation.

In past times, the milk production was not on economic or industrial basis, because there was no market for milk and lack of transport facilities. When urbanization grown up and transport facility made available, milk produced in rural area transported to these urban areas which were now new market. Milk produced in rural area was transported in these cities by middleman. The profit went to middleman and producer was always in loss so there is need for organization in dairy industry.

In this critical condition some local leaders in Gujarat came together to think over and found out the solution. They formed society at Anand in 1947 it was beginning of co-operative movement in dairy industry. This co-operative AMUL (Anand Milk Union Limited) dairy was established.

After the success of AMUL in 1964 the existing Prime Minister Mr. Lalbahadur Shastri visited to Anand and decides to implement such system throughout India and establish a committee under the chairmanship of Dr. V. Kurian. It is beginning of co-operative movement in India and established board is known as NDDB (National Dairy Development Board)

Network of co-operative in dairy industry



1. Milk producer

Milk producer is the heart of dairy co-operative, his play an important role in the production of milk

2. Village milk producer's co-operative society

To establish the village co-operative society minimum 11 milk producer should come together to form society. They have collect some amount of money as a share capital (Rs. 10) and admission fees (Rs.1). They nominate one member as a chief promoter. He is looking after the registration of the society. For registration the application should be submitted to district co-operative registration through taluka co-operative register. After fulfilling all requisites & legal requirements district co-operative registrar hand over the certificate of registration and thus village milk producer co-operative society come into existence.

After registration the members of society elect a governing body. The elected members of governing body elect one as a chairman and this chairman look after the working of society.

Functions of primary cooperative milk producer's societies:

- Producer becomes member of society.
- Collection of milk at village level from milk producers.
- Producer supply milk t the society.
- Testing of milk for fat and snf.
- Transport of milk to allotted collection center.
- Milk payment to producer.
- Financial assistance to the individual producer.
- Provision of bilateral services on no profit no loss basis eg. Veterinary services, supply of feed, etc.
- Act as a guarantee for the milk producer against the bank loan for dairy unit.
- Organization of seminars, cultural programmers, shows etc.

3. District Milk Union

All village milk produces societies in the district come together to form district milk union. District milk union looking after all dairy activities in the district.

Functions

- Collection of milk: District milk union collects milk from village society.
- Transportation: It provides facility of transportation of milk from village society to processing plant.

- Processing of milk: Various type of milk process like standardization, pasteurization is given to milk to increase keeping quality.
- Product manufacturing: District milk union manufactures different dairy products to maintain demand, supply and price of milk and milk products.
- Payment of milk: Payment of milk is given to respective society regularly in the district.
- Supply of milk cans & ice for transportation of milk.
- Veterinary facilities makes available to producer through village co-operative society.
- Supply bilateral service like cattle feed or subsidy.
- Organization of seminars, cultural programs, shows and competition.

4. State Dairy Federation

All district milk union in the state elects some person as a member of state federation for example MAHANAND in Maharashtra. The state ministry of dairy also nominates some person as a member. These persons are MLA (Member of legislative assembly), ecologist, technical person etc. the minister of dairy is chief of the federation. He is assisted by secretary.

Function

- Provide guidelines regarding milk procurement (collection, transportation and storage.) and processing.
- Maintain co-ordination between government, dairy department and district milk union.
- For selling milk & milk products out of state, districts milk union should receive permission from the state federation.
- It supply equipment to district milk union and milk co-operative societies like milk tester, pasteurization plant, storage tank, etc.

5. NDDB (National Dairy Development Board)

It is registered society by the government of India at Anand since 1965. This nonprofit institute provides technical services to various dairy projects in India. These services area designed to help and built up a rural milk producer in co-operative lines. Functions of NDDB are directed towards establishment of co-operative infrastructure on Anand pattern.

Following are the functions of NDDB:

1. Planning: Different plans regarded with dairy and dairy development are worked out by NDDB under five years plan.
2. Implementation: These plans are implemented in a country through is co-operative lines/infrastructure.
3. Research: In different sections of dairy NDDB does the research work like transportation, processing, packing, collection market etc.
4. Technical Guidance and Consultancy: The guidance regarding the development of dairy business is given by NDDB; they also consult regarding the problems in dairy business.
5. Man Power Development: The technical persons appointed by NDDB at district milk union. This persons looking after working of milk union. Meanwhile they also train the persons working at this union and thus develop manpower.
6. To conduct exhibition at national level.
7. To advice the price fixation.
8. Implementing all these through cooperative organization.

6. IDC (Indian Dairy Corporation)

IDC is complete organization by the government of India and established in 1970. it is a specialized institution to promote and finance dairy development activities in India. IDC gave the finance to dairy cooperative organizations after recommendation of NDDB. Proved input as;

1. Veterinary first aid box,
2. Artificial insemination.
3. Balance cattle feed.
4. Seed for fodder production.
5. Training for animal management.
6. Generation of the funds.

Role of Co-operation in Dairy Industry

Dairying is an instrument of socioeconomic changes. Dairy business organized through co-operative sector brings about economic and social changes in society.

a. Economic Role or Change

1. Co-operative moment eliminated the middle man from the sale and purchase of milk and milk products.
2. It built the dairy industry by offering market to milk and gaining consumers faith towards the quality of milk and milk products.

3. Due to promising good price to milk the dairy industry settle as a profitable business which increases income of agricultural family.
4. To project financial condition of milk producer's co-operative moment offer free extension and veterinary services which help to increase milk production.
5. Through the co-operative movement big market achieve for milk in state interstate and abroad.
6. Latest knowledge and technique provide to producer.
7. Establishment of processing plant like AMUL.
8. Financial assistance to milk procedures and save needs of producer.
9. Reduce the adulteration.
10. Regular payment to producer.

b. Social Role of Change:

1. Development of social leadership.
2. Elimination of untouchability.
3. Elimination of inequality.
4. Solve the problem of unemployment.
5. It gives the health facility to producer children and their family.
6. It gives educational as well as cultural facilities to milk producers.
7. It helps to produce clean and safe milk and control milk bone diseases.

Study of Government Milk Supply Scheme

Role of Government Milk Supply Scheme:

1. Collection of milk: Milk is collected at village through milk produces co-operative society controlled by government milk supply scheme by sampling, testing, grading, weighing etc.
2. Transportation of milk: Collected milk at society is transported to government milk supply scheme. It is organized by government milk scheme.
3. Chilling of milk: Chilling centers are established at distant places by government milk supply scheme where raw milk is chilled and increase the keeping quality.

4. Processing of milk: Collected milk at government milk supply scheme is pasteurized milk to increases the keeping quality before marketing.
5. Production of milk products: The surplus milk is converted into different milk products, i.e., Butter cream, ghee, SMP, WMP, etc.
6. Marketing of milk and milk products: Government milk supply scheme is making milk and milk product in distant market. They also conduct market survey, establish new market. It also tries to meet the demand for milk & milk products in different market.
7. Organized distribution: Distribution of milk & milk products is organized properly and successfully.
8. Ensure better returns to the primary milk products
9. Enhance more production, better quality & cheaper milk.
10. Saving: Save the seasonal problem i.e. in winter season it convert surplus milk into milk products and in summer it maintain the demand and price of the market.
11. Store milk & milk products: Milk & milk products are perishable so cannot store long so facility for the storage is provide by government milk supply scheme.
12. Intensive cattle development scheme implementation.
13. Payment of milk: Payment is given to taluka sangh & thus to society. Payment is given on basis of minimum standards and an additional payment is given to higher quality milk while less price to poor quality milk.

Organizational set up/network/line diagram/infrastructure

At the different places government milk supply schemes are established to perform processing of milk and to convert milk into milk products. In the processing there may be standardization, sterilization, pasteurization. In the milk products there may be cream, butter, ghee, ice-cream, SMP, WMP.

When there is surplus production of milk it is converted into milk product or skim milk powder or WMP. SMP or WMP is utilized when there is scarcity of milk production. Before the processing government milk supply scheme collect milk, grading of milk, chilling of milk, transportation of milk. After processing it is stored into storage tank or distribute as per the requirement.

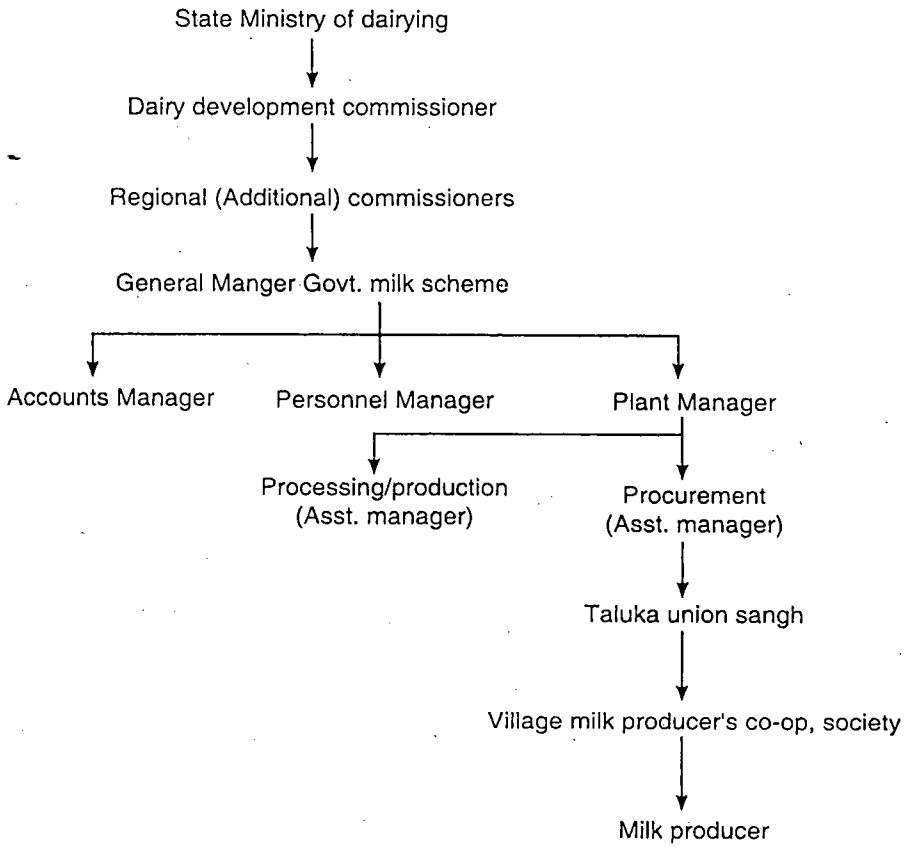


Figure 1: *Working of government milk supply scheme*

Anand Pattern

Anand Pattern is a world famous 4 tier system of dairy development in India. In foundation, there are a large number of small milk producers in villages, who are the members of village level Co-operative dairy (Generally called Society). Such societies form a district level union, having a processing plant. The third tier is a state level co-operative milk marketing federation, which looks into aspects of marketing of milk and milk products, with the aim of giving better benefits to the producer and ensuring supply of quality products to the consumers at reasonable rates. The profits gained goes in giving social and monetary benefits to the producers. The fourth components added recently is a cluster of state level milk marketing federations called as National Co-operative Dairy Federation of India (NCDFI).

Technology Mission on Dairy Development

The technology mission on dairy development was launched mainly to tackle the problems thrown up by the operation flood programme such as lack of professional management, and viability of dairy co-operative to prevent retardation of progress. This mission was launched in September 1989 to achieve the following objectives:

1. To accelerate the pace of increasing rural income employment through dairy development.
2. To consolidate the achievement of the co-operatives dairy sector.
3. To provide new technology for larger more energy efficient dairies processing facilities and to reduce the cost of operation at every stage, from farms to the market.

Dairy industry is indebted to various national and international organization which have given their whole hearted support for the development of this sector like WHO, FAO (Food and Agricultural Organization), EEC (European Economic Community), UNICEF, DANIDA, TCM and have actively particularly in the dairy development programmes in India. Various governments particularly New Zealand, USA, UK, Denmark, Holland, Germany and Australia have been liberal in providing financial assistance to improve dairy industry in the country.

CHAPTER 2

Definition, Composition of Milk and Factors Affecting Composition of Milk

Milk is an important article of daily diet of vast population of earth due to its high nutritive value. Milk is complex chemical substance produced naturally by all mammalians. It is a liquid food secretion by the mammalian for the nourishment of their young ones. However since ancient time, humans have exploited this natural food for their own benefits. The utilization of milk has become a business and hence it is required to be legally defined. Each country sets its own standards for major constituents of milk for safe trade.

In India Prevention of Food and Adulteration (PFA) act defines the milk. According to this milk can be defined as the whole, fresh, clean, lacteal secretion obtained by complete milking of udder of one or more healthy milch animals, excluding that obtained with in 15 days before or 5 days after calving or such periods as may be necessary to render the milk practically colostrum free and containing the minimum prescribed percentages of milk fat and milk solids not fat (SNF).

(Milk serum is defined as milk plasma minus casein micelles, which is close to the composition of whey.)

Market milk refers to fluid whole milk that is sold to individuals usually for direct consumption. It excludes milk consumed on the farm and that used for the manufacture of dairy products.

There are certain terms used in connection with the above constituents when referring to them singly and in groups. Milk fat or butter fat refers to the fat of milk. The protein, sugar and ash of salt of milk are termed milk solid not fat or solid not fat. Those in dairy work often refer to this group of solids as the term serum solids of milk. The dry matter is commonly designated as total solids.

Composition of Milk

The major constituents of milk are: water, fat. protein, lactose, ash or mineral matter. The minor constituents are: phospholipids, sterols, vitamins, enzymes, pigments etc. The true constituents are: milk fat, casein, lactose.

Composition of milk of different species

Sr. No	Species	Water %	Fat %	Protein %	Lactose %	Ash %
1	Cow	86.6	4.6	3.4	4.9	0.7
2	Buffalo	84.2	6.6	3.9	5.2	0.8
3	Goat	86.5	4.5	3.5	4.7	0.8
4	Ewe	79.4	8.6	6.7	4.3	1.0
5	Camel	86.5	3.1	4.0	5.6	0.8
6	Human	87.7	3.6	1.8	6.8	0.1
7	Ass	90.0	1.3	1.7	6.5	0.5
8	Mare	89.1	1.6	2.7	6.1	0.5
9	Dog	75.4	9.6	11.2	3.1	0.7
10	Cat	84.6	3.8	9.1	4.9	0.6
11	Elephant	67.8	19.6	3.1	8.8	0.7
12	Sow	89.6	4.8	1.3	3.4	0.9
13	Whale	70.1	19.6	9.5	--	1.0
14	Porpoise	41.1	45.8	11.2	1.3	0.6
15	Reindeer	68.2	17.1	10.4	2.3	1.5
16	Llama	86.5	3.2	3.9	5.6	0.8
17	Guinea pig	82.2	5.5	8.5	2.9	0.9

Sources: Chemistry of milk by Davies (1939)

Major Constituents of Milk

1. **Water:** This water is not different from ordinary water and serves to hold in solution the soluble constituents of the milk. The percentage of water varies from 84.0 to 89.0 per cent although, occasionally, an individuals sample of authentic milk may exceed these limits.
2. **Milk fat (lipid):** Milk fat often called butter fat is commercially the most valuable constituents of milk. It has a great importance from the standard point of the food value of the milk. The bulk of the fat in milk exists in the form of small globules, which average size of approximately 2 to 5 microns in size (range 0.1 to 22 microns). During the first part of the lactation period, the fat globules are relatively large while as the lactation period progresses the globules become smaller. Fat exist in oil in water type emulsion. Chemically milk fat is composed of a number of glyceride esters of fatty acids; on hydrolysis, milk fat furnishes a mixture of fatty acids and glycerol.
3. **Milk proteins:** Proteins are among the most complex of organic substances. They contain carbon, hydrogen, oxygen, nitrogen, sulfer and some times phosphorus. Proteins are composed of a large number of amino acids, some are essential and others are none essential amino acids. The essential amino acids are necessary in the diet for the formation of body proteins. The proteins of milk consist mainly of casein and whey proteins (Beta-lactoglobulin, alpha-lactalbumin, etc). Casein exists only in milk and is found in the form of a calcium caseinate phosphate complex.
4. **Lactose/Milk sugar:** Lactose, a disaccharide sugar exists only in milk. It is in true solution in the milk serum.

Minor Constituents of Milk

1. **Mineral matter of ash:** The mineral matter or salts of milk, although present in small quantities, exert considerable influence on the basic chemical properties and nutritive value of milk.
2. **Phospholipids:** In milk three types of phospholipids exist, viz., lecithin cephalin and sphingomylin,
3. **Cholesterol:** This appears to be present in true solution in the fat, as part of the fat globule membrane complex and in complex formation with protein in the non fat protein of milk. Milk contain 105-176 ppm of cholesterol.
4. **Pigments:** These are (i) fat soluble, such as carotene and xanthophylls which is responsible for the yellow colour of milk fat and (ii) water soluble, such as riboflavin which is contributing a yellowish green tint colour to the serum or whey.
5. **Enzymes:** These are biological Catalysts which can hasten or retard chemical changes without themselves participating in the reactions. The important milk enzymes and their specific actions are as follows. Analyses (diastase) is a starch splitting enzyme. Lipase is a fat splitting enzyme leading rendering rancid flavour. Phosphatase is capable of splitting certain phosphoric acid esters (basis of phosphates test for checking pasteurization efficiency). Protease is capable of splitting protein. Peroxidase and catalase decomposes hydrogen peroxide.
6. **Vitamins:** Although present in foods in very minute quantities, these are vital for the health and growth of living organisms. Fat soluble vitamins A, D, E and K, and water soluble vitamins of B complex group (such as thiamine or B1, rioboflavin or B2, Pantothenic acid niacin pyridoxine or B6 biotin, B12 folic acid etc. and vitamin C (ascorbic acid).

7. **Gases:** Milk contains from 7 to 10 percent by volume of gas, Carbon dioxide is present in milk as it comes from the udder, while nitrogen and oxygen are taken up by the milk during milking.
8. **Non-protein Nitrogen Substances:** Milk contains a group of nitrogenous substances in addition to the proteins. These substances do not occur in large quantities but they are known to be present.

Chemical composition of milk of Indian breeds

Breed of Cow/ buffalo	Percentage of composition				
	Water	Fat	Protein	Lactose	Ash
Sindhi	86.07	4.90	3.42	4.91	0.70
Gir	86.44	4.73	3.32	4.85	0.66
Tharparkar	86.58	4.55	3.36	4.83	0.68
Sahiwal	86.42	4.55	3.33	5.04	0.66
Murrah	83.63	6.56	3.88	5.23	0.70

Sources IDRI Annual Report (1948)

Chemical composition of milk of foreign breeds of cow

Breed of Cow/ buffalo	Percentage of composition				
	Water	Fat	Protein	Lactose	Ash
Holstein	87.74	3.40	3.22	4.87	0.68
Shorthorn	87.19	3.94	3.32	4.99	0.70
Ayrshire	87.10	4.00	3.58	4.67	0.68
Brown Swiss	86.59	4.01	3.61	5.04	0.73
Guernsey	85.39	4.95	3.91	4.93	0.74
Jersey	85.09	5.37	3.92	4.93	0.71

Sources: The fluid Milk Industry by Henderson (1971)

Factores Affecting Composition of Milk

Milk differs widely in composition. All milks contain the same kind of constituents, but in varying amount. The variation is always greater in small herds than in large ones. Dairy manager of every dairy farm or owner should know the different factors which affect the quality and quantity of the milk for economic dairy farming. In general, milk fat, shows greatest daily variation than comes protein followed by ash and sugar. There are various factor affecting the quality and quantity of the milk and these are as follow

1. **Species:** The milk from various species of mammals have different composition.

Species	Water%	Fat%	Protein%	Lactose%	Ash%
Cow	87.54	3.71	3.31	4.7	0.76
Buffalo	82.90	7.5	4.1	4.7	0.80
Goat	85.58	4.93	4.11	4.78	0.89
Human	88.50	3.3	1.3	6.8	0.20

Thus each species yields milk of characteristic composition. Different species of mammals yields different amount of milk.

2. **Breed:** The composition of milk is strongly influenced by breed. In general, breeds producing the largest amounts of milk yield milk of lower fat percentage. Generally milk containing a higher percentage of fat is also rich in solid-not fat and vice-versa.

Species	Water%	Fat%	Protein%	Lactose%	Ash%
Jersey	85.43	5.14	3.8	4.87	0.76
Sindhi	86.07	4.90	3.42	4.47	0.70
Sahiwal	86.42	4.55	3.33	5.04	0.66

3. **Individuality:** Even in herds kept under identical condition of management and feeding each cow tends to yield milk of a composition that is characteristics of the individuals.
4. **Interval of milking:** When milking is done at longer interval, the yield is also more with a corresponding small percentage of fat where as milk drawn after short intervals yield smaller quantities with higher amount of fat. The variation due to interval between milking appears to apply only to the fat content.
5. **Completeness of milking:** If the cow is completely milked the test is normal while incomplete milking shows usually low tests. The first milk drawn from the cow is low in fat while the last milk is very in fat (stripping).
6. **Irregularity of milking:** Frequent changes in the time and interval of milking result in lower tests.
7. **Frequency of milking:** Weather the cow is milked two, three or four times a day, it has no great effect on the fat test.
8. **Day to day milking:** Day to day milking may show variation for the individual cow. The percent composition of milk as well as milk

yield vary considerably from day-to-day. In general daily variations of milk yield are caused by excitement, estrus, incomplete milking, disease, under feeding and related factors.

9. **Portion of milking:** Foremilk is low in fat content while striping is highest. The other milk constituent are only slightly affected on the fat free basis.
10. **Stage of lactation:** The yield of the milk increases to a maximum in early lactation and the fall to normal. Towards the end of lactation milk contents higher amount of minerals. The first milk is colostrum and change from colostrums to milk takes a few days. Colostrums is lower in water, sugar and fat and higher in casein, albumin, globulin and ash than normal milk. The size of the fat globules of a fresh cow secretes relatively large fat globules. There is considerable decrease in size until about the second month.
11. **Age:** The effect of age of animal is more important with regard to yield than the composition of milk. As the age of the animal advances the quantity of milk produced by the animal decreases. The fat percentage in milk declines slightly as cow grows older.
12. **Condition of cow at calving:** If the cow is in good physical condition at calving, it will yield milk of higher fat percentage than if its physical condition was poor.
13. **Diseases and abnormal condition:** Diseases and abnormal condition of cows causes a marked reduction of milk yield and change in composition of milk. e. g. Mastitic milk.
14. **Yield:** For a single cow, there is a tendency for increased yields to accompanied by a lower fat percentage and vice versa.
15. **Season:** The percentage of fat and SNF show well defined variation during the course of year. An increasing temperature favors slight decrease in fat and vice versa. The amount of vitamin D in milk directly related to the exposure of the cows to sunshine. Carotene and riboflavin is maximum when green feeds are available. The green feeds are plenty in rainy season. Normally in mansson and winter fat content in milk is lower as compare to that in summer season. Milk production is usually less during the summer because of the higher environmental temperatures and the prevalence of green-forage scarcity.
16. **Feeding:** The feeding has temporary effect only. The feed affects the composition of milk to a great extent. Mineral matter of milk will increased by feeding mineral supplements. The ration containing high crude fiber produces milk with higher fat percentage. Feeding of green fodder results into yellowish colour in milk particularly in case of cows.

17. **Excitement:** During the periods of sexual excitement, storms, worrying affects the yield as well as composition of milk.
18. **Body weight:** The general relationship exists between the body weight of the cows and level of the milk production. Larger cows have more udder secretary tissue for alveoli and larger digestive system.
19. **Gestation:** A mark drop in milk production occurs towards the end of pregnancy.
20. **Dry period:** A dry period is important for replenishing body supplies. If the cow is in poor body condition at calving and also to regenerate milk secretary tissue an increase of highest milk yields results with dry period between 50 to 90 days.
21. **Administration of drugs and hormones:** Administration of certain drug or hormone has temporary effect on yield and composition. Prolactin injection increases yield and fat percentage. The synthetic hormones like thyroprotein (iodinated casein), synthetic hormone have been in common practice for feeding lactating cows. Provate injections prepared from gonadotropin hormone are in use to increase the milk production. This practice is not legal; hence it is prohibited by some countries of the world.
22. **Effect of oestrus cycle:** When the animal is in heat, considerably abnormal milk is obtained from such animals.
23. **Stress:** Recently, more attention has been focused on the role of stress in the secretion of milk. As animals are selected to secrete higher levels of milk, any sort of stress will play an increasingly important role in lactation.

CHAPTER 3

Physico-Chemical Properties of Milk

Milk is not only of a complex chemical nature, but , due to the various constituents, its physical nature also is very complicated. Water is the dispersion media for other solids. It is in continuous phase of liquid to which the constituents are dispersed. The chemist considers that matter exists in three general states of dispersion the limits of which have been set arbitrarily as follows; Coarse dispersion; particles greater than 0.0001 millimeter; Colloidal dispersion; particles between 0.0001 millimeter & 0.000001 millimeter and Molecular dispersion where the particles are less than 0.000001 millimeter in diameter. All three of these states of dispersion are known to exist in milk. The proteins of milk, in general, are dispersed colloidally through milk in finely divided particles, of which casein is the largest and albumin and globulin are smallest. The calcium phosphate is associated with the proteins and also colloidally dispersed. The fat of milk is the only constituent which occurs in the form of coarse dispersion or an emulsion. In milk, water is the continuous phase and for this reason the emulsion is said to be of the fat-in-water type.

Chemical Properties

1. **Acidity of milk:** Freshly drawn milk is amphoteric in nature i.e. it turns red litmus blue and blue litmus red. However, it a shows certain acidity as determined by titration with an alkali (sodium hydroxide) in the presence of an indicator (phenolphthalein) this acidity is called as titratable acidity (T.A) is usually expressed as a percentage of lactic acid. The titratable acidity of cow milk varies on an average from 0.13 to 0.14 per cent and buffalo 0.14 to 0.15 per cent. Titratable acidity of milk is also know as natural or apparent acidity. It is caused by the presence of casein, acid-phosphates, carbon dioxide and citrates in milk. Developed acidity or real acidity is due to lactic acid, formed as result of bacterial action on lactose in the milk. Hence, the titratable acidity of stored milk is equal to the sum off natural acidity and developed acidity. The natural acidity of fresh milk vary with species, breed, individuality, stage of lactation, health of the animal etc. Natural acidity in milk is considered important from the heat stability point of view. Colostrum is high in acidity but in a few days following calving the acidity of milk becomes normal.
2. **pH of milk:** pH is defined as negative logarithm of hydrogen ion concentration. pH of normal fresh cow milk ranges from 6.4 to 6.6 and buffalo 6.7 to 6.8. It may vary with species, breed, individuality, stage of lactation, health of the animal etc. Higher pH values for fresh milk indicates udder infection (Mastitis) and lower value, bacterial infections.
3. **Buffering action milk:** Fresh milk acts as a complex buffer because of the presence of carbon dioxide, proteins, phosphates, citrates and a number of other minor constituents This property of the milk is considered important form for the curdling and also from heat stability point of view. Buffer value of milk is 0.024 at pH 6.00 and 0.18 on pH 6.6.

Physical Properties

1. **Colour:** The normal milk ranges in colour form yellowish creamy white to creamy white. Normal cow milk has yellowish creamy white colour due to the presence of carotene pigment which is soluble in butter fat. Buffalo milk is white in colour due to the absence of carotene pigments which is efficiently converted to vitamin A. After removing milk fat, the skim milk shows greenish tint which is due to the presence of riboflavin or lactochrome. Intensity of colour of milk it depends upon the kind of feed, breed, amount of fat and solids present in milk. Dilute acid or rennet when added results in coagulation of casein and fat and the separated whey will be having a distinct greenish yellow colour due to the pigment riboflavin.
2. **Taste:** Normal freshly drawn milk has slightly sweet in taste. As lactation advances, lactose declines while chloride increases; the taste is deflected towards salty. A similar effect is caused by udder infections. When odouriferous substances like garlic are fed, milk gets the characteristic taste, by its passage to the milk from the blood. Milk also absorb the volatile vapours present in the atmosphere.
3. **Odour/smell/flavour/aroma:** A freshly drawn milk has a characteristic mild, pleasant flavour. Some research workers attribute the rich flavour of milk to the fat present in milk. The change in the flavour of milk occur due to type of feed, season, stage of lactation, condition of udder, sanitation during milking, handling of milk during storage, and sour due to developed acidity. The sulfydryl compounds significantly contribute cooked flavour of heated milk.

4. **Consistency:** Generally cow milk has thin consistency while buffalo milk has thick consistency. The consistency may change due to microorganisms, addition of starch, sugar, water etc. The ropy consistency of milk may due to formation of enzymes produced by bacteria.
5. **Density/Specific gravity:** Density of milk is its mass (weight) per unit volume. Specific gravity is defined as unit volume weight of milk divide by unite volume of water. It is ratio of unit volume weight of milk to unit volume of water. Specific gravity is determined by pycnometer or hydrostatic balance and lactometer. The common types of lactometers are Zeal and Quevennes. The average specific gravity of normal cow milk ranges from 1.028 to 1.030 at 60°F and normal buffalo milk ranges from 1.30 to 1.32 at 60°F and 1.035 to 1.037 for skim milk . Milk is heavier than water. The variation in specific gravity of different individuals milk is due to the flocculation in water, fat, protein, lactose and minerals of milk. There will be difference in specific gravity of liquid fat and solid fat. Addition of water lowers the specific gravity. High fat milk has low density. Removal of fat results in separated milk or skim milk or defatted milk, which has a higher specific gravity. The specific gravity of individual constituents in milk is approximately as follow. Water 1.0; fat 0.93; protein 1.346; Lactose 1.666, Minerals (salts) 4.12 and SNF 1.616
6. **Boiling point:** The boling point of milk is 100.17°C or 212.3°F. It is slightly more then water due to higher percentages of salt and sugar in milk. Water boils at 100°C under normal atmospheric temperature and pressure.
7. **Freezing point:** This is the temperature at which the liquid phase may freeze or crystallize and the solid phase may melt or liquefy. Pure distilled water freezes at 0°C under normal atmospheric pressure. Milk freezes at temperature slightly lower than that of water due to soluble constituents which raise the boiling point, viz., lactose, soluble salts, etc. The average freezing point depression of Indian cow milk may be taken as 0.547°C (31.02°F) and buffalo milk 0.549°C (31.01°F) . Most bulk milk samples have a freezing point depression of 0.530°C (31.05 °F). It is determined by the instrument known as Hortvet cryoscope or cryoscope. The addition of 1% water of milk raise the freezing point by 0.006°C.
8. **Surface tension:** Surface tension is a phenomenon attributed to the attractive forces or cohesion between the molecules of liquid exposed to surface areas. These forces of attraction converge to the centre of molecules from all directions. The force required to break 1cm of the surface is called as surface tension. The surface tension of milk at 20°C is 54.5 (ranges from 40 to 60) dynes/cm. It decreases as the temperature is raised or fat per cent is increased. Whole milk has a slightly lower surface tension than skim milk and that of cream is still lower. Milk and cream on aging undergo a slight decrease in surface tension. Colloidal constituents like proteins also lower the surface tension along with fat globules.

9. **Viscosity:** The viscosity of a substance refers to its resistance to flow caused by intra-molecular attraction. At normal temperature the viscosity of milk varies from 1.5 to 2 centi poises at 20°C. The viscosity of milk determined by viscometer. The viscosity of milk is always higher than viscosity of water due to the presence of dissolved solids in milk. Aging, souring and the development of certain type of bacteria in milk are responsible for increase in viscosity. Increase in temperature causes reduction in viscosity. Viscosity is exceedingly important in marketing of few of the milk product. Homogenization increases the viscosity of milk.
10. **Electrical conductivity:** Electric conductivity is measure of ability of solution to carry current. It is 0.005 ohm^{-1} cm^{-1} at 25°C. The electrical conductivity increases with increase in temperature. About 80% of current in milk is carried by chloride ions [chloride content increases during mastitis]. Electrical conductivity is determined by electric conductivity bridge.
11. **Specific heat:** Specific heat is the amount of heat required rise a unit mass through a unit temperature rise. Specific heat of milk with of normal constituents ranges 0.92 to 0.93 it changes with change in temperature.
12. **Refractive index:** Refractive index is obtain by the ratio of angel of incidence to angel of reflection. The refractive index of milk is 1.35 at 20°C while that of water is 1.33.
13. **Coefficient of thermal expansion:** Milk expand when heated and contracts when cooled, the ratio of increase in volume per degree of increase in temperature is known as "coefficient of expansion".
14. **Oxidation Reduction Potential:** Oxidation is defined as the uptake of oxygen or loss of hydrogen. In the same manner, reduction may be defined as the process of loosing oxygen or gaining hydrogen. In ionic system, it can be demonstrated that phenomenon may involve loss or gain of electrons. In practice, the potential difference created by platinum electrode in a solution of an oxidant or reductant is measured by completing a circuit through calomel half-cells and a potentiometer. The voltage measured under these conditions reflects the oxidizing or reducing capacity of the solution. This potential is called as oxidation-reduction potential or O-R potential (or) redox potential and is designated by Eh.

15. **Adhesiveness of milk:** Milk is more sticky than water. Milk possesses adhesive property. When a piece of paper is moistened with milk, it sticks to surface of wood or glass or metal due to casein glue. This property of milk doubtless is due to the casein of milk.
16. **Coagulation/effect of acid, heat, enzymes and rennet:** Milk is coagulated by heat, acidity, rennet etc. Milk coagulates when there is addition of acid or increased acidity of milk. Coagulation of milk proteins and minerals due to heat and acidity. Milk subject to many changes by enzymes which are produced by microorganisms in milk.
17. **Recknagel phenomenon:** Recknagel found that the specific gravity of freshly drawn milk was lower than the specific gravity subsequently obtained, after an hour or later. He found that the rise in specific gravity to be regular, more rapid at lower temperatures than at higher ones and to amount on an average of 0.001. This is called Recknagel's phenomenon.

CHAPTER 4

Layout of Milk Processing Plant

A good layout reduces (a) The effort in normal material handling of a worker, (b) Cost of handling (c) Indirect manufacturing cost by decreasing spoilage under difficult handling situation. It uses only essential space required and eliminates congestion and accidents, permits flexibility for methods improvement and future expansion besides making supervision easier. It also facilitates scheduling and dispatching operations.

Layout of milk processing plant may be straight through plan, the 'V' plan and the 'T' plan.

- **Straight through plan:** In this, reception of milk is designed at one end, the processing machine in the middle and the cold storage at the other end to dispatch the final products at the opposite side of reception.
- **The 'V' plan:** The raw milk reception will be at a corner of one end the processing plant at the opposite end but the dispatch dock is planned at the same end as reception, but at the opposite corner.
- **The 'T' plan:** In this plan boiler, compressor and maintenance room will be on one side and the milk reception, processing and loading out is planned as in case of straight through plan.

Role of Plant Manager

Building and equipment are important component of dairy processing plant [D.P.P.] Professional engineers and equipment manufactures assist in designing and constructing these units. The dairy processing plant manager must be able to evaluate the engineers implications of both aspects so as to plant enable to easy handle milk & production of milk products according to his capacity.

A dairy processing plant handles bulky, valuable & highly perishable food product. Milk in processing plant must be protected from contamination by odours, chemicals, microorganisms and other filth. Standards of sanitation of building site and building interior must be followed.

Factors considers for selection of site for dairy processing plant

1. **Milk shade area:** Building site must be at the center of row milk supplying area. The milk production from milk shade area must be more than the plants capacity so it will work regularly without loss. Milk production is so much that there will be scope for feature expansion. Milk shade area- it is a geographical area from where milk-processing plant receives its raw milk supply.
2. **Topography:** Dairy plant should be located on high elevation of proper drainage, for good slop of rain fall, to avoid stagnation & to maintain attractiveness. Low laying area with improper drainage is unsuitable for dairy processing plant
3. **Land and Price:** Land price is also important factor for site selection. Higher purchase cost of land will increase the fixed investment. As access to good road and rail line. Two acres of land is required for a milk processing plant handling about 10000 liters of milk per day (8 hours).
4. **Access to good road & rail line:** Building site must be near to good road and rail line which help quick transportation with economic. Availability of rail line for quick, bulky transportation of processed milk to distant market.
5. **Proximity of good market:** Building site must be near to good market, which reduces cost of transportation resulting in increasing profit. Plant site should be located where heavy demand for milk and milk product. Easy availability of spare parts of machine and equipments.
6. **Suitable water supply:** Water is an absolute requirement for dairy processing plant. In general for processing of 1 lit. of milk there is need of 8 lit. of water. Water must be fresh, clean, safe and soft. Hard water supply is unsuitable for washing the dairy equipments.
7. **Independent electric supply:** Electricity is requiring for various purposes in dairy processing plant i.e. lighting, running the different machine etc. milk processing is a continuous process, so it requires constant electric supply. Interrupted electric supply will disturb the processing. For constant electric supply plant should have facility of independent generators.
8. **Adequate labor supply:** Surrounding of processing plant should able to supply more labor than the plant requirement, shortage of labor increases cost of production. Labor should be honest, economical, skilled etc.
9. **Sewage (drainages) disposal facility:** Water is use constantly in dairy processing plant mainly for washing the equipments, cans, floors, bottles etc. the waste water contains residues of milk detergent, chemicals and other filth. This is a source of contamination and also damages the floor. Therefore speedy disposal of sewage is essential. Building site should provide facility for sewage disposal. Their may be arrangement with corporation disposal facilities.
10. **Building cost:** Construction of building should be strong, high utility, proper area & arrangement but at low cost.
11. **Availability of equipment & spare part:** Dairy processing plant requires numbers of equipment & the spare part. These equipments and spare part should be available easily, quickly & economical, in nearby market with ease, which is needed for constant processing.
12. **Advertisement:** It is easy to possible to make effective and economical advertise.
13. **Area for future expansion:** There should be chance for vertical and horizontal expansion for increase the business and production.

Functional design of layout of dairy processing plant

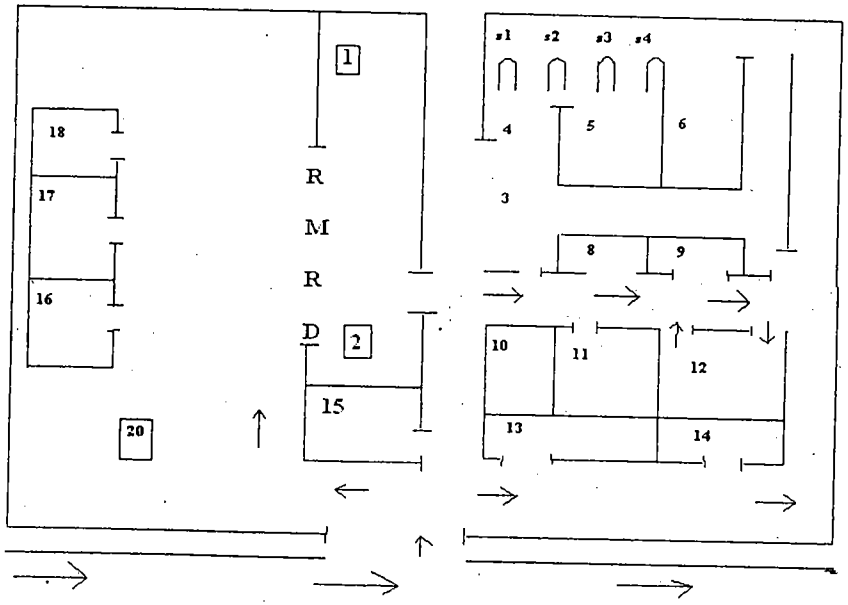
It requires separate rooms for receiving raw milk, washing and sterilization, pasteurization cooling and filling operations, quality control laboratory, dry storage, canteen, toilet, and washes room facilities for staff, separate sales area, workshop, and office. Besides ancillary boiler, refrigeration and connected service lines, fire protection systems are to be located as close as possible. Point considers to designing lay out of dairy processing plant are

1. Boiler unit should away from processing plant.
2. Pasteurization, separation, homogenization plant, clarification should be near to each other.
3. Storage tank should be near to these units.
4. If there is separate milk products there should be separate building.
5. Bottling, capping should be near to storage and processing.
6. Laboratory should be centre of the dairy processing plant
7. Office should be at the entrance of the plant and financial department should be close to office.
8. Account and financial department should be close to office.
9. Canteen and W.B.C. should be away from dairy processing plant
10. There should be proper arrangement of chain convey.

Essential sections of a milk processing plant

1. **Raw Milk Receiption Dock (RMRD):** Consisting of can conveyor, can washer, weighting balance, dump tank etc.

2. **Processing Hall:** Cream separator, chiller, homogenizer, pasteuriser and other related machinery.
3. **Storage area:** For milk storage tanks.
4. **Products manufacturing area:** Depends upon the type of products and the quantity of milk handled.
5. **Packing area:** For packing of liquid milk and other products.
6. **Cold storage:** For keeping the milk and milk products before sending to market.
7. **Quality Control Laboratory:** For testing the quality of milk and milk products.
8. **Utilities area:** For installing boiler, generator set, water treatment plant, maintenance and store area for spares.
9. **Waste water treatment plant area:** For treating the dairy effluents before releasing to the fields.
10. **Quarters and office area:** For all the essential staff.
11. **Vehicle parking area:** Both for the milk procurement and distribution vehicles.
12. **Input supply area:** For providing veterinary service, supply of feed, fodder seeds, etc.



1. Milk receiving plat form
2. Can washer
3. Chilling unit
4. Pasteurization plant
5. Cream separator
6. Bottle washer

7. Receiving of empty bottle
8. Bottle filler
9. Capping unit
10. Butter Churman
11. Butter cold storage
12. Milk bottle cold storage
13. Office
14. Canteen and W.C.
15. Quality control lab
16. Workshop
17. Refrigation
18. Heating section
19. Quartors
20. Weaighing balance

SI, S2, S3, S4: Storage plant

Advantages of Good Layout

1. Increase out put.
2. Reduce risk to help
3. Improve moral and production saving floor shape.
4. Reduce delay in production saving floor shape.
5. Reduce handling of worker.
6. Proper utilization of man power, machine and services
7. Easy and better supervision
8. Proper appearance and sanitary condition
9. Less conjunction and confusion

Infrastructural Facilities

1. Raw Material

a. The principal raw material is milk.
b. The method of procurement, transportation of milk.
c. The availability of other inputs such as packing materials, disinfectants and consumable.

2. Power

a. Normally a three phase electricity supply is required for milk processing plants.
b. The power requirement depends upon the load to be connected.
c. Depending upon the position of power supply, standby generators may be considered for connecting the essential sections.

3. Water

a. A milk processing plant requires the water in the ratio of 2:1 (2 liters of water for 1 liter of milk processed) for cleaning of equipment, cold storage and drinking purposes.

4. Steam

a. The steam requirement (kg/hr) depends upon the processes involved and the source of steam may be met by coal/oil/gas fired /electric boiler.

5. **Fuel**
   a. Diesel/petrol /coal/gas/ requirement and the availability to be ascertained.
6. **Compressed Air**
   a. It will be required for various pneumatic operations flow control operations as well as for cleaning purposes.
7. **Vehicles**
   a. The vehicles required for procurement and distribution of milk depends on the quantity of milk to be handled.
   b. The number of vehicles required, source of supply, rental charges etc.
8. **Other Services**
   a. Although a maintenance workshop is an integral part of milk processing for carrying out repairs and maintenance of equipment.
9. **Communication**
   a. Proper communication facilities are essential.

Construction of Milk Processing Plant

Floor:

Sufficient floor space is a primary necessity for efficient layout. Location of equipment either too close to each other or near to the walls and ceiling should be avoided.

A. Floor is Required in Dairy Processing Plant for

- Space for working.
- Space for equipment.
- Space for additional equipment.

B. Allotment of Floor

One meter distance between two equipments in processing hall. For the store and office the area should be 25% of the total floor area. Space may be reserved for empty cans or bottle.

C. Characters of Floor

Following characteristics require for dairy processing plant floor

1. **Floor must be durable:** The foundation & structure of the floor must be capable of large equipment like storage tank, workload and traffic. It should tolerate physical friction and thermal shock.

Loose and iron sheets are used as floor covering in the receiving plant form.

2. **Floor must be acid resistance:** Different acids use in dairy processing plant may affect floor. So floor should be acid resistance. The milk solid in the cracks and pores of floor ferment to produce acid such acid may damage to floor.
3. **Floor must be properly drained:** Floor must be sufficiently sloping for speedy drainage. Too sloppy floor cannot give protection to worker. Floor which drain well are less slippery. Drain for dairy plant floor should be square type with flush joint in the bottom so that no liquid is retained in the drain and they easily cleaned.
4. **Floor should be non sleepery:** Floor must be non slippery as it gives protection to workers.
5. **Floor should be easy to clean:** Rough floor surface prevent slipping but it is difficult to clean. So floor surface must be smooth as easy to clean but non slippery it should be strong.

D. Material for Flooring

Various type of material used in flooring of dairy processing plan as follow:

1. **Cement concrete:** It is the very common material for flooring of dairy processing plant it should be relatively cheap. Non crackable and strong and it should be easily repaired.
2. **Terrazzo:** It is combination of cement concrete with marble. Marble chip to make strong, airproof and smooth flooring in dairy processing plant
3. **Kotha stone:** A thick stone is use and dressed on floor side. The thickness of Kotha also 50 mm.
4. **Iron grill:** These are use for covering drainage' unloding, processing or heavy impact of temperature and chemicals.
5. **Still plate:** Up to 6 mm thick plates are use at heavy traffic, heavy load.
6. **Tiles, breaks/metal tiles:** These are use as per requirement particularly.

Windows and Doors

Sufficient number of door and window are provided for the better expose to sunlight and properly ventilation. The panels, frames and screens must be easily and regularly cleaned. Care should take that doors and window should be opposite to each other. It should not be directly open toward the direction wind and rains.

Roof or Ceiling

Best ceiling is on well with plane, smooth, horizontal surface with light fixtures inset in dealing. Such ceiling is more sanitary than those suspended from it. Exposed beams and girders of iron or reinforced concrete may harbors dust and dirt which may contaminate both products and equipments.

Rodent Control Beat Stations

Rodent, bird and insect control is also an important aspect for control of plant sanitation and hygiene. Earlier harborage must be eliminated and baiting and traps should be used inside the plant with due care. All incoming materials, including machinery boxes, may be examined to prevent entry of rodent inside the plant.

Plant and Machinery

The plant operation efficiency depends on arrangement of equipments to produce the desired products with minimum efforts, expense and waste of time and high labour efficiency. Square design of building is preferred for compactness and economy in material used. Floor space one meter to be provided for every ten liters of milk processed. Most of the dairy machinery are manufactured in the country by ALFA-LA-VAL, L&T, HMT, Nichrome Pvt. Ltd., Samarpan Fabricators , Goma Engineering Ltd. etc. Equipments to be procured are

Milk Reception Tank	Plate Type Chiller
Milk Cans,	Quality Control Lab.
Equipments	Cream Separator
Bottle Filter	Milk Bottles or Tetra Pack
Butter Churn	Packing Machine
Can Washer	Dump Tank
Steam Boiler	Can Trolleys
Surface Cooler	Bottle Capping Machine
Crates	Ghee Kettle
Weighing Machine	Exhausts
Milk Pump and Pipe Lines	Storage Tank
Refrigeration Unit	Power Generator
Plate Type Pasteurizer	Bottle Washer
Homogenizer	Plunger

Hygienic Requirements of Processing Plant, Equipments and Personnel

1. Cleanliness of plant surroundings is essential and should be maintained. The first impression regarding the sanitation and hygiene of the plant would be quite accurately gauged by examining its surroundings. Clean surrounding should be maintained by removal of litter, waste; refuse and other unwanted materials from the vicinity of the plant. If the lawn or the garden is maintained around the plant, the lawn should be trimmed and kept free from insects, flies etc. it is important to keep check on rodent burrows and keep the dogs, cats etc. away from the plant.
2. The floors, walls, ceilings, doors and windows must be kept clean.
3. Proper cleaning of drains and removal of waste and waste water from the plant should be carefully ensured.
4. Frequent cleaning of fixtures, fittings etc. at regular interval inside the plant should be undertaken.
5. Furniture, fixtures and equipments should be moved periodically to clean accumulated dirt below them.
6. No unwanted material and machinery or its parts should be allowed to remain in processing and packing areas to avoid abstraction in work and cleaning.
7. Doors of the processing and packing areas should be self closable and windows be covered with wire net.
8. Wherever possible air-curtains should be provided as this will keep flies, insects, dirt and dust away from the processing areas.
9. Efforts should also be made to ensure rodent and pest control.
10. Proper chemical cleaning of manufacturing, storage, transferring and packing equipments be it CIP or after dismantling should be done.
11. A close monitoring on chemical cleaning by bacteriological analysis of surface swabs and rinses should be undertaken.
12. Monitoring of cleaning and sanitizing solutions is equally important.
13. The flow of milk and products should be logical and a proper balance between the inflow of products and processing capacity need to be ensured.
14. Piping system should be as simple as possible and many interconnections, dead-ends and cross flows should be avoided. This would greatly help in proper cleaning and sanitation of lines and equipment and avoid continuous contamination.

15. Good material storage and movement is another important aspect of maintenance of plant cleanliness, sanitation and hygiene. In order to keep the materials storage areas under control, the materials should be stacked on plates and away from the wall for easy cleaning. Sufficient movement space between the storage stacks for personnel, equipment and materials should be provided.
16. Proper lighting and ventilation of the storage area is essential.
17. Provision should also be made to spray insecticide and fumigants in the storage areas to keep the pests away.
18. Supply of adequate amount of clean and potable water, proper drainage and sewerage, adequate toilet facilities possibly away from processing areas, and its regular cleaning and provision of waste bins etc. should be made to ensure proper hygiene and sanitation in the plant.

The Important Activities Related to the Personal Hygiene

1. Careful screening of personnel at the time of recruitment and subsequent periodical check up for detection of infectious diseases and the like
2. Continuous monitoring of personnel for personal cleanliness.
3. Provision of good hand washing and drying facilities not only in the toilets and wash rooms but also inside the sections where solutions of sanitizer such as chlorine water or potassium permanganate solution can be kept in a tube or bowl for washing hands just before start of work.
4. Provision of sufficient pairs of clean uniform. Uniform include head cap, masks and gloves.
5. Provision of basic first-aid kits in work areas.

Environmental Aspects and Pollution Control

1. There are no hazardous effluents generated from a milk processing plant.
2. Construction of effluent treatment plant is necessary for treating the effluents before discharging for proper disposal.
3. The final effluent should meet the requirements of Pollution Control Board and is necessary to get clearance from them.

5 Microbiological Deterioration of Milk

Milk and milk products are popular because of their unique flavour and texture so much so that if these attributes are not attractive the product is rejected by the consumers. As soon as the milk is drawn from the cow, the biochemical reactions of bacteria present in milk are initiated leading to flavour deterioration and other defects which ultimately cause milk spoilage. Apart from the gross manifestation of spoilage such as lactic acid production and souring, milk and milk products also undergo a number of other changes like proteolysis, lipolysis, sweet curdling etc.

Mechanisms of Microbial Spoilage/Changes

The microbial changes in milk and milk products are produced as a result of fermentation of one or more of the milk constituents by the causative micro-organisms. Following three types of interactions among causative organisms.

Synergism

In this type of association the two organisms being about a change neither of them could carry out alone.

Ex. The production of blue discoloration in milk by pseudomonas syncyanea is possible only in association with *S. lactis*.

Metabiosis

In this type of association among micro-organism a " food chain" is formed which means the metabolic end products of one are utilized as food staff by the other for producing the final changes.

Ex. In swiss cheese making the lactic acid bacteria convert lactose to lactic acids which in turn is utilized by propionibacteria to produce propionic acids.

Antibiosis

In this phenomenon the growth of one organism inhibits as suppresses the growth of the second organisms.

Ex. The growth of lactic acids bacteria causes the inhibition of proteolytic organisms (manly spore formers) due to production of lactic acids by the former group.

The various types of microbiological spoilage with regards to their causative organism, significance and preventive measures are described individually in the following.

1. Natural Souring/Curdling (Lactic Acids Production)

If raw milk is held for sometime under ambient conditions, the immediate effect noticed is souring followed by curdling. This is primarily due to the production of acidity (chiefly lactic acid from lactose) by certain groups of bacteria already present in raw milk.

Fresh milk has normal acidity ranging from 0.14 to 0.19 percent. The acidity at which milk sources is usually in the range of 0.2 to 0.25 percent, the milk usually curdles when acidity further increases to 0.50 to 0.65 percent. Generally the acidity continues to increases even after coagulation of casein till the lactic acids producing flora is inhibited by accumulation of acids or till whole of lactose is exhausted when more acids tolerant organisms predominate.

The mechanism of acid coagulating involves interaction of lactic acid produced with the calcium bound to casein thus leading to precipitation of casein which appears as curd. The casein starts precipitating in the pH range of about 4.64 to 4.78.

Causative Organisms

(i) Lactic Acids Bacteria

a. *Lactic streptococci:* This group includes *Streptococcus lactis, Streptococcus cremoris* (which produces lactic acid at room temperature). However, this group gets inhibited beyond 1% level of lactic acid accumulation.

b. *Lactobacilli: Lactobacillus casei* which produce lactic acid at room temperature,

c. *Lactobacillus acidophilus* and *Lactobacillus bulgaricus* (which produce lactic acids optimally at around 40°C). Lactobacilli continue to produce lactic acid beyond 1% lactic acids but get inhibited at 2 % level.

d. *Leuconostocs: Leuconostocs dextranicum* and *leuco. citrovorum* etc. This group is primarily responsible for flavour development and produce lower levels of lactic acids.

(ii) Other streptococci

a. *Streptococcus thermophilus:* It produces lactic acid in milk optimally at around 45°C though it is produced slowly even at lower temperature range. This organism is capable of souring higher heat treatment such as pasteurization.

b. *Streptococcus liquefaciens:* It produces lactic acid optimally at about 37°C. Milk is rapidly coagulated followed by proteolysis. The latter phenomenon causes the curd to shrink from the walls of the container which is followed by whey separation.

c. *Bacillus coagulans:* (lactic acid producing aerobic sporeformers). This organism unlike most of the other lactic acid producers can survive heating conditions during processing of milk and can multiply and produce lactic acids at 37 to 55°C.

d. *Coliforms:* These include *E. Coli* and *Enterobacter aerogenes.* This group produces both acids and gas at around 37°C. The coagulum appears shrunken and week.

The coagulum formed by lactic streptococci, *S. thermophilus* and *lactobacilli* is smooth and with typical clean flavour.

Sources

Different microbes enter raw milk at the farm level from sources like utensils, coat of the cow, feed, fecal matter and environment.

Significance of Souring

It is the most commonly encountered spoilage under tropical Indian condition. The increased acidity in raw milk makes it unfit for subsequent processing into fluid market milk and other products. That is why raw milk giving a positive Clot on Boiling (COB) test is rejected at the receiving platform of the dairy plant. The producer enter into malpractices (like neutralization with caustic soda) to mask the developed acidity and escape the rejection of milk.

Control Measures

1. Milk production should be practiced under hygienic conditions to minimize entry of acid producing organism.
2. Immediate chilling of raw milk after production.
3. Proper cleaning and sanitized the utensils.

Adequate pasteurization of milk should be done followed by cooling.

2. Gas Production

In case of fluid dairy products like milk, cream etc. the production of gas (mainly CO_2) by certain micro-organisms is responsible for a defect called gassiness. In cases where it is associated with acids production.

In source cream or high fat milk, there is foaming as the gas escapes the partially coagulated mass and the defects is called "frothiness". Frothiness is therefore due to associative action of acids producing bacteria and gas producing yeasts.

Causative Organisms

(i) Lactose Fermenting Yeasts:

Candida Pseudotropicalis and Torulopsis Sphaerica: These yeasts produce CO_2 and small amounts of ethyl alcohol in milk, cream, whey etc. at or below 37°C. Since they are acidic tolerant, their growth is possible under acidic conditions.

(ii) Coliforms:

Escherichia Coli, Enterobacter aerogens. These microbes ferment lactose of milk/cream into gas and acid. These organisms usually grow in a temperature range of 10 to 37°C.

(iii) Anaerobic Sporeforming Bacteria

Clostridium butyricum, Cl. Pasteurianum, Cl. Sporogenes etc. These organisms grow and produce gas under anaerobic conditions.

Sources:

Microbes may be entering milk from soil, manure, feed or utensils. The Coliforms are mainly associated with fecal contamination where as clostridia come through feed (Silage) and manure.

Significance in Dairy Products

The gas production is by and large undesirable in dairy products e.g. In milk and cream. The gassy cream is rejected for butter making.

In canned dairy products e.g. in processed cheese the production of gas may leads to early or late blowing of cans.

Control Measures

1. Avoid contamination of milk at the farm level.
2. Minimized holding of milk & cream under ambient temperature.
3. Adequate heat treatments should be given

3. Ropiness/Sliminess

Ropy fermentation of milk, cream etc is brought about by the growth of bacteria leading to change in consistency of the product that forms threat

or viscous masses when poured. The change is so much pronounced that the milk can be drawn into long threads.

Causative Organisms

A large number of organisms are responsible for rapines in milk and milk product. (*Alcoligenes viscolactis*)

1. **Gram-negative rods:** *Alcaligenes viscolactis, Achromobactor lacticum, Flavobacterium tremelloides* these organisms do not form gas from lactose and glucose and only occasionally develop slight acidity in milk.
2. **Coli-aerogenes group:** *Enterobacter aerogenes, Citrobacter freundii and Serratia marcescens* etc. generally produce gas in milk but it is not detectable in ropy milk.
3. **Micrococci:** These organisms are relatively infrequent in ropy conditions.
   Examples: *Micrococcus pituitoparus, M. Viscosus and M. Freudenreichii*
4. **Aerobic sporeformers:** Certain strains of Bacillus species like *B. cereus, B. subtilis and B. circulans.* Have also been infrequently involved in ropiness.

Mechanism of ropiness: There are two types of material which are responsible for ropiness.

1. True gums or gum like substances-Polysaccharides.
2. Mucins-Nitrogenous mucous like substances.

Gums are derived from lactose which is fermented by *Coliforms, S. lactis* var. *hollandicus*, other *streptococci, leuconostocs* and *lactobacilli mucins*, which are combinations of proteins with a carbohydrate radical may be produced by peptonizing bacteria (micrococci, gram-negative bacteria, aerobic sporeformers). Ropiness is related to capsule formation.

Sources

1. Water supplies: Primary sources of infection may often be infected water used for washing and rinsing the dairy equipments.
2. Dairy equipments: Ropy organisms are practically always found in rinses of dairy utensils.
3. Coat of cows: it is one of the common routes of infection as organisms may fall into the pails from the coats of the cows.

Significance of Ropiness

Ropiness has been found to occur quite often in farm milk supplies and its liquid products like cream and affect the acceptability of the product.

Ropiness has practically been observed in milk held at low temperatures as many of the causative organisms show active growth in milk at temperature around 10°C or even lower (3-7°C) in some cases.

Control Measures

Once ropy milk bacteria have become established only the most drastic method of cleaning and sterilization will be also to eradicate them and efficient pasteurization will kill ropy milk organisms.

4. Proteolysis

It is the process by which casein or some insoluble casein derivatives (Complex milk protein) are broken down to water soluble compounds through the action of organism or their enzymes.

The following scheme summarizes the general conception:

Protein
↓
Proteases
↓
Peptones
↓
Polypeptides
↓
Peptides
↓
Amino acids

Milk contains different proteolytic enzymes which may bring about alteration of milk constituents. Some of them are naturally presents in milk, while others are elaborated by microbes.

Changes in milk protein as a result of psychrotrophic quality growth or enzymatic action are important in the keeping quality of milk and its products at refrigeration temperatures. Release of various nitrogen components or degradation of individual protein fractions has been observed while studying proteolysis caused by psychrotrophic enzymes. These proteases can attack caseins and whey proteins leading to coagulation of milk and development of bitter flavour.

Proteolytic flavour (unpleasant) flavour are attributed to the breakdown of proteins and amino acids and thus proteolysis is often accompanied by development of a bitter flavour due to the presence of protein decomposition products like peptides which are bitter.

Causative Organisms

1. Psychrotrophic bacteria: *Pseudomonas fluorescens, Ps. fragi, Ps. putrefaciens, Ps. aureofaciens, Ps. putida, Ps. aeruginosa* etc.
2. Thermoduric bacteria: Bacillus spp. e.g. *B. subtilis, B. cereus, B. stearothermophilus.*

Significance of Proteolysis

Changes in milk proteins affected by bacterial action are important in various aspects of proteolytic spoilage of fluid milk and dairy products in refrigerated storage. With the extended periods of refrigerated storage, psychrotrophs have become important to the quality of fluid milk. The proteolytic action of psychrotrophs in raw milk is particularly important to the sterilization of milk at ultra high temperature.

5. Sweet Curdling

Sweet curdling is a common defects associated with milk and cream particularly the moderately or highly heat treated products. The defect is occasionally encountered in raw cream which is delivered from the farm or in pasteurized milk which has been held for considerable longer periods of time. In severe cases, the whole of the milk in the containers become solid as of it has soured, although no acid has been produced. That is why this defect is referred to as sweet curdling. It is visible in the cream layer at the top of milk during early stage because the organisms require air for their growth.

The defect is due to the production of an extracellular enzymes by bacteria similar to rennin which causes to precipitate in the form of small specks of curd. This enzymes is produced generally by aerobic sporeforming bacilli. The pH of sweet curdled samples was between 6.2 and 6.6. The defects started from the bottom of the bottle as was evident by higher counts.

Causative Organisms

1. Aerobic sporeformers: *Bacillus cereus. B. subtilis and B. mycoides.* They causes sweet curdling in pasteurized milk or cream held at higher temperatures.
2. Psychrotrophic sporeformers: The Psychrotrophic sporeforming bacteria mainly to the genus *Bacillus* are capable of surviving commercial heat treatments of raw milk like pasteurized, sterilized (UHT) and aseptically packaged milk resulting in sweet curdling defects. Ex: *B. cereus, B. licheniformis, B. albolactis* and certain Microbacterium spp.

Significance of Sweet Curdling

This defect is not so common though it does occur in milk and cream on certain occasions. Usually sweet curdling is prevalent in heat treated products. Particularly when held at higher temperature as in summer months.

Control/Preventive Measures

1. Prevention form contacts with sources of contamination.
2. Milk should be cooled on the farm as well as after pasteurization.
3. Milk should be delivered to the customers with the minimum delay.
4. Addition of inhibitory organisms like acid producing bacteria suppress the growth of *B. mycoides.*

6. Lipolysis

Lipolysis of milk fat and the subsequent production of off flavour in dairy products is a potentially important problems in fat rich dairy products especially in cream and butters. Lipolysis is the hydrolysis of milk fat by lipolysis resulting into the accumulation of free fatty acids. The lower chain fatty acids particularly butyric and caproic are responsible for the lipolysis off flavour also referred to as rancidity. In fluid milk the flavour impairments is rarely a very distinct rancidity but rather a sharp aroma or sopy taste.

Causes of lipolysis: There are multiple causative factors of lipolysis i.e., mechanical, physiological and microbial or enzymatic.

(i) The intrinsic milk lipase: It is present in sufficient quantities to cause hydrolysis of milk fat. However, the fat globule membrane protects the milk triglycerides from attack by the lipolytic enzymes, whose activity is further impaired by their association or occlusion with casein' micelles and possibly by the presence in milk of inhibitors to lipolysis. Thus, the native milk lipase is well partitioned from the triglyceride substrates. Hence, little or no lipolysis Occurs in normal milk.

(ii) Lipolytic micro-organisms or their enzymes:

a. Certain heat-resistant lipases produced in bulk cooled milk/ cream by psychrotrophic bacteria survive HTST pasteurization and UHT sterilization and cause lipolysis during subsequent storage of heat processed milks or their derivatives even at low temperatures.

b. Lipolytic organisms growing in milk and milk products are listed below:

(a) Psychrotrophs

- Pseudomonas species mainly *Ps. fragi. Ps.fluorescens*
- Achromobacter spp. e.g., *A. lipolyticum*
- Alcaligenes spp. e.g., *A. viscolactis*
- Serratiamarcescens
- Flavobacterium spp.
- Acinetobacter spp.

(b) Other Types

- *Staphylococcus aureus*
- *Clostridium perfringens*
- Mycoplasma spp.
- *Micrococcus freudenreichii*
- Bacillus spp. like *B. cereus, B. subtilis, B. coagulans*

(c) Yeasts and Molds

- Candida spp. like C. pseudotropicalis, C. lipolytlca
- Geotrichutn candidum
- Penicillium spp.
- Aspergillus spp.

Control Measures

1. Milk should be cooled to 30°F or less when received at the processing plant.
2. Milk should be processed prior to microbial growth reaching the log phase.
3. Effective sanitation of milk production facilities is required to control contaminating micro-organisms capable of producing lipolytic enzymes.
4. Sensory evaluation of ingredients and stored dairy products must be conducted by trained personnel.

7. Bitty Cream

Bitty or broken cream is a well known example of the effect of sporeformers on milk. This defect, which occurs in raw and pasteurized milk, is characterized by the appearance of flakes in the cream layer

which do not mix again when milk is shaken. If such milk is used in tea, the flakes float on the surface making it unacceptable to many people. Flaking normally occurs before the changes in flavour or heat stability. The flakes are of two types. One of which is of mechanical/ physical origin and when the same is pronounced, it occurs as 'cream plug'. It is formed when the fat globule membrane is partly disrupted and the globules stick together. The second type is of bacterial origin and is produced partly by the lecithinase enzyme of *Bacillus cereus* and *B. cereus var. mycoides* attacking the phosholipid part of the fat globule membrane and partly from the coagulation of casein associated with the membrane.

Cause of 'Bitty' Cream

The causative organisms belong to the genus Bacillus. They occur naturally in soil and find their way on to the skin and hair of animals and into water supplies. First of all, nearly all the spores present in milk collect in the cream layer where they develop microcolonies. Flocks form around these and at the beginning of the appearance of bitty cream, each micro-colony has formed one flock.

Significance in Different Milk and Milk Products

(i) Pasteurized milk: 'Bitty' cream is the chief spoilage problem of pasteurized milk. The extent of this problem is related to different factors which influence the degree of spoilage, like failure of refrigeration, seasonal variations, prolonged storage etc.

Working on these lines, Franklin (1969) indicated the widespread potential incidence of bitty cream and concluded that dairy bulk milk held overnight or transported long distances before pasteurization might account for its poor quality. Later, Barkley and Delaney (1980) found development of bittiness and cream plug in pasteurized milk held at 7 or 30°C for 1-14 days.

(ii) Sterilized milk: The sterilized milk is characterized by cooked flavour and dark appearance. The incidence of bitty cream is low but problems may arise due to coagulation by spores. Franklin (1969) reported that a significant proportion of sterilized milk in bottles contains viable bacterial spores but these do not give rise' to spoilage except at higher ambient or incubation temperatures.

(iii) UHT milk: During UHT treatment, there is about 99.99 per cent destruction of Bacillus spp. but the small number of survivors is capable of multiplying rapidly in the milk stored at 35°C. The risk associated with heavy spore contamination in raw milk is, therefore, possible.

However, it is pointed-out that under similar spoilage conditions, it is not always established whether the spoilage organisms survive UHT treatment or if they are post-UHT contaminants.

Control Measures

1. Contamination of milk with *B. cereus* cells may be minimized by ensuring that milk is produced and processed in a relatively dust free atmosphere.
2. Use of bedding and fodder should be avoided and ventilated milking house should be located far from bedding, fodder and feed storage barns.
3. Milk should not be transferred to or allowed to stand in open pails.
4. Milking areas should be kept clean at all times.
5. In filling areas the use of air conditioning and filtration may reduce the level of contamination.

8. Off Flavours

Consumer acceptance and preference for milk and milk products is influenced by their flavour more than any other attribute. The flavour of milk is influenced by genetic as well as numerous environmental factors from production to consumption. Fresh normal milk as produced should have a pleasing, slightly sweet flavour, little aroma and a pleasant mouth feel and after taste. Any deviation in flavour or consistency is sensed readily by the consumer. Microbial off flavour are invariable encountered in raw milk through occasional neglect of failure in refrigeration.

A. Microbially Induced Flavours

(a) Malty Flavour

Causative Organisms

The malty flavour defect that commonly develops in raw milk produced in certain areas of North America and Europe has long been known to be due to the metabolic activity of *Streptococcus lactis* var, *maltigenes.*

Sources:

The main causative organism, *S. lactis* var. *maltigenes* has been isolated from cow's faeces and water in a cooling tank as well as from numerous cream and butter samples and was found to be widely distributed. The organism mainly gains entrance to milk from improperly sanitized equipment. Multiplication of the organism is favoured by delayed cooling and holding of milk at 50°F or above. Some earlier reports, however,

indicated the presence 'of this organism in milk even before leaving the udder. Mechanism of malty flavour or components responsible for malty flavour:

The volatiles from fully developed milk Cultures of *S. lactis* var. *maltigenes* contain a number of aldehydes which are formed from certain amino acids by way of an enzymatic transamination with alpha-ketoglutaric acid and decarboxylation of the resulting keto acids. The corresponding alcohols are also produced as a result of reduction of aldehydes.

(b) Fruity Flavour

Causative Organisms

The organisms which produced fruity flavour (strawberry-like aroma) in milk and butter held at low-temperatures were first described as psychrophilic by Eicholz (1902) and Gruber (1902). In 1932, Hussong proposed the name *Pseudomonas fragi* for the organisms, which produced an ester-like aroma from defective dairy products held at low temperatures. The aroma resembled the aroma of the flower of *May* Apple. Although the aroma produced by *Ps.fragi* in milk and dairy products has been variously described as strawberry-like, ester-like resembling that of the flower of the May Apple and fruity, the last term appears to be more acceptable.

Sources: The habitat of the organism is soil and water. It is widely distributed in the environment of milk production and processing. The organism is sensitive to heat and thus its presence in pasteurized products is most likely due to post-pasteurization contamination.

Mechanism of Fruity Flavour

The fruity aroma of milk cultures of *Ps.fragi* was found to be due to the production of ethyl esters. However, ethyl-isovalerate was erroneously identified as the principal aroma compound. Reddy et al. (1968) have confirmed that although the organism produced some ethanol in milk cultures, production of the fruity aroma was enhanced by addition of 0.2% ethanol to the milk medium. Such a Culture having developed a strong fruity aroma was found to contain ethyl butyrate and ethyl hexanoate in concentrations of 0.35 and 0.50 ppm respectively.

(c) Musty Potato Flavour

Causative Organisms

The mustiness of eggs was first observed by Levine and Anderson in 1932 and was ascribed to an organism and a short time later in 1934,

Olson and Hammer found an identical organism responsible for musty Potato aroma in milk. Since that time, there have been sporadic reports of musty Potato flavour defects in various refrigerated foods and the organism is described as *Pseudomonas taetrolens*. *Pseudomonas mucidolens* has also been isolated from cream that had developed a potato odour at low temperature.

(d) Unclean Flavour

This defect may be the result of an increase in the production of dimethyl sulphide caused by the contamination of milk with rod-shaped gram-negative psychrotrophic bacteria which may develop at 40°C.

(e) Phenolic Flavour

Phenolic flavour may develop occasionally in the in-bottle sterilized milk because of contamination with spores of certain types of *Bacillus circulans*. Probably, cresols are responsible for this defect.

(f) Bitter Flavour

It is caused by the breakdown of proteins by proteinases and usually develops in UHT milk. It has already been described under proteolysis.

(g) Rancid Flavour

It is due to the liberation of fatty acids by milk lipases or bacterial lipases and has been discussed under lipolysis. The various other terms used for this defect are 'hydrolytic' rancidity, 'oxidative' rancidity, butyric, goaty, bitter and soapy.

(h) Medicinal Flavour

In 1943, an outbreak of medicinal flavour was reported in pasteurized market milk that was caused by *Aerobacter aerogenes* (now called Enterobacter aerogenes). The Source of contamination was found to be hand-washed bottles.

(i) Acid Flavour

Because of the universal distribution of *Streptococcus lactis* in the environment of milk production, most milk is unintentionally inoculated with this organism immediately after milking. If the milk is not cooled to 4.4°C or below, it eventually develops acid taste due to proliferation of the organism and its conversion of lactose to lactic acid as described under souring/curdling due to lactic acid fermentation. The speed with which this defect develops is related to the degree of contamination and the temperature history of the milk.

B. Non-Microbially Induced Flavours

(a) Light Induced Flavours

Many investigators have found that a large percentage of milk packaged in translucent plastic containers exposed to fluorescent light in merchandising cabinets has this flavour. Two main components have been identified, one attributed to degradation of serum protein fractions and the other from photooxidation of lipid components.

(b) Oxidized Flavour

Milk fat contains unsaturated fatty acids oxidizable by molecular oxygen resulting in the flavour defect termed 'oxidized'. Principal catalysts are-light, copper and iron of less importance. These days, there has been a marked decrease in copper-induced oxidized flavour because of replacement of copper bearing metals with stainless steel. Descriptive terms include cardboard, metallic, tallow, oily and fishy.

(c) Heat-induced Flavours

The most common flavour defect in this category has been characterized as cooked and is from volatile sulfides originating from heat denaturable proteins of milk. A complex of heat-induced flavours is associated with UHT processing of milk. These include cooked and cabbagey flavours. The typical cooked and cabbagey flavours diminish during storage but a heated flavour described by some as a 'sterile milk flavour' usually persists during aseptic storage for several months.

(d) Transmitted Flavours

Off flavours usually considered as transmitted include feed, weed, cowy and barny. The components responsible are benzyl mercaptan, methyl mercaptan, dimethyl sulphide, indole, skatole and trimethylamine.

(e) Miscellaneous Flavours

This category includes those flavours that either can not be attributed to a specific cause or specifically defined in sensory terms.

Control Measures

Most of the flavour defects in milk can be prevented if proper precautions are taken during production, processing and handling.

9. Abnormal Colours (Discolourations)

The colour of foods is important aspect of their marketability. The foods, even if, good and nutritious, will not be liked by Consumers unless the colour is analogous to that naturally associated with the product. The

three aspects associated with colour are (a) tint, (b) intensity, and (c) uniformity. Variations in intensity can occur practically but deviation from the traditional tint or mottling in a product is a serious defect. The colour of milk is largely controlled by feed but the milk of certain breeds is always more coloured due to the fact that the milk fat contains more carotene and less vitamin A. However, certain microbes may cause changes in the colour of milk. But, with the advancement of technology and hygienic production of milk, colour defects are of little practical significance.

(A) Blue Colouration

a. ***Bacterial causes:*** The organism responsible is *Pseudomonas Syncyanea*. Blue colour is generally produced in the presence of acid, i.e., associative action is important. Certain lactic streptococci, e.g., *S. lactis*, may be responsible for the production of acid which may become stimulatory for *Ps. Syncyanea* if present as contaminant. However, addition of acid from outside does not substitute for *S. lactis* associative action.

b. ***Non-bacterial causes:*** Blue tinge may be seen in the milk from cows eating certain plants.

B. Yellow Colouration

a. ***Bacterial causes:*** The causative organism is *Pseudomonas synxantha,* As the organism is aerobic in nature, colour is more pronounced in the cream layer of milk. It causes proteolysis of milk and the liquid may show the colour. However, in case of udder infections (mastitis) milk becomes yellowish in colour but it is not due to the growth of organisms after the milk has been drawn.

b. ***Non-bacterial causes:*** The milk of various animals shows a yellow tinge which varies in intensity depending on factors like season, feeds (carrots and mangels), udder infection etc.

C. Red Colouration

a. ***Microbial causes:*** The organism responsible is *Serratia marcescens*. It is generally present in the aseptically drawn milk, but the rapid growth of other bacteria prevents their development. The other causative organism may be a yeast, Rhodotorula glutinis, which develops in sour milk or cream and forms pink or red colonies. However, it does so only occasionally because of its slow growth.

b. ***Non-microbial causes:*** Sometimes the blood may be present in the milk due to ruptured vessel or udder infection as in acute mastitic cases.

D. Greenish Colouration

a. ***Bacterial causes:*** The organism *Pseudomonas fluorescens* may develop greenish colouration in the milk held at low temperatures.

b. ***Non-bacterial causes:*** The milk drawn from certain cattle show greenish tinge because of certain feeds consumed.

6 Collection, Chilling, Standardization and Homogenization of Milk

Collection of Milk

Production of milk is confined to rural areas, while demand is mostly urban in nature. Hence milk has to be collected and transported from production point in the milk-shed areas to processing and distribution points in cities. The collection of milk in India was started by Mr. Rews at Nadiyal of Gujrat state in 1940. Collection of is the first step of the market milk industry particularly in India where specialized dairy farms do not exist.

Methods / Sources of Milk Collection or Procurement

The various sources through which milk is procured by the dairies could be classified under the following categories:

1. Directly from the producer,
2. Through the middlemen,
3. Through agents or contractors,
4. Through cooperative societies,
5. Directly from the dairy/processing plant.

1. Directly from the milk producers: In this system of milk procurement individual milk producers supply milk to the dairy independently. There is no obligation of any kind on either party. Dairy has the right to fix the price for the milk and producers have no say in the matter.

The system has the following drawbacks:

a. Milk producers are normally exploited.
b. Price offered to producers is normally low.
c. There is no assured market for the milk through out the year.
d. All the milk supplied by the producers is not accepted by the dairy because of the seasonal variations.
e. Producers do not get incentive for increasing milk production.
f. The profits earned by the dairy are not shared by the milk suppliers.
g. Dairy does not provide facilities to increase milk production nor takes care of social and economic needs of the producers.

2. Through the middlemen: In this system of milk procurement dairies receive milk through the middle men who may be milk producers as well. These people buy milk from the milk producers in the rural areas and supply to dairy. Dairy has no contact with the producers directly. Sometimes, milk producers are given loan for the purchase of the animals with the agreement that the milk produced by the animal will be sold to the money lender only. This system also has the following disadvantages:

a. Price for the milk received by the producer is very low.
b. Prices are fixed by the middlemen.
c. Producers do not get reasonable amount for the supply.
d. All the profits go to middlemen only.
e. Producers do not get any incentive for milk production.
f. No help is given to the producer for increasing milk production.
g. Milk producer generally remains a debtor.

3. Through the Agent/Contractor: In some dairies milk is received by the authorized agents or contractors. Dairy enters into an agreement with the agent/contractor for the supply of certain quantity of milk for a specific period. Normally the price for the milk is depend on its quality and quantity in different seasons is mutually agreed upon. Other details such as mode of payment, timings, transportation of milk, settlement of dispute if any and penalties for failures etc. are decided at the time of making agreement. This system also suffers from the following shortcomings.

a. Milk producer is exploited by the agent/contractor.
b. Producer does not get the real price for his produce.
c. Profits are not shared by the producers.
d. Prices for the milk are fixed by agents or contractor.
e. Often milk producers remain debtor to the agent or contractor.
f. No facilities are provided to the producers for increasing milk production.
g. Producers do not have freedom to find better market for their produce.
h. No efforts are made for the social upliftment of the milk producers.
i. This system encourages red-tapism/corruption.

4. Through Cooperative Societies: In this system of procurement milk producers form a cooperative society at the village level and several societies join together to form a milk union at the district level. Unlike the other systems of procurement, the milk producer is the owner of the

dairy. He has the right to decide the price of milk and also to get the share of profit made by the dairy. This system apart from eliminating the defects, the defects of the other systems already discussed above, gives self-reliance to the producers for solving their problems in a collective manner.

5. Directly from Dairy/Processing Plant: Due to improvement in refrigeration and transport facility it has now become possible to procure milk in bulk quantities from distant places. One dairy situated thousand kilometers away can get milk from another dairy. The example of this type of dairy is Worli Dairy in Bombay and Calcutta Milk Supply Scheme getting their supplies from Amul Dairy located in Anand (Gujarat). Mother Dairy, Delhi is another example which procures milk from the feeder balancing dairies of Uttar Pradesh, Madhya Pradesh, Haryana, Rajasthan and Punjab. There are several other dairies where this system of procurement exists.

6. Milk collection cum chilling centers/depots: Normally attached to city dairies.

Objectives: These are (i) To preserve the quality of raw milk supplies, and (ii) To provide easy transport to the processing dairy.

Location: This is guided by

1. Adequate milk production;
2. Adequate (potable) water supply;
3. Proximity to a good road or railway station;
4. Electric supply and
5. Sewage disposal facilities.

Major Items of Equipment

1. Milk weigh tank/pan and weighing scale;
2. Drop (dump) tank with cover;
3. Cash washer;
4. Milk pump (sanitary type;)
5. Surface/plate cooler;
6. Refrigerating unit (of suitable capacity);
7. Cold room (of suitable capacity);
8. Milk testing unit, etc.

Operational procedure: Essentially this is the same as a in a small dairy. On arrival, the milk is graded for acceptance/rejection, weighed, sampled for testing, cooled and stored at a low temperature until dispatch to the processing diary.

Chilling of Milk

Milk contains some micro-organisms when drawn from the udder; their numbers increase during subsequent handling. The common milk micro-organisms grow best between 20 to 40°C. Bacterial growth is invariably accompanied by deterioration in market quality due to development of of-flavors, acidity, etc. One method of preserving milk is by prompt cooling to a low temperature. Therefore, the freshly drown milk should be promptly cooled to 5°C or below and also held at that temperature till processed.

Methods of Chilling: There are several methods of chilling which are discussed below:

1. Can Immersion
2. In can Cooling
3. Surface Cooler
4. Tubular Cooler
5. Plate Chiller
6. Bulk Milk Cooler.

(1) Can immersion: In this method milk cans are immersed in water through containing ice water, the milk inside the cans is stirred with the help of plunger for uniform cooling. The system is less expensive, simple and suited for a small scale operation.

Disadvantages

a. Cooling process is not very efficient.
b. Encourages contamination of milk.
c. Require more time and labour.
d. Cannot be adopted for large operations.

It is possible to immerse milk cans in an insulated tank having chilled water. The water can be cooled with an air cooled condensing unit and Chilled water can be sprayed on milk cans by using an electrical pump or an agitator can be mounted on the cabinet to hasten cooling. They are quite suitable for handling from 4 to 12 numbers of milk cans with mushroom type lids.

(2) In can cooling: This system of cooling involves use of metallic cylindrical container called ice gum or ice cone. Ice pieces are put in the ice-gum and it is placed in the milk can. The temperature of milk in the can is reduced due to ice present in ice-gum. When ice is completely melted in the ice-gum and there is no more heat transfer, the water is thrown and fresh ice pieces are put in ice-gum. The process of cooling milk by this method continues even during transportation

from collection centers to processing unit. For this purpose ice-gums are closed with tight lids and then they are placed in the milk can. If power supply is available immersion coolers operating automatically is useful for chilling milk. This needs no water for cooling purpose. This unit has an air cooled wall mounted Freon condensing unit with evaporator which can be immersed in milk can. Within hour milk can be cooled from 38 to 40°C. This procedure is more efficient as compared to can-immersion method, less expensive and requires less labour.

Disadvantages of in can Cooling

1. Encourages contamination
2. Cooling process is slow
3. Maintenance of ice cone becomes difficult
4. Sometimes ice water is mixed with either due to leakage of ice-gums or careless handling
5. Cleaning of ice-gums involves extra costs
6. Suited only for small operation.

(3) Surface Cooler: Several hollow pipes are horizontally welded and they are joined at both the ends so that the water can flow through the pipes without interruption. These pipes are mounted on a strong iron frame. On top a milk supply tank is provided with faucet which delivers milk on to a perforated tray. At the lower end of the pipes a milk collecting trough is fixed which has an outlet through which milk is collected in the milk cans. Chilled water is produces separately in an insulated tank by using air cooled Freon condensing unit. In this method of cooling ice water or chilled water is used for cooling the milk. Chilled water is pumped through the pipes and milk flows over the pipes in the form of thin film. As the milk travels downward, its temperature is being reduced progressively. If the flow of milk is uniform as chilled water is properly regulated, then, the temperature of chilled milk is nearly the same as that of chilled water and the chilled milk is collected in the cans.

Advantages

1. More efficient cooling
2. Cost of operation is low
3. Less labour is involved
4. More milk can be handled
5. Initial cost is low as compared to plate chiller and bulk milk cooler
6. Less chances for contamination.

Disadvantages

1. Regular supply of chilled water is required
2. Milk is exposed to atmosphere during cooling process
3. Maintenance of pump and refrigeration plant is involved
4. Process can not be adopted for a large operation.

4. Tubular Cooler: In this method of cooling milk is passed through a hollow pipe which is placed inside another pipe having larger diameter. The length and diameter of both the tubes are determined according to capacity of the plant. In one of the pipe milk flows, while in the other chilled water flows. The flow of milk and chilled water is in opposite direction. This arrangement is called counter-current method. The cooling efficiency is more when compared to co-current method where the cooling medium and the liquid to be cooled, flow in the same direction.

Advantages

a. Cooling efficiency is more
b. Milk is not exposed to atmosphere
c. Less labour is required, and space requirement is also less.

Disadvantages

a. Regular supply of chilled water is required
b. System requires use of electricity
c. Maintenance of separate pump is required
d. cleaning of equipment is difficult
e. Some milk is lost due to incomplete drainage/flashings.

5. Plate Chiller: This is the most widely used equipment for chilling of milk by the commercial dairy plants. Several stainless steel plates are mounted on a solid stainless steel frame in which the flow of milk and chilled water is so arranged that in the alternate plates chilled water flows and yet it does not mix with water. The flow is always in opposite direction (counter-current). The chilled water circulates though the plate chiller which is constantly pumped from the ice bank tank or chilled water tank. Milk is pumped directly from the dump tank which passes through the plate chiller and goes to the raw milk storage tank. The number of plated and size of plates in the chiller depends upon the capacity of the plant. The capacity of plate chiller may vary from 1000 to 10000ph. This method of chilling is more efficient, more hygienic, involves less manual labour, low cost (when plant is operated to its capacity). There are several disadvantages of this system of chilling:

a. Cannot be adopted without electricity
b. Requires technically trained personnel,

c. Regular supply of chilled water is required
d. Initial cost is high.
e. Suited only for large operation.

6. Bulk Milk Coolers: Bulk milk cooler consists of a double jacketed vat fitted with a mechanical agitator. It also has provision for circulation of chiller water which comes from the chilled water tank. As the name itself suggests it is meant for bulk quantities. Normally milk is chilled and subsequently stored at low temperature until transported to processing units for further processing. The agitator is used for stirring the milk which helps in uniform cooling. Bulk milk coolers are generally installed in chilling centers.

In small chilling centers a bulk milk tank with plate type cooler for chilling milk are useful. It consists of an insulate dice bank tank operated by a Freon condensing unit and an insulated stainless steel milk storage tank with agitator, thermometer, thermostat etc. it is also fitted with a milk dump tank, strainer, milk pump and a plate chiller. This unit is ideal for small milk handling units.

Main Advantages of Bulk Milk Cooler are

a. They are used of cooling as well as storage of milk.
b. Contamination of milk through external agencies is avoided.
c. Less labour is involved.
d. Operational costs are low.

Disadvantages are

a. Requires electricity.
b. Initial cost is more.
c. Suited only for large scale operation.

Standardization of Milk

Definition: Standardization of milk refers to the adjustment, i.e., raising or lowering, of the fat and /or solid-not-fat percentage of milk to a desired value, so as to conform to the legal or other requirement prescribed.

Procedure: Milk is standardized by the addition of milk or cream with a higher or lower fat-percentage than that of the material to be standardized; sometime the addition of skim milk will do. To solve the problem, it is necessary to find the relative amount of the original material and the standardizing material to be mixed together to give a product with the desired fat contain. Once these relative amount/ proportion have been determined, it is easy to calculate the exact

amount of each which must be mixed together to give a certain weight of the finished product or the exact amount of standardizing material needed to use up a given weight of milk or cream.

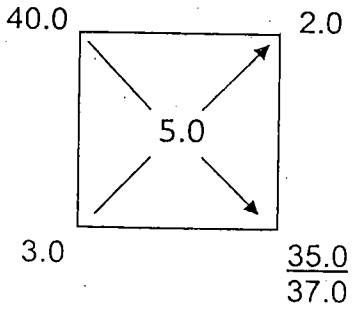
A simple scheme, the Pearson's Square, can be used to calculate the relative quantities of the materials involved in a standardization problem. It should be remembered that all measurement based on these calculation are by weight and not by volume.

The Pearson's square method is as follows: draw a square and place in the center of it the fat percentage desired. Place at the left-hand corner of the square the fat percentage of the materials to be mixed. Next, subtract the number in the centre from the larger number on the left-hand side of the square and place the remainder at the diagonally opposite right-hand corner. Subtract the smaller number on the left-hand side from the number in the centre and place the remainder at the diagonally opposite right-hand corner.

The number on the right-hand side now represent the number of parts of each of the original material that must be blended to make a product with a fat test given by the number in the middle of the square. The number at the upper left corner and the number at the lower right corner refer to the parts of material whose fat test was placed at lower left corner. If the number on the right is added, the sum obtained will represent the parts of the finished product, with the fat test given by the number obtained in the middle of the square.

1. Problem: How many parts by weight of 40% cream and 3% milk must be mixed to make milk testing 5% fat?

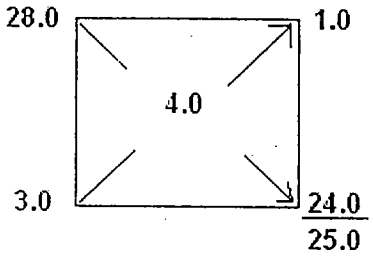
Solution



Hence; 2.0 parts of 40% cream when mixed with 35 parts of 3.0% milk will give 37 parts of 5% milk.

2. Problem: How many kg each of 28% cream and 3% milk will be required to make 500 kg of a mixture testing 4% fat?

Solution



1.0:25.0 = C : 500

Where C is cream required in kg.

Or 1/25 = C/500

C = 500/25 = 20 kg. (28% Cream required) Ans.

Milk of 3% fat = 500 – 20 = 480 kg. (3% Milk required) Ans.

Proof

500 kg of 4% milk contains 500 × 4 / 100 = 20 kg fat.

(A) 20kg of 28% milk contains 20 × 28 / 100 = 5.6 kg fat.

(B) 480 kg of 3% milk contains 480 × 3/ 100 = 14.4 kg fat.

Total = A + B

= 5.6 + 14.4 = 20 Kg

Under the PFA rules (1976) the standardized milk for liquid consumption should contain a minimum of 4.5% fat and 8.5% solids-not-fat throughout the country.

Merits of Standardized Milk

1. It insures a milk of practically uniform and consistent composition and nutritive value to the consumer.
2. The surplus fat can be converted into butter or ghee.
3. It is possible to supply cheaper milk.
4. It is more easily digestible (because of reduced fat content).

Homogenization of Milk

Definition: According to the United States Public Health Services, homogenized milk is milk which has been treated in such a manner as to insure breakup of fat globules to such an extent that after 48 hours quiescent storage no visible cream separation occurs on milk.

The process of making a stable emulsion of milk fat and milk serum by mechanical treatment is termed "homogenization" suggesting that

the mixture is homogeneous. The machines used for this purpose are known as homogenizer. Milk is an oil-in-water emulsion, with the fat globules dispersed in a continuous skim milk phase. If raw milk were left to stand, however, the fat would rise and form a cream layer. Homogenization is a mechanical treatment of the fat globules in milk brought about by passing milk under high pressure through a tiny orifice, which results in a decrease in the average diameter and an increase in number and surface area, of the fat globules.

The net result, from a practical view, is a much reduced tendency for creaming of fat globules. The size of the fat globules in milk is between 0.1 and 10 microns. The average size is close to 4 to 6 microns. When milk and cream is passed through the homogenizer, there is a decided reduction in the size of the fat globules. Most of them are reduced to 2 microns or less. Naturally, the number of individual globules is increased several hundred times.

The pre-heated (60°C) milk is homogenized at 2500-3000 psi pressure in a single-stage homogenizer. This causes sub-division of the original fat globules to less than 2 micron size (diameter) which is considered satisfactory 'Pasteurization' to storage.

Types of Homogenizers: There are on the market three distinct types of homogenizers. These may be generally classified as the

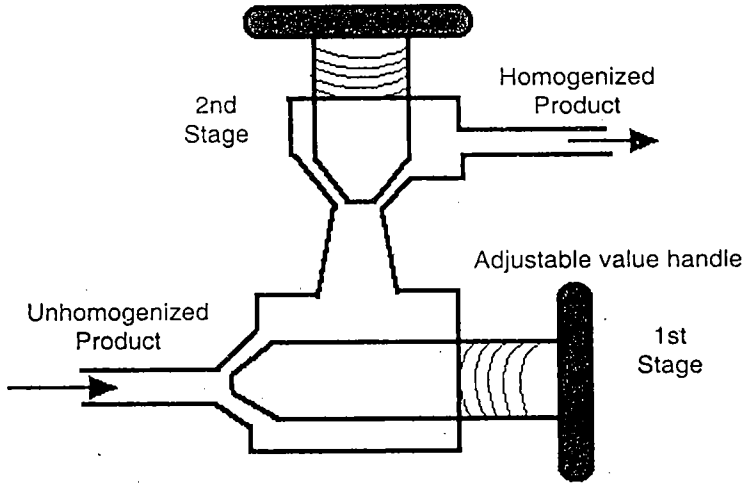
- High-pressure type
- The low-pressure rotary type
- The sonic Vibrater

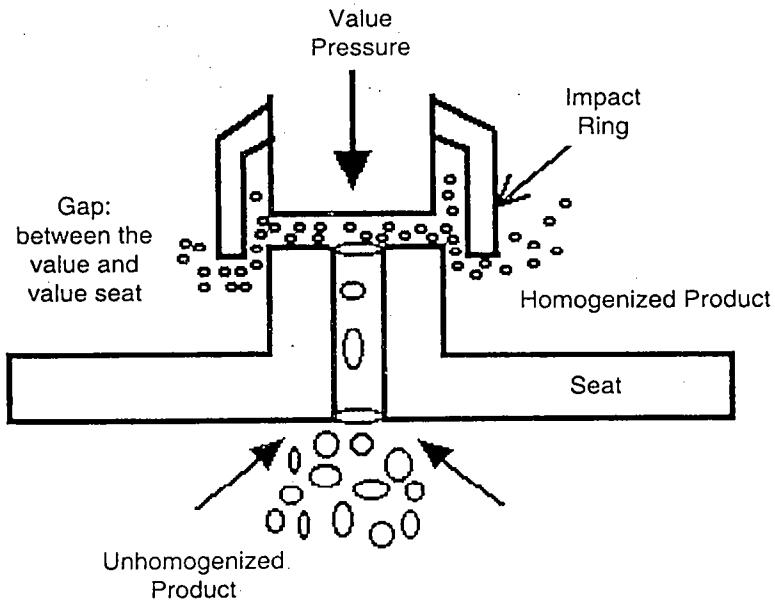
1. High-pressure Homogenizers (piston type): In general, the high-pressure homogenizers on the market consist of a single acting triplex pump with each cylinder having a section and discharge valve. The discharge valve of each pump implies into a common discharge pipe in which is placed a special valve. This valve is similar to a safety or release valve and so constructed that the operator of machine may vary the pressure required forcing a liquid through it.

It is the construction of this valve that accounts for the chief difference between different makes of high pressure homogenizers one of these machines is equipped with two valves, commonly designated as a "two-stage valve" homogenizer which makes it possible to operate the machine at two different pressures.

These machines function by forcing the milk fat under pressure through a very small orifice with the result that the fat globules are broken apart and remain dispersed in a stable emulsion in the milk serum. Homogenizer milk or cream intended for market milk or cream may be passed through the homogenizer when it is desired that the

milk fat remain in a more permanent emulsion when the mixture used in making ice cream is passed through the homogenizer, the finished ice cream has a smoother and more velvety texture.





2. Low-Pressure (Rotary Type): In this type of homogenizer milk fat is subjected to a grinding and shearing action under low pressure. Pressure homogenizer may be operated at a pressure ranging from 500 to 5000 per square inch.

3. Sonic Vibrater: In this process of homogenization milk is subjected to high frequency vibration in a device called sonic Vibrater. The machine is not widely used in dairy industry at present but its use may increase.

Factors Influencing Homogenization

1. Temperature: For the efficient homogenization of milk fat should be in the liquid state the temperature should be above 33°C this for proper sub-division. The enzyme lipase is active in the temperature range 38 to 49°C. To avoid this lipase activity preferable prior to homogenization the temperature should be more than 55°C. In routine practices, the milk is heated to 65-70°C for homogenization.

2. Pressures: In single stage when fat % is up to 6%, generally 2000 to 2500 psi is sufficient. When fat % is more than 6% two stage homogenization is needed to prevent fat clumping, in first stage pressure is 2000 psi and in second stage 500 psi.

Effect of Homogenizer on Milk

1. Color: By application of homogenization cow milk appears less yellowish due to increase no. of fat globules and area covered by fat but more opaque, but buffalo milk remains white.

2. Curd tension: The force required cutting/breaking the curd under atmospheric pressure and temperature. It reduced by homogenization i.e. less than 25 G in normal milk it is ranges 50 to 90 G.

3. Viscosity: Viscosity of milk slightly increases by homogenization of milk, because clumping of fat and absorption of casein increase surface of fat globule.

4. Titrable acidity: It is slightly increases when raw milk is homogenized and no change in heat treated milk due to destruction of lipase.

5. Surface tension: It is due to force of attraction between molecules. It is slightly increases.

6. Specific Gravity: The specific gravity of milk is not affected by homogenization, except the presence of air in the milk.

7. pH: it is slightly decreases of raw milk.

Advantages of Homogenization of Milk

1. No formation of cream layer.
2. Product of uniform composition can b obtained.
3. Better adopted for bulk dispensing mixing not necessary.

4. Milk is more palatable due to perhaps bright appearance, heavy body and rich flavor.
5. It produces soft curd and is better digested, hence recommended for infant feeding.
6. It is used for preparation of better tea, coffee, soft cheese, and reconstitute milk.
7. It increases body and texture of ice-cream.
8. Milk looks more viscous.
9. Less susceptible to oxidized flavor development.
10. Fat in milk does not churn due to rough handling or excessive agitation.

Disadvantages of Homogenization

1. Increase cost of production.
2. Returned homogenized milk is problematic.
3. Sediment appearance to a greater degree.
4. Chemical compositions are altering.
5. Cream separations cannot do efficiently.
6. Grater tendency of milk seepage through bottle caps.
7. Curdling in cookery.

7 Pasteurization and Bactofugation of Milk

Pasteurization is one of the universally adopted processes for making the milk safe for human consumption and increases its keeping quality. The process of pasteurization is named after a French Scientist, Louis Pasteur, who worked on wine during the years 1860-1870. He observed that heating process improve process improved the quantity of wine. Since then application of heat to improve the keeping quality of milk products became a practice and subsequently shortly after 1900 a commercial plant for pasteurization of milk was established in Germany and Denmark.

At the present time, the practice milk in homes is a common practice in India, almost all the dairies pasteurize milk before it is supplied. According to the prevention of Food Adulteration Act, the term pasteurization, when used in association with milk and milk products, mean heating milk and milk products at 63°C for 30 minutes or 72°C for 15 seconds and cooling to a suitable temperature before distribution. Pasteurized milk and milk products shall show a negative phosphate test.

Definition: The term pasteurization as applied to market milk today, refers to the process of heating every particle of milk to at least 63°C for 30 minutes, or 72°C for 15 seconds or to any temperature time, which is equally efficient, in an approved and properly operated equipment. After pasteurization the milk is immediately cooled to 5°C or below. From the above definition, it is clear that the process of pasteurization involves essentially three steps:

1. Heating-to a required temperature;
2. Holding-to a required time,
3. Cooling-to a sufficiently low temperature.

The Objectives of Pasteurization are

a. To render the milk safe for human consumption by destruction of cent percent pathogenic microorganism.
b. To improve the keeping quality of milk by destruction of almost all spoilage organisms (85-99 per cent).

Disadvantages/ Objections

a. The nutritive value of milk is to some extent adversely affected.
b. It affects the cream line.
c. Adversely clotting of milk with rennet.
d. Increase the cost of milk due to processing charges.
e. This process does not destroy bacterial toxins in milk.
f. The process masks inferior quality milk.
g. In India pasteurization is not necessary, as milk is invariably boiled on receipt by the consumers.

Basis for Fixing Temperature-time Combination for Pasteurization

The temperature-time combination for pasteurization process is principally based on the thermal death point of the pathogenic organisms particularly *Mycobacterium tuberculosis*. This organism is one of the most heat resistant pathogenic organisms. If this organism is destroyed by a particular temperature-time combination then all the pathogenic organisms which might be present will be destroyed.

The phosphates enzyme requires slightly higher temperature-time treatment for its inactivation than the *Mycobacterium tuberculosis* does for its destruction. The temperature-time combinations should therefore be slightly higher than required for inactivation of phosphates enzyme. The various temperature-time combinations for pasteurization process as effective against *Mycobacterium tuberculosis* and phosphates enzyme as given below:

63°C for 30 min	94°C for 0.1 s
72°C for 15 s	95°C for 0.05 s
89°C for 1 s	100°C for 0.001 s
90°C for 0.5 s	

Methods of Pasteurization

Two general methods of pasteurization are in use. One is known as the holding system and the other is the high-temperature short-time.

1. In -the-bottle pasteurization
2. Batch Pasteurization / Holding Pasteurization / Low-Temp-Long-Time (LTLT)
3. High Temperature Short time Pasteurization (HTST)
4. Ultra-high Temperature Pasteurization (UHT)
5. Vacuum Pasteurization (Vacreation)

6. Uperization
7. Stassanization process
8. Electric Pasteurization

1. In-Bottle Pasteurization: In this method of pasteurization, raw milk is filled in glass milk bottles which are tightly sealed with special caps. These bottles are subjected to heat treatment ranging from 63-66°C for a period of 30 min. Later on, they are passed through a spray of cold water which cools the milk to 5-7°C. The method has the merit of eliminating the chances of post pasteurization contamination and ensures the quality of the milk to the consumers as filled before pasteurization. In bottle pasteurization is not commonly practiced because of the following reasons:

1. More chances for bottle breakage.
2. Heat transfer during heating as well as cooling in very low.
3. Special caps/capping mechanisms are required.
4. Over sized bottles, are required to allow expansion of milk during heating.

2. Batch or holding pasteurization (LTLT): This is also called the Low-Temperature –Long Time –method. The milk is heated to 63°C for 30 minutes and promptly cooled to 5 °C. The pasteurizers may be of three types

a. Water-jacketed vat: This is double walled around the sides and bottom in which hot water or steam under partial vacuum circulates for heating and cold water for cooling. The outer wall is usually insulated to reduce heat loss. The heat exchange takes place through the wall of the inner lining. The milk is agitated by slowly moving paddles or propellers. When heating, the vat cover is left open for escape of off-flavours, and when holding, the cover is closed.

Advantage: Flexibility in use

a. Water spray type: A film of water is sprayed from a perforated pipe over the surface of the tank holding the product. The product is agitated as above. A rapidly moving continuous film of mater provides rapid heat transfer.

b. Coil vat type: The heating /cooling medium is pumped through a coil placed in either a horizontal or vertical position, while the coil is turned through the product. The turning coil agitates the product.

Disadvantages: Coils are difficult to clean, which accounts for the decline in their use.

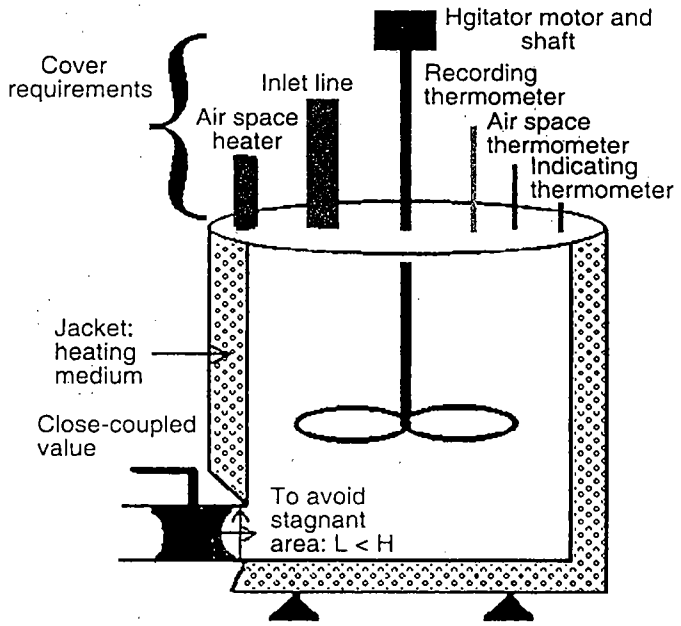
Advantages

1. It is suitable for small operation
2. It does not involve sophisticated equipment
3. This process is simple
4. Technically trained personnel are not required
5. No milk is lost during pasteurization process.

Disadvantages

1. Time taken for pasteurization is too long.
2. There are no automatic controls to check the operation.
3. The growth of thermophilic organisms is encouraged.
4. Involves cleaning of vat after pasteurization of each lot.
5. In-place-cleaning is not convenient.
6. Operation cost per liter of milk handled is more.
7. Cannot handle large quantity of milk.

Batch Pasteurizer



3. High temperature short time (HTST) pasteurization: This was first developed by A.P.V.Co. in the United Kingdom in1922. It is the modern method of pasteurizing milk and is invariably used where large

volumes of milk are handled. The HTST pasteurizer gives a continuous flow of milk which is heated to 72°C (161°F) for 15 seconds and promptly cooled to 5°C or below. Process involves heating of milk to 71.5°C. This process of pasteurization of milk is most commonly practiced in dairies particularly where large volume of milk is handled. This process is also called Continuous process/ Plate pasteurization process/ Flash pasteurization.

The steps involved in milk the milk passes through HTST pasteurization system are in this process of pasteurization, milk is pumped either from the dump tank or from the storage tank to float control tank from where it is pumped to the pasteurizer.

Pasteurizer has four different sections through which milk passes through during the process of pasteurization. Milk enters into the regeneration section where its temperature is raised to about 45°C by out going heated milk on the other side of the plates and then it goes either to filters or clarifier as the case may be for removal of extraneous materials if present.

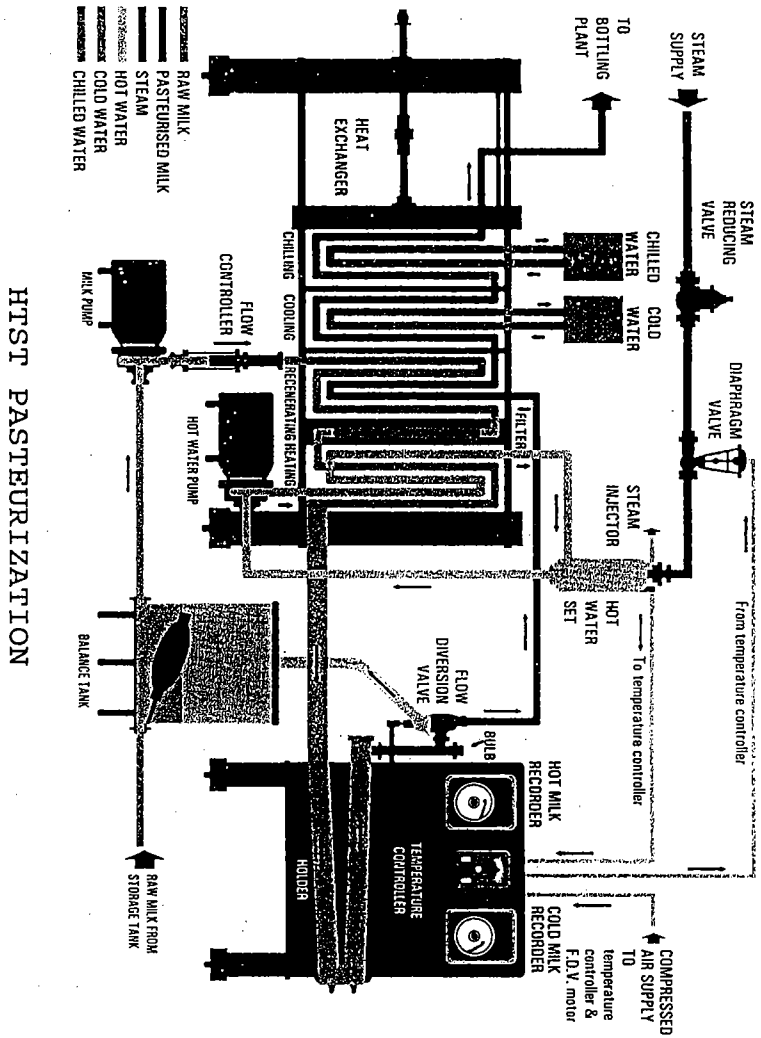
Then milk comes to the heating section where the temperature of milk is raised to at least 71.5°C by hot water or intermittent steam and this hot milk enters in the holding section of the pasteurizer. In some plants instead of having a holding section, a holder tube is provided whose length and the diameter is so adjusted that the milk takes at least 15 sec. to pass through. At the end of the holding section, a Flow Diversion Valve (FDV) is fixed to control proper heating and holding and milk passes through the valve.

Flow diversion valve automatically checks the flow of milk; if it has not been properly heated to the required temperature then FDV diverts the milk to the float control tank and milk comes to the regeneration and follows the same path again. In case milk has been properly pasteurized then through the FDV milk enters the regeneration section where it heats the incoming raw milk and its temperature is reduced (partial cooling) and finally this passes to the cooling section where it is cooled to about 4.5°C. Thus, the milk is pasteurized and it is now ready for packing, storage or conversion into products. The flow diagram of High Temperature Short Time pasteurization (HTST) is shown in Figure.

Functions of the Important Parts

1. Float-Controlled Balance Tank (FCBT): Maintains a constant head of milk for feeding the raw milk pump; also receives any sub-temperature milk diverted by FDV.

2. Pump: Either a rotary positive pump between the regenerator and heater (USA), or a centrifugal pump with a flow control device to ensure constant output, after FCBT (UK and Europe) is used.

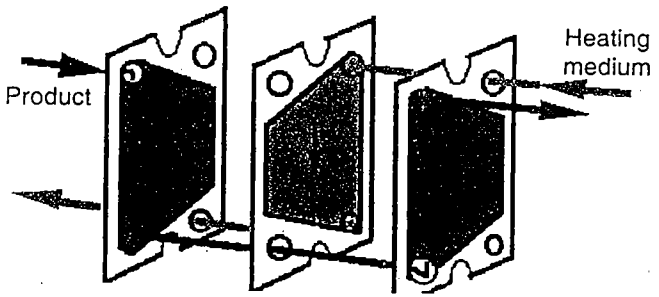


3. Plates: Plates are supported in a press between a terminal block in each heating and cooling section. The heat moves form a warm to a cold medium through stainless steel plates. A space of approximately 3 mm. is maintained between the plates. A space of approximately 3 mm. is maintained between the plates by a non-absorbent rubber gasket or seal which can be vulcanized to them.

The plates are numbered and must be properly assembled. They form but not excessively turbulent flow of products with rapid heat transfer. Raised sections (corrugations) on the plates in from of the knobs, diamonds and channels, help provide the turbulent action

required. Grater capacity is secured by adding more plates. Ports are provided in appropriate places, both at the top and bottom of the plates, to permit both the product and the heating / cooling medium to flow without mixing.

Flow Pattern in Series of Plates



4. Regeneration (Heating): The (raw) cold incoming milk is partially and indirectly heated by the hot outgoing milk (milk-to-milk regeneration). This adds to the economy of the HTST process, as the incoming milk requires less heating by hot water to raise its temperature for holding.

- Milk entering at 4°C
- Heated in regenerator to 34°C
- Heated in heating section to 74°C
- Cooled in regenerator to 44°C
- Cooled in cooling section to 4°C

5. Filter: Variously shaped filter units to connect directly to the HTST system are placed after the pre-heater or regenerative (heating) section. These units, using 40-90 mesh cloth, are usually cylindrical in shape. Usually two filters are attached but they are used one at a time. This permits continuous operation, the flow being switched from one to the other while replacing a filter.

6. Holding: The holding tube or plate ensures that the milk is held for a specified time, not less than 15 seconds, at the pasteurization temperature of 72°C (161°F) or more.

7. Flow diversion valve (FDV): This routes the milk after heat treatment. If the milk has been properly pasteurized, it flows forward through the unit; that which is unpasteurized (i.e., in which the temperature does not reach the legal limit) is automatically diverted back to the FCBT for reprocessing. It is usually operated by air pressure working against a strong spring.

Should the temperature fall, air pressure is released and the valve snaps shut immediately. When the temperature is regained, air pressure builds up and the valve opens to forward flow. The system is so arranged that any failure of air or electricity moves the valve in the diverted position.

Note: The flow of unpasteurized milk can also be stopped with a pump stop, which automatically stops the milk-pump motion if the product temperature drops below the desired level. When the proper temperature is reached, the pump stop restarts the operation and allows the flow of milk to continue.

8. Regeneration (cooling): The pasteurized hot outgoing milk is partially and indirectly cooled by the incoming cold milk (milk-to-milk regeneration). This again adds to the economy of the HTST process. In fact, when pre-cooled (raw) milk is received, the high degree of regeneration (72 to 85%) allows water cooling to be dispensed with entirely.

9. Control panel: Contains instruments, controls, FDV mechanism and holding system, all centralized in one moisture proof panel. The lower half of the panel forms an air-insulated chamber which carries the holding tube.

10. Hot water set: Circulates hot water through the heating section of the machine to maintain the correct milk temperature within very fine limits.

11. Automatic control devices: These include:

a. **Steam Pressure controller:** Maintains a constant hot water temperature for heating milk accurately to the required pasteurization temperature. (Acts as a reducing valve in the steam supply line, so as to give a constant steam pressure.)

b. **Water Temperature controller:** Regulates the amount of steam entering the hot water circulating system.

c. **Milk Temperature Recorder:** Records the temperature of milk leaving the holding tube/plate. This is an electric contact instrument that operates either a FDV or a milk pump, automatically preventing milk from leaving the holding section at sub legal temperatures. Both the frequency and duration of the flow diversion and the temperature of milk leaving the holder are recorded on the thermograph (recording chart) by means of two separate pens. (The check thermometer is placed near the milk temperature recorder.)

12. Pressure in the System: The normal pressures maintained in the HTST system are:

1. Pasteurized milk	15psi
2. Raw milk	14psi
3. Heating/cooling medium	12 to 13psi

13. Holding time test: The holding time of a HTST pasteurizer is the flow time of the fastest particle of milk at a prescribed temperature through the holding section. The holding time is calculated between the points at which the heated milk leaves the heating section and reaches the FDV.

The efficiency of pasteurization in the HTST system depends as much on the correct maintenance of temperature as on the holding time. Hence the latter should be checked periodically. Several methods are used for determination of the holding time, viz. the electrical conductivity method (of a salt solution); the dye injection method; the electronic timer method; etc.

Advantages HTST Pasteurization

1. The process is highly suited for large operations.
2. Large volume of milk can be pasteurized without interruptions.
3. Milk is readily available for packing or further processing, once the operation has started, milk can be filled in bottles or sachets within 15 min.
4. Less floor space is required because this equipment is compact.
5. There are control devices which make the process more efficient.
6. The capacity of the plant can be increased or decreased by increasing or decreasing the plates. This is easy as well as less expensive.
7. In place cleaning can easily be followed.
8. Involves less labour because the process is automatic.
9. Heating holding and cooling are done in the same unit and no further equipment is necessary.
10. The process of pasteurization is carried out within the pipe lines and the plates which reduce the chances of contamination. Milk is not exposed to atmosphere.
11. Provides Economy of operation because of regeneration.

Disadvantages HTST Pasteurization

1. Initial cost is more.

2. This system is not well adopted for small quantities of several liquid milk products.
3. Requires an experienced and technical operator.
4. Some milk is lost in the pipe lines which gets diluted due to flushing.
5. Gaskets used in pasteurizer (HTST) are damaged and require replacement more frequently.
6. Maintenance cost is more.
7. Due to higher heating temperature more chilled water is required for cooling of milk which in turn increases the cost of refrigeration.
8. Greater accumulation of milk stone in the heating section.

4. Ultra-high Temperature Pasteurization (UHT): Ultra high temperature pasteurization was developed in the 1950s; this usually encompasses temperature time combination of 135°C to 150°C for no hold(a fraction of a second). The success of UHT heat treatment of milk depends on immediate aseptic packaging. Milk pasteurized by UHT process has additional advantages such as maintaining the nutritive value, colors and flavour similar to that raw milk.

5. Vacreation (Vacuum pasteurization): This refers to pasteurization of milk /cream under reduced pressure by direct steam. This process removes feed and other volatile flavors from cream and to pasteurize it for butter making. The equipment used in this process is called "vacreator" and the process is called "vacreation."

The vacreator consist of three stainless steel chambers connected to one another for steam heating and vacuum treatment with continuous product flow.

The product in the form of droplet enters the first chamber of the vacreator where pasteurization occurs. The chamber is operated under a vacuum of 5 inches Hg, which maintains a temperature of 90-95°C, while steam is fed from the top and falls by gravity to the bottom of the chamber and then the product and free steam is removed from the bottom of the first chamber to the top of the 2nd chamber.

The temperature of 2nd chamber is maintained at 71- 82°C under a vacuum of 15 – 20 inch Hg. A portion of the steam previously added is removed and the product moves down through the chamber. Some of the tainting substances and off-flavours are removed by heat and vacuum treatment.

The product then moves on to the 3rd chamber at 43°C by maintaining a vacuum of 26- 28 inches of Hg and here more water and off flavours

are removed. A multistage centrifugal pump removes the product from the 3rd chamber. This process takes about 10 seconds to pass through the three chambers.

Advantages

1. Long keeping quality.
2. Removal of feed and other volatile off-flavour.
3. Appreciable homogenization effect.
4. Reduction in acidity.
5. Efficient destruction of micro-organisms.

6. Uperization: This is also called as "ultra–pasteurization" and it was developed in Switzerland. In this process, milk is heated with direct steam up to 150°C for a fraction of a second. This is a continuous process.

The first step in the uperization process is fore warming of milk to 50°C and intended to remove most of the dissolved oxygen and volatile off-flavours by vacuum treatment. In the second part, the milk is first preheated to about 80-90°C and then heated on the uperization chamber with high-pressure steam to around 150°C for ½ to ¾ th of a second. After this heating, the product moves into an expansion chamber at near atmospheric pressure, thereby forcing some evaporation of moisture. The product is then moved to a cooler and then into storage chamber.

The advantages of uperization are

1. Long keeping quality.
2. Removal of feed and other volatile off-flavours.
3. Appreciable homogenization effect.
4. Reduction in acidity.
5. Efficient destruction of microorganism.
6. Effect of uperization on nutritive value and flavour are not greater than that of pasteurization.

7. Stassanization : The process is now in use to a considerable extent in milk plants of France, Denmark, Italy and certain other countries. The principle of its operation is heating of milk to the desired temperature by passing it through two water heated pipes, through narrow space of 0.6 to 0.8 mm. The milk is forced through the narrow space which comes in contact with heating surface. The temperature of milk is raised to about 74°C for 7 sec. and immediately cooled to 4-5°C. The credit of developing this process of pasteurization goes to a French Scientist Henri Stassano who invented the process in France.

8. Electric pasteurization: The most modern form of electric pasteurizer consists of a regenerative pre-heater where the milk is brought to about 120°F (48.98°C) and the pasteurizer proper, consisting of a rectangular, vertical chamber 2 feet high and 3 by 4 inches in cross-section, with two sides made up of carbon electrodes about 3 inches apart.

The milk flows between these electrodes and the temperature of the milk is raised to 161-163°F (71.76-72.78°C) by the resistance offered by the milk to the passage off a 110-volts alternating current. The milk is exposed at this temperature for 15 to 20 sec. after which it is cooled.

Bactofugation of Milk

To increase the efficiency of thermal processing of milk, certain modifications have been attempts based on a pretreatment before pasteurization. This pretreatment could be a short incubation (to incuse germination of spore formers) or a brief heat treatment (to give heat shock to spores to facilitate germinations) or a physical removal of micro-organisms including spores by centrifugation.

Bactofugation process is found to be a practicable approach especially for milk destined for cheese making, although it is also known to increase the processed as market milk. It has been known to increase the shelf life of market milk by at least to pasteurization as it is still necessary to destroy the bacteria not removal by centrifugation.

Definition: According to one of the definitions, Bactofugation is the process to remove and reduce the bacterial count of fluid milk by 99.98% by using mechanical forces.

Process: The basic principle of the process involve preheating of milk (to decrease viscosity) followed by centrifugation (to remove micro-organisms) and pasteurization (to further reduce the remaining microflora). The centrifuged milk is then pasteurized. The heavier sludge collected during centrifugations, which is rich in bacteria is sterilized at 130-140°C after which it may be reunited with the main lot.

The high speed centrifugation step (20,000rpm) has been found to remove 85-93% of the bacterial cells depending on the types of bacteria present. The degree of removal is further dependent on the size of bacterial cells. For examples, the following order of removal is practically observed: *Bacillus megaterium, B. subtilis, Streptococci, staphylococcus hemolyticus, staph. aureus, E-coli.*

The bacterial counts in sludge have been recorded to range from 6 × 106 to 750 × 106 per gm.

Effect of Bactofugation on reduction of cells and spores of bacteria in pasteurized milk

Sr.No.	Counts	Reduction in counts (%)	
		Pasteurized milk	Pasteurization after bactofugation
1.	Total bacterial count	99.17	99.83
2.	Aerobic spore count	13.0	77.0
3.	Anaerobic spore count	0.0	79.0

The removal of bacteria as well as spore by including Bactofugation is especially important in milk for cheese making as it increase the cheese quality by eliminating organisms like *E. coli, Enterobactor aerogenes* and spores which may produce spoilage during products manufacture or storage.

8 Principles of Dehydration of Milk

Dehydration of Milk

The milk can be preserved by dehydrating to various degrees. The dehydrated milk has a longer "shelf-life" than the milk preserved by pasteurization and refrigeration. Milk from which part or all of the water has been withdrawn is termed concentrated or dried milk (milk powder). More than 80 percent of milk is water. The best way to preserve milk is by removal of water. It can be done by drum drying or spray drying or by freeze drying.

Condensed Milks

(A) Definition: Condensed milks are the milks obtained by evaporating part of water of whole milk, or fully or partly skimmed milk, with or without the addition of sugar. The term 'condensed milk' is commonly used when referring to "full cream sweetened condensed milk" while the term evaporated milk is commonly used when referring to 'full cream unsweetened condensed milk'. Skimmed milk products are known as "sweetened condensed skim milk" and "unsweetened condensed skim milk". The ratio of concentration of milk solids is about 1: 2.5 for full cream products and 1: 3 for sweetened condensed skim milk.

According to the PFA (1976) the various condensed milks have been specified as follows:

- **Sweetened condensed milk** is the product obtained from cow or buffalo milk or a combination thereof, or from standardized milk, by the partial removal of water and after addition of cane sugar. Sweetened condensed milk should contain not less than 9.0 percent milk fat, and not less than 31 per cent milk solids and 40.0 per cent cane sugar.
- **Unsweetened condensed milk** (Evaporated milk) is the product obtained from cow or buffalo milk or a combination therefore, or from standardized milk, by the partial removal of water. Unsweetened condensed milk should contain not less than 8.0 percent milk fat, and not less than 26 per cent milk solids.

Viscosity → Viscous → Having a thick, sticky, consistency b/w Solid & Liquid.

- **Unsweetened condensed skim milk** (evaporated skimmed milk) is the product obtained from cow or buffalo milk or a combination thereof, by partial removal of water. Unsweetened condensed skim milk should contain not less than 20.0 percent total milk solids. The fat content should not exceed 0.5 per cent by weight.
- **Sweetened condensed skim milk** is the product obtained from cow or buffalo skimmed milk or a combination thereof by the partial removal of water and after addition of cane sugar. Unsweetened condensed skim milk should contain not less than 26.0 per cent total milk solids and not less than 40 per cent cane sugar. The fat content should not exceed 0.5 percent by weight.

Composition of condensed milks (%)

Type of condensed milk	Water	Solids	Fat	Milk solids not Fat	Protein	Lactose	Ash	Total milk solid	Sucrose
Condensed milk	26.0	74.0	9.0	22.0	8.3	12.2	1.5	31.0	43.0
Evaporated milk	31.0	69.0	9.0	22.0	8.3	12.2	1.5	31.0	-
Skim sweetened	29.0	71.0	0.5	25.5	9.3	14.0	2.2	26.0	45.0

Indian Standard Specifications for condensed milks

Characteristics	Requirement for	
	Condensed milk	Skim sweetened
Total milk solids (%wt.) Min.	31.0	26.0
Fat (% wt)	Not less than 9.0	Not less than 0.5
Sucrose (% wt.) Min.	40	40
Acidity(% lactic) Max.	0.35	0.35
Bacterial count (per g.)	500	500
Coliform count (per g.)	Negative	Negative
Yeast and mould count (per g.) Max.	10	10

(B) Food and Nutritive value of condensed and evaporated milks: Both have high nutritive value, both are rich in fat and fat-soluble vitamins A, D E and K, body building proteins, bone forming minerals and energy-giving sucrose, evaporated milk is suitable for infant feeding since it makes a soft curd which is easily digested.

(C) Physico-Chemical Properties

- Specific gravity/density: Evaporation of water in the manufacture of condensed milks raises their specific gravity/density, which is universally employed to control their composition. Baume hydrometers are widely used for this purpose. Specific gravity of condensed milk is 1.303 and evaporated milk is 1.083 at 16°C.
- Freezing point: The freezing point of condensed milk is – 14.9°C and that of evaporated milk is - 1.3°C.
- Colour and flavour: Heat treatment during manufacture of condensed milks tend to darken their color and develop cooked flavour, the darkening/browning-discoloration results from the interaction of the amino-compounds with sugar (casein and lactose) and is called as Maillard-type browning. The brown pigment is called as melanoidin.

(D) Role of Milk Constituents in Condensed Milks

- **Milk fat:** Imparts a rich and pleasing flavour, soft body and smooth texture to both condensed and evaporated milks. Affects viscosity. Significant in flavour problems, such as rancidity, tallowiness, etc.
- **Milk-proteins:** Technologically of great importance and their physico-chemical reactions to processing-heat largely determine the heat stability and viscosity of the condensed milks.
- **Milk sugar:** Plays an important role in the successful control of the texture of condensed milk. The size of the lactose crystals determines the relative smoothness of the condensed milk, and is controlled by the procedure used for the cooling and crystallization of this product.
- **Mineral salts:** These-particularly calcium and magnesium together with citrates and phosphates control the salt balances and heat stability of milk. A disturbed salt-balance causes objectionable heat-coagulation of milk.

(E) Method of Manufacture of Condensed Milk

Receiving milk
↓
Filtration /Clarification (38-40ºC)
↓
Standardization
↓
Forewarming/preheating (115-118ºC / No Hold)
↓
Addition of sugar
↓
Condensing (2.5:1)
↓
Homogenization
↓
Quickly cooled to 30ºC
↓
Seeded with 0.1 - 0.3% lactose
↓
Vigorous stirring and slow cooling for 1 hour
↓
Temperature of 15ºC reached
↓
Stirring continued until packed.
↓
Storage at 10ºC

Details of Manufacture

The basic principle in the production of condensed milk and evaporated milk is that high quality milk is filtered /clarified, standardized, forewarmed and condensed /evaporated to the desired level. The concentrated product is preserved by the addition of sugar for condensed milk and by heat sterilization for evaporated milk.

(a) Receiving of milk: Good quality milk is received at the plant, its temperature should be at 10°C or below. The milk should be clean, sweet, and free from off-flavours and odours and reasonably free form extraneous material. Contamination by antibiotics, pesticides and other chemical residues and metals is highly undesirable; abnormal milk should not be accepted. Acid development is objectionable for not only does this indicate an excessive bacterial count, but it also reduces the heat stability of milk. The various platform tests and laboratory tests usually performed on the intake milk to determine its acceptance for condensing. These tests are discussed below.

1. Alcohol test: To make this test, 5 ml of milk is placed in a test tube and an equal amount of solution with 68 % alcohol added. The mixture

is shaken and any formation of clot or flake denotes a positive test, i.e., the milk is susceptible to heat coagulation. A disturbed salt balance affects alcohol coagulation in the same manner as heat coagulation. The test detects: abnormal milk including colostrum.

2. Alcohol Index: Determined by placing absolute alcohol in the burette and 10 ml of milk in a beaker. The number of ml of alcohol required for flake formation is known as the Alcohol Index (AI). An AI of 7 is indicative of good stable milk for acceptance, while 3 or less shows that the milk is fit for rejection.

3. Alcohol Alizarin test: This test not only determines the heat stability of milk but also the pH. (Milk is coagulated if the pH is 5.6 or below).

4. Clot on Boiling test: In this test, 5 ml of milk is placed in a test tube and kept in a boiling water bath for 5 minutes, afterwards it is removed and examined for precipitation, if curd is observed, the milk is said to fail the COB test and should be rejected.

(b) Filtration/Clarification: This is done in order to remove the visible foreign matter. The milk is generally preheated to 35-40°C to increase the efficiency of the operation.

(c) Standardization: Standardization of the raw milk is carried out in 3 stages.

1. The first standardization which establishes the desired fat / SNF ratio (usually 1:2.44, done in raw milk)
2. The second, which establishes the desired ratio of added sugar to the total milk solids (done prior to condensing)
3. The third, which adjust the concentration of the finished product to the desired percentage of total solids (prior to homogenization).

(d) Forewarming/Pre-heating: Which is done for making the finished product free from microorganisms and enzymes, to ensure uninterrupted boiling in pan and to control age thickening in the finished product. The time temperature combination varies over 82-93°C for 5-15 minutes. Tubular heat exchangers are commonly used.

(e) Addition of sugar: Sugar is added for preserving the condensed milk, without resorting to sterilization process, and cane or beet sugar is the commonly added as sweetening agent.

1. *Amount of sugar to be added:* This ranges from 40-45 % in the finished product, which requires 18-20 percent sugar on milk basis. As per Hunziker, (scientist), advocated a sugar ratio (sugar-in –water concentration) of 62.5 to 64.5 per cent.

2. *Sugar Ratio Determination (SR).* This can be calculated either by using

$$\% \ SR = \frac{\% \text{ Sugar}}{100 - \%TMS} \times 100$$

or

$$\%SR = \frac{\% \text{ Sugar}}{\%\text{Sugar} + \%\text{Water}} \times 100$$

Where TMS stands for Total Milk Solids.

3. Quality of sugar: Sucrose in granules or syrup of good quality is added. Liquid sugars (65 %) should be subjected to high pasteurizing temperatures to destroy the microorganisms before they are added to condensed milk.

4. Method of adding sugar: Sugar should be added towards the end of the condensing process. The dry sugar is dissolved in the least possible quantity of water. If added before condensing, an increase in viscosity and greater difficulty in the evaporation of moisture result. The sugar syrup should be filtered or passed through a centrifugal clarifier before adding to the product.

(f) Condensing: The basic principle consist in the removal of water from the standardized milk by boiling it under partial vacuum at a low temperature till the desired concentration is reached. This operation is carried out in an evaporator, which should preferably be of the single effect type (also known as a vacuum pan).

Advantages of Condensing Milk in Vacuum

1. Economy to operation
2. Rapidity of evaporation
3. Protection of milk against heat damage
4. Obtaining a finished product which is free from any cooked flavour
5. Readily reconstituted in to the original milk.

(A) Types of Evaporators: The different types of evaporators can be classified as recirculation, falling film and plate evaporators, working either on a batch basis or continuously. There can be heat recovery (in single-stage plants by preheating installations or multistage plants by using the energy of the vapor from one stage to heat the following stage). Residual heat is used to preheat milk.

Batch evaporators are used in smaller plants whereas, continuous evaporators are used in larger plants. Evaporation installations

can be either single-stage or multistage, which reduces the steam consumption up to 0.08 kg steam/kg evaporated water. (The normal steam requirement is 0.3 – 0.6 kg steam/kg of evaporated water).

1. Single stage recirculation evaporator: This evaporator has a capacity of 1000 Kg/ hr of water. The product is pumped into a tubular heating section from the bottom, where heat is added to the steam-vapour mix through a pipe. The product rises in the tube to the top, reaches boiling point and goes to a centrifugal separator, where the concentrated product and steam are separated.

If desired concentration has not been reached, it is sent into the bottom of the evaporator through the pipe again and is re-circulated. The separated vapor (steam) is again re-used in the evaporator and balance transferred to a condenser and sent out.

2. Falling film evaporator: This installation for manufacturing of evaporated milk has a water evaporation capacity of 6000 kg/h, where the dry matter of milk is increased to 24%. Milk that is to be concentrated, is first heated to 70-80°C, and then again pre-heated in tubular pre-heaters to 110-130°C. The steam pressure of the hot milk, due to heating, is first reduced in a flash vessel and then is evenly distributed into tubes installed on the upper part of the equipment.

The product now, flows downward in the tubes in a very thin film, which is heated by the steam vapour and part of water is evaporated here. Now, the product-steam (vapour) mixture goes to the centrifugal separator, where they are separated. From here, the separated product is again sent to the next evaporator stage and the process is repeated, where further concentration takes place.

Thus recirculation of the product is done to have a more uniform product with desired concentration. The steam (vapour) that is separated and extracted are condensed back and utilized for further heating. The whole plant is put under vacuum by using steam jet-operated vacuum nozzle.

Advantages are

1. Product passage with short contact times on the heated tubes
2. Good heat conduction due to thin product
3. Uniform residence time and therefore no different thermal load
4. More economical from energy point of view

3. Plate evaporator: The main differences between this system and the one mentioned above are that, this is a continuous operating system and moreover, here plate heat exchangers are employed instead of the tubular heating section for heating of the product.

Advantages are

1. Low installation height
2. Short contact time between the product and plates
3. Flexible capacity by increasing or reducing plates
4. Easy cleaning

4. Striking the batch: This term indicates that the correct concentration of the product has been reached. When the boiling milk approaches the desired concentrations with visual indications of the final stage, i.e., settling of milk to a quite boil, assuming of glossy and glistening luster, forming of heavy roll from the periphery towards the centre etc., these signs should warn the operator as to the right time for "striking the batch". The striking is confirmed with standard tests like determination of specific gravity/density etc., using

1. Pycnometer test
2. Hydrometer test
3. Refractometer test
4. Viscosimeter test

The Baum'e hydrometer test is the commonly used for condensed milk to find out the density. The specific gravity of the condensed milk is obtained by the formula:

$$\text{Specific gravity of condensed milk at } 16°C(60°F), = \frac{100}{\frac{\%\text{Fat}}{\text{Sp. gr. (0.9)}} + \frac{\%\text{MSNF}}{\text{Sp. gr. (1.6)}} + \frac{\%\text{Sugar}}{1.6} + \%\text{Water}}$$

The specific gravity is then converted to Baum'e degree by the formula:

$$\text{Baume at } 16°C = 145 - \frac{145}{\text{Sp.Gr. at } 16°C(60°F)}$$

This is then corrected to the standard temperature used for the hydrometer test of the pan sample by the temperature correction factor 0.03 for each °F of deviation from the standard. For condensed milk the Baume Hydrometer value ranges from 30′ to 37′ Be.

(B) Finishing the batch: On 'striking the batch' when the desired density has been reached, the condensing process is stopped. All the steam to the pan is shut off, the valve in the water line to the condenser is closed, the vacuum pump is stopped and the vacuum relief is opened. The above operation should be carried out in the order stated to prevent milk from burning on to the heating surface and condenser water from flooding the pan. When the vacuum has been dissipated, the condensed milk is drawn from the pan. This should be done promptly.

(C) Third standardization: Some manufacturers tend to prefer slightly over-condense the milk and then standardize it back to the exact concentration desired by the addition of the correctly calculated amount of water.

(g) Homogenization: Hot condensed milk is invariably homogenized before it is cooled and crystallized with the objective of getting a uniform fat emulsion and reduces fat separation to minimum during storage. A total pressure of 2500 psi, with 2000 psi in the first stage and 500 psi in the second stage are used.

(h) Cooling and Crystallization

1. Importance: Cooling process is an important step in the manufacture of condensed milk as this delays the tendency of age-thickening and discoloration in the product and moreover, the method of cooling decides the smoothness of texture of the finished final product.

2. Role of lactose: Lactose plays an important role in the control of texture of condensed milk, as it constitutes a highly concentrated solution in the product. The size and the number of the lactose crystals determine the smoothness of the product, which is once again controlled by the cooling processes, which the product receives, during its manufacture.

3. Mechanism of lactose crystallization: Under the existing hot conditions of the condensed milk manufacture only α -lactose-hydrate will crystallize. This proceeds slowly due to the slow mutation of the highly soluble β –lactose anhydride to the less soluble α -lactose-hydrate from its super saturated form. For normal condensed milk of average composition, the temperature of mechanism of maximum rapidity of crystallization is approximately 30°C.

4. Importance of mass crystallization: As the smooth texture of the finished product is not to prevent the formation of lactose crystals during the cooling progress, but to prevent the crystals that are present at the end of this process from subsequently growing larger.

5. The force crystallization period: The purpose of this is to produce mass crystallization of lactose. The mass crystallization is achieved by what is known as 'forced crystallization' where the product is held at a particular temperature, which helps in mass crystallization, which is optimal for the 'seeding'. The optimum temperature for forced crystallization is in the range of 30-40°C.

6. Seeding of condensed Milk: Seeding refers the introduction of lactose in very fine powder form during the cooling process to provide nuclei for crystallization. Seeding at an optimum temperature is also

done for mass crystallization and more importantly to yield uniform sized crystals. The seed material commonly used is powdered lactose (a -lactose-hydrate) or dried milk or sweetened condensed milk from the previous batch that should be able to pass through a 200-mesh screen with sharp edges.

7. Amount of seed material to use: The amount of seed material to be used is about 375-500 gm per 1000 kg of the original fluid milk or 0.1 – 0.3 % of condensed milk.

8. Method of adding seed lactose: The seed lactose should ensure uniform dispersion, hence blending the 'seed' into a small amount of condensed milk and then adding this to the whole batch of the product with vigorous agitation helps to stimulate crystallization and that too formation of desired numerous small crystals is favored. Direct addition of the seed to the batch in dry form should always be avoided, as this causes lumpiness and aggregate masses in the product.

9. Lactose crystal formation: After 'seed' lactose is uniformly dispersed, cooling of the product may be continued slowly to 24°C over one hour and finally to 13-18°C with continuous agitation. Rapid crystallization leads to the formation of a large number of small crystals, giving a smooth texture to the condensed milk on the other hand slow crystallization creates a small number of large crystals which produce sandy of gritty texture.

10. Finishing the cooling process: After seeding and forced crystallization, cooling is resumed under constant agitation, as rapidly as possible until the final temperature is reached. Agitation is then generally continued for another hour or longer, and is sent for packaging.

(i) Systems of cooling: Batch, consisting of a vat with water jacketed sides and bottom and a powerful rotary agitator Continuous-flow, represented by internal tube arrangement with counter-current mechanism, and is used in large scale manufacture. Combined batch and continuous, a combination using internal tube cooler and a crystallizer tank. Vacuum cooling, which uses high vacuum as the cooling medium in other word the temperature of milk in this cooler is reduced by evaporation under vacuum.

(j) Packaging and storage: Packaging is done in barrels, drums with polythene liners, or tin-containers. For retail market, fillers are used to pack in can. After filling the cans are sealed, labeled, and packed in cases for storage and distribution. The recent trend is to store the condensed milk at 10°C or slightly below.

Evaporated Milk

Evaporated milk is commonly used when referring to full cream unsweetened condensed milk.

Flow Diagram of Evaporated Milk

Receiving milk
↓
Filtration/Clarification (35-40ºC)
↓
Standardization
↓
Forewarming /Pre-heating (115-118ºC / No Hold)
↓
Evaporation (1: 2.5)
↓
Homogenization I stage-2000psi
↓
II stage-500psi
↓
Cooling (7ºC)
↓
Pilot Sterilization Test
↓
Packaging
↓
Sterilization (116-118ºC/ 15 min)
↓
Cooling (27-32ºC)
↓
Shaking
↓
Storage (5-15ºC)

Details of Manufacture

(a) Receiving of milk: Good quality milk is received at the plant, its temperature should be at 10°C or below. The milk should be clean, sweet, and free from off-flavours and odours and reasonably free form extraneous material.

(b) Filtration/clarification: This is done in order to remove the visible foreign matter at a temperature of 35-40°C, and then cooled.

(c) Standardization: Standardization of the raw milk is carried out for desired fat/SNF ratio, (usually 1:2.44, done in raw milk).

(d) Forewarming/Pre-heating: Which is done for making the finished product free from microorganisms and enzymes, to ensure uninterrupted boiling in pan and to control age thickening in the finished product. The time temperature combination varies over 82-93°C for 5-15 minutes or 115 –118°C for no hold. Tubular heat-exchangers are commonly used.

(e) Evaporation: Evaporation means the concentration of milk or liquid products. The water has to reach a boiling point and the steam must have a pressure equivalent to ambient pressure. Vapour (steam) is removed by the pressure difference. No sugar is added here and normally a continuous operation takes place. It is customary to slightly over-condense the milk to facilitate standardization later.

(f) Homogenization: After evaporation and before cooling, the evaporated milk is thoroughly homogenized to obtain a uniform fat emulsion and reduce separation of fat to a minimum during storage. The temperature is usually 49 °C as the product is removed from the last part of the evaporator. Two-stage homogenization is generally recommended, 2000 psi in the first stage and 500 psi in the second stage.

(g) Cooling: After homogenization the evaporated milk is cooled to 7 °C.

(h) Pilot Sterilization Test

a. *Purpose:* To determine the amount of chemical stabilizers (trisodium citrate or disodium phosphate) to be added to any given batch of evaporated milk for the most satisfactory heat stability.

b. *Apparatus:* 1. Pilot sterilizer with accessories. 2. Viscosimeter. 3. Sample can filler 4. Weighing balance. 5. Glassware.

c. *Preparation of stabilizer solution:* An approximate 10 percent solution of stabilizer is prepared for use, so that 0.1 ml. of the solution is equivalent to the addition of 1 g. of dry salt for every 16 kg of evaporated milk.

d. *Methods of adding stabilizers and sterilizing the sample:* The amount of stabilizers to be added to any batch is determined by trial. The sample cans containing measured amount of stabilizer solution and a fixed quantity of evaporated milk are sterilized at 117 °C for 15 minutes and then cooled rapidly to 24°C. As soon as they have been cooled the cans are opened, examined for smoothness and colour and tested for viscosity. According to Mojonnier and Troy, a viscosity of 150°R on the Mojonnier-Doolittle viscosimeter represents the correct viscosity of evaporated milk as it comes out of the sterilizer.

e. *Determining the amount of stabilizer to be added:* If the viscosity tests for pilot sterilized can show that no stability correction is necessary, the batch is ready to be filled into the cans. However if the tests show otherwise, stability corrections becomes necessary.

f. *Adding stabilizer to the batch:* Based on the pilot sterilization test, the calculated quantity of stabilizer should be added to the evaporated milk in the form of a solution using just enough water to dissolve it. The stabilizer solution should be added slowly through agitating the evaporated milk both during addition and for several hours thereafter.

(i) Packaging: The cans are now mechanically filled with volumetric fillers, the types of one used for this purpose are the sanitary can, the can with a solder seal, and the vent hole can. The temperature of evaporated milk when filling the cans should be neither high nor low, but around 5°C. a higher temperature cause foaming, while a lower temperature increases the tendency towards 'flipping'. (This refers to the sudden snapping of can ends from their slightly concave contour to a convex outward bulge, which is usually accompanied by an audible sound). The cans should be filled as nearly full as possible.

(j) Sterilization: Immediately after sealing and before sterilization, each can is tested by means of a dependable leak checker /detector, where the cans pass submerged in a hot-water bath. In case of leaky cans, air bubbles rise to the top: these cans are removed for repair. The filled and sealed cans, which have passed the test for leaks, are now ready for sterilization. The purpose of sterilization is to destroy all germ life and enzymes present, there by preserving the product permanently. During sterilization, the temperature is raised to 116-118°C and held at this temperature for 15 minutes. If the cans cannot be sterilized within an hour or two, they should be then held under refrigeration.

Equipments: There are two types of sterilizers in use batch and continuous.

1. Batch method: This consists of a water steam boiler-like, horizontal steam drum, with hollow interior having a revolving frame, into which the cans are loaded. A perforated steam distributing pipe near the bottom, with a steam inlet at both ends, extends over the entire length of the sterilizer drum. The sterilizer rotates at 6-12 rpm, with uniform distribution of heat.

A successful sterilizing routine consist of:

Coming –up time	Between 15 and 20 min.
Holding temperature	116°C and 118°C

Holding time	Not less than 15 min.
Cooling time	Not more than 15 min.

2. Continuous: Here the cans are pre-heated gradually by slow moving them through progressively hotter hot-water chambers to a few degrees below the boiling point, and finally enter the sterilizing area proper where they are heated to sterilizing temperature by steam under pressure. Later the cooling process starts with the cans gradually moving through progressively less hot-water chambers and finally through cold water.

(k) Cooling: Immediately after holding time is over, the evaporated milk is cooled within 15 minutes to 27-32°C. Rapid and uniform cooling is important. Further, delayed cooling may causes the cans to bulge badly. Bulging of the cans can be avoided by using cold water and in continuous system, cooler operates at 10 psi.

(l) Shaking: This is done to break mechanically, any curd, which might have formed during the process of sterilization to a homogenous smooth consistency.

(m) Storage: Though evaporated milk can be stored at room temperatures. A storage temperature of 5-16°C is generally used, which helps to keep the product acceptable even upto 2 years. Inversion of cans once in 3-6 months during storage will help to minimize fat separation.

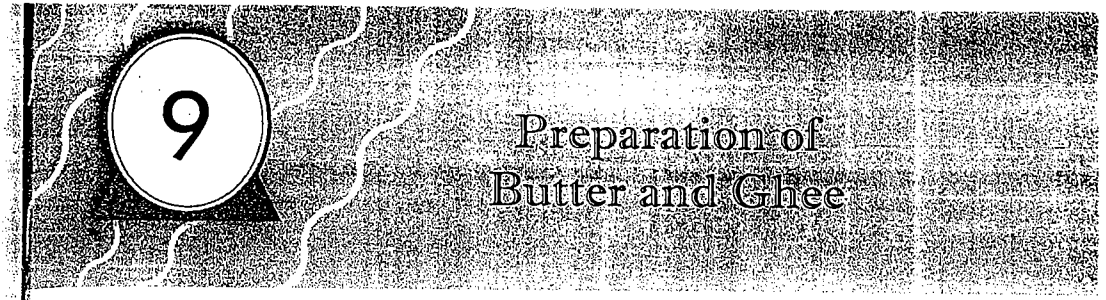
Uses of Condensed and Evaporated Milks

(a) Condensed Milk

1. For reconstitution into sweet milk drinks.
2. In tea or coffee.
3. In ice-cream preparation.
4. In candy and confectionery.
5. In prepared foods.

(b) Evaporated Milks

1. For reconstitution into milk for feeding infants and persons with weak stomach etc.,
2. In ice-cream.
3. In tea and coffee.



9 Preparation of Butter and Ghee

Butter

Butter leads amongst the milk products manufacture in developed dairying countries of the world today. The worlds recorded annual butter production in 2008 is 9605 thousand tones; and India contribution in the same was 3813 thousand tones (39.7%) and export of butter in 2008 is 18 thousand tones (2.2%) (European Commission, Agriculture and Rural Development DG).

Definition: Butter may be defined as a fat concentrate, which is obtained by churning cream, gathering the fat into compact mass and then working it.

According to the PFA rules (1976), table (creamery) butter must contain not less than 80% by weight of milk fat, not more than 1.5 % by weight of curd, and, not more than 3% by weight of common salt. Diacetyl may be added as a flavoring agent but, if so used the total diacetyl content must not exceed 4 ppm., calcium hydroxide , sodium carbonate, sodium polyphosphate may be added, but must not exceed the weight of butter as whole by more than 0.2 %.

Classification

Many kinds of butter are found in the market. This differs with the type of cream from, which they are made, and with variations in the manufacturing process. The types of butter are

(a) *Pasteurized cream butter:* Made usually from pasteurized sweet cream. Such butter usually has a milder flavour than that made from similar cream not pasteurized.

(b) *Ripened cream butter:* Butter made from the cream in which butter culture is added and incubated till the desired acidity and flavour are produced. Properly made ripened cream butter has a delicate flavour which is referred to as 'real butter flavour'.

(c) *Unripened cream butter:* Made from unripened cream.

(d) *Salted butter:* Butter to which salt has been added

(e) *Unsalted butter:* Contains no added salt.

(f) *Sweet cream butter:* Butter produced from cream in which the acidity does not exceed 0.20% in the churned cream.

(g) *Sour cream:* Made from cream in which the acidity exceeds 0.20%.

(h) *Fresh butter:* Butter that does not undergo cold storage.

(i) *Cold storage butter:* Butter stored at a temperature of about (–18°C) for some time.

(j) *Dairy butter (USA):* Made on farm. It is usually made from unpasteurized sour cream, which has not been standardized for acidity. This butter generally has a sour flavour due to the high acid content of the cream.

(k) *Creamery butter:* Made in a creamery or dairy factory. It is more uniform in quality than dairy butter.

Composition

According to PFA Rules (1976) Table/Creamery Butter should contain not less than 80% fat, not more than 1.5 % curd and not more than 3% common salt. The typical composition of Indian butter has been given below.

Composition of Indian Butter

Sr. No	Constituent	Percentage
1.	Butter fat	80.2
2.	Moisture	16.3
3.	Salt	2.5
4.	Curd	1.0

Food and Nutritive Value

Butter is very rich in fat and fat-soluble vitamins A, D, E and K.

Method of Manufacture, Packaging and Storage

Flow Diagram of Manufacture

Receiving milk
↓
Preheating (35-40°C)
↓
Separation of Cream
↓
Receiving cream
↓
Neutralization
↓
Standardization for Fat (cow milk – 40%) (buffalo milk – 35%) (around 40% Fat)
↓
Pasteurization
82–88°C No hold / 71°C for 20 min.
↓
Cooling (Room Temp.)
↓
Butter culture (0.5 to 2.0%)
↓
Ripening (21°C/15-16 hrs.)
↓
Ageing (5-10°C) (at least 2 to 4 hours)
(Preferably 15 to 16 hours)
↓
Churning and Washing (9-11°C)
↓
Addition of butter colour
(3%) Salting and Working
↓
Packaging and storage (–23 to –29°C)

Details of Manufacture

(a) Receiving Milk/Cream

1. Unloading of cream: The cans of cream brought to the creamery/ butter-making factory are unloaded on the receiving platform, which is at truck height, and arranged according to the route and the patterns.

2. Grading: Purpose of grading is to pay for the product on the basis of its quality. Cream is graded on the basis of smell, taste, appearance, touch, acidity and sediment. The technique of grading cream consists in removing the lid of each can, inverting it and smelling it. Next the cream is examined for appearance and the presence of any extraneous matter. Cream for butter making is graded in accordance with the grades of butter that can be made from it.

The cream grades are:

First grade cream	Sweet or slightly sour
Second grade cream	Sour, coagulated
Reject grade cream	Markedly sour, fermented.

3. Sampling: Before sampling, the cream in the cans is thoroughly mixed by a combined rotary and vertical movements of the plunger/stirrer. If required the lidded cans of cream may be sprayed with hot water to reduce cream viscosity, and facilitate mixing. Then a representative sample may be drawn for testing purposes.

4. Weighing: The cans of cream, which have been accepted, are then weighed and the weight recorded for accounting and other purposes. First grade creams are tipped directly into the main neutralization vats. The cream adhering to the inner walls of the cans is recovered by inverting the cans over steaming jet for short periods.

5. Testing: The cream samples drawn are tested for fat, solids –not fat, acidity, etc., by standard methods.

(a) Pre-heating of milk: To increase efficiency of cream separation.

(b) Separation of milk: By centrifugal methods.

(c) Neutralization of cream: Neutralization of sour cream for butter making refers to a partial reduction in its acidity.

Objectives

1. To avoid excessive fat loss in buttermilk that results from churning highly acid pasteurized cream. When pasteurizing sour cream, the casein curdles; thereby entrapping fat globules as the bulk of curd goes to butter milk this cause high fat loss.
2. To guard against the production of an undesirable off-flavour in cream.
3. To improve the keeping quality of butter made from high acid cream. Salted acid-butter develops a fishy flavour during commercial storage at –23 to –29°C. Correct method should be followed for neutralization of cream.

Procedure

1. Adoption of a definite standard of churning acidity: Butter for long storage, the cream acidity should be reduced to 0.06-0.08% before churning. Butter for early consumption the cream acidity should be reduced to 0.25-0.30 % before churning.

2. Testing correctly for acidity: Testing correctly for acidity take a sample of cream after through mixing. First partially neutralize acid cream with a known quantity of standard alkali, and then treat as above. Determine the lactic acidity of cream by titration of a fixed weight with a standard alkali using phenolphthalein as an indicator.

3. Correct amount of neutralizers to be added: The quantity of neutralizers to be added to a vat of cream varies with the acidity of cream, the final acidity desired in pasteurized cream and with the neutralizer compound used. It is necessary, first, to calculate the quantity of lactic acid to be neutralized per 100 kg cream. Then the amount of neutralizer required must be calculated. There are two groups of neutralizers available for use viz., lime (calcium hydroxide, and magnesium hydroxide) and soda (caustic soda, sodium carbonate, sodium bicarbonate and sodium sesquicarbonate).

Amount of neutralizers required

Sr. No.	Types of neutralizers	Theoretical (kg)	Practical (kg)
1	Sodium carbonate	0.93	0.83 up to 0.30 % cream acidity. 0.91 beyond 0.30 % cream acidity.
2	Calcium hydroxide	0.41	0.94

4. Correct procedure for adding neutralizer: The neutralizer should never be dry when added, but dissolved in clean, potable water and properly diluted, mixed with 10-15 times of its weight in water. The temperature of cream when adding the neutralizer should be preferably is 29- 32°C. The cream acidity should be determined to check to whether it has been correctly neutralized.

5. Role of carbon dioxide in neutralization of cream with sodium bicarbonate: Fresh cream contains some dissolved carbon dioxide; acidic creams contain, more of it. The carbon dioxide reacts with sodium hydroxide during titration and shows a higher acidity test. But the carbon dioxide does not react with sodium bicarbonate neutralizer and consequently over-neutralization results.

6. Double neutralization with lime and soda: The following sequence is recommended

1. First use lime neutralizer to bring the cream acidity down to 0.3 to 0.4 %.
2. Next use soda neutralizer to bring the cream acidity to the desired level.

The Objectives of Double Neutralization

1. To avoid the intense effect on flavour of a large amount of any one neutralizer with high acidity cream
2. To avoid production of excessive carbon dioxide by use of sodium bicarbonate with high acid cream.

(e) Standardization of cream: This refers to the adjustment of the fat level in cream to the desired percentage, conforming to standard requirements in cream 35-40% fat.

Objective: To obtain the lowest fat loss in buttermilk during churning.

(f) Pasteurization/ Vacreation of Cream

1. Holder pasteurization: The cream is heated in a batch/holder pasteurizer to 71°C for 20 minutes and then promptly cooled.

2. HTST (Plate) Pasteurization: This is a continuous process used for large scale handling. The maximum heating temperature may be 95-100°C for 15-16 seconds.

(g) Cooling and ageing of cream: Cream is cooled by lowering its temperature, and aged by holding it at this low temperature for a few hours. Under Indian conditions, the optimum temperature for the cooling and ageing of cream should preferably be 5-10 °C. The ageing periods should be at least 2-4 hours, preferably 15-16 hours. During ageing, crystallization of fat in the fat globules should satisfactorily complete.

Objective

1. To reduce losses of fat in butter milk
2. To obtain satisfactory product with firm body.
3. To provide a degree of solidification of fat in cream that may yield a normal churning period.

(h) Ripening of cream: This refers to fermentation of cream with the help of desirable starter cultures. Fermentation means strictly the composition of carbohydrate (sugar) by micro-organisms only.

Objectives

1. To produce butter with a pleasing pronounced characteristic flavour, aroma, uniformly from day to day
2. To obtain an exhaustive churning, i.e., a low fat loss in buttermilk.

Procedure: The butter starter culture containing lactic acid producers such as *Str. lactis* and/or *Str. cremoris* together with aroma (diacetyl) producers such as *Str. diacetilactis*, *Leuconostoc dextranicum* or *L. citrovorum*, in correct proportions, is added to the standardized, pasteurized and cooled (20-22°C) cream at 0.5-2.0 per cent. After being thoroughly mixed, the cream is incubated at 21°C for 15-16 hours.

Role of Diacetyl: The typical flavour of butter from ripened cream is mainly the effect of diacetyl, and to a smaller extent, of acetic acid and propionic acids. There is no diacetyl in sweet cream. The normal diacetyl content of ripened cream butter is on average 2.5 PPM and very rarely over 4 PPM.

Synthetic flavours: Cream ripening is expensive and time consuming and exacting. Further, most flavors enter into the buttermilk and wash water, and are loss to the butter, Hence the use of starter, starter –distillate or synthetic flavours, which are mixed with sweet-cream butter during the working process to impart the characteristic flavour to the ripened cream butter or the finished product. One way of improving flavour is to increase the citric acid content of cream or milk before fermenting begins.

(i) Churning

(1) Definition: Churning of cream consists of agitation at suitable temperatures until the fat globules adhere, forming larger and larger masses and until a relatively complete separation of fat and serum occurs.

(2) The factors contributing towards the stability of this fat in skim milk emulsion: The object of churning is to produce butter from cream. The fat exists in the form of emulsion i.e. a continuous phase. This emulsion is fairly stable. As long as it remains intact, there is no formation of butter. The factors contributing towards the stability of this fat in skim milk emulsion are:

(a) *Force of surface tension:* This causes the fat globules in milk/ cream to retain their individuality and prevent butter formation.

(b) *Phenomenon of adsorption:* The surface layer of the fat globules contains an adsorbed, phospholipids protein complex, which resists de-emulsion.

(c) *Electric charge:* The fat globules have negative charge and repel each other. The charge decreases as the cream acidity increases.

(d) *Viscosity:* Increased viscosity retards churning.

(3) Churning cream vs. whole milk: A greater concentration of fat globules in cream promotes a more profuse and rapid coalescence and aggregation than milk.

(4) Churning of cream: Good churnability refers to

(a) *Easy of churning:* Clear breaking stage churning until the grains of butter are of the correct size.

(b) *Completeness of churning:* Exhaustiveness of churning (refers to fat losses in buttermilk)

(c) *Satisfactory washing:* Which is possible if the grains of butter are sufficiently firm.

(d) *Optimum churning period:* Turn effects the firmness of body and exhaustiveness of churning.

(5) Factors Influencing Churnability of Cream and Body of Butter

(a) *Chemical composition of fat:* An increase in the proportion of soft fat shortens the churning period, diminishes the firmness of butter and increases the fat losses in butter milk and vice versa. Fresh green succulent feeds increase the proportion of soft fat and dry hard feeds increase the proportion of hard fat.

(b) *Size of the fat globes:* The higher the proportion of the small sized fat globules, the longer the churning time and the greater the fat loss in buttermilk and vice versa.

(c) *Viscosity of cream:* The greater the viscosity of cream, the greater the churning period and vice versa.

(d) *Temperature of cream at churning:* Under Indian conditions the optimum churning temperature ranges from 9-11°C. A higher churning temperature causes a shorter churning time, higher fat loss and a weak body in butter, which is difficult to wash and from which it is difficult to remove curd particles properly. A lower churning temperature prolongs the churning period.

(e) *Fat percentage of cream:* The higher the percentage of fat in cream, the lower the churning period. The fat percentage under Indian conditions will be 40 % for cow milk and 35 % for buffalo milk.

(f) *Acidity of cream:* According to Hunziker, acid cream churns more rapidly and exhaustively than sweet cream. However Mc Dowel believes that the reverse is the case.

(g) *Load off churn:* The butter churn should be filled with one-half to one –third of its total capacity with cream. Over loading prolongs churning time, while under loading reduces total capacity of the churn.

(h) *Nature of agitation:* This is influenced by the size, type, and r.p.m of the churn, and affects the churning period.

(i) *Speed of churn:* Since this maximum agitation and shortens the churning period.

(j) *Pre-churning holding period:* Refers to cooling and ageing.

(6) Churning Operation

(a) *Preparation of churn:* A new churn requires careful pre-treatment before use. An old churn requires proper sanitation and cooling, to render it clean.

(b) *Filling the cream into the churn:* After the stained the amount of cream filled should be slightly below the rated capacity.

(c) *Addition of butter colour:* This is done to maintain the uniformity of yellow colour in butter throughout the year for consumer satisfaction. The amount of standard color added varies from 0 to 250 ml or more per 100 kg of butterfat. The butter colour should preferably be added to the cream in the churn. Butter colour should have the following properties: it should be harmless, free from off flavours, concentrated, permanent, and oil soluble. Butter colours are of following types:

1. Vegetable origin: Annatto and carotene. Annatto is obtained from the seeds of the annato plant (Bixa orellana). Carotene is extracted from carrots and other carotene rich vegetable matter. Its use growing for it increases vitamin A potency.
2. Mineral origin: Harmless oil soluble coal tar dyes are mixed with neutral oil as above. Examples: Yellow AB (benzene Azo-β-naphthyl-amine), Yellow OB(Ortho Tolune Azo- β -naphthylamine).

(d) *Operating the churn:* After initially rotating the churn for 5 –10 min., the liberated gas is removed once or twice by opening the churn vent. Then the cream sample is drawn for the fat test. During the churning process there is invariably a rise in temperature from 1 to 3 °C. Churning is accompanied by foaming. Then comes the "breaking stage". When the cream breaks away from the spy glass, which becomes clear. At this stage the fat in the skim milk emulsion breaks and very small butter granules of the size of pin heads make their appearance, it is sometimes necessary especially in the tropics, to add break water at this stage to reduce the temperature of churn contents, and thereby

control the body of the butter. The amount and temperature of breakwater depends on the temperate reduction required. After the breaking the churning is continued until the butter grains are of the desired size (viz.,' pea size' in large churns). In the tropics, addition of break water can be avoided by providing an air-conditioned butter making room and /or chill water spray over the butter churn.

5. Factors Affecting Fat Loss in Buttermilk

1. **Fat percentage of cream:** Lower the fat percentage of cream, the lower the fat percentage in butter milk, but the greater is the percent total fat loss in buttermilk, vice versa.
2. **Size of fat globules:** The greater the proportion of small sized fat globules, the greater the fat loss ,and vice versa.
3. **Acidity of cream at churning:** According to Hunziker, sour cream causes a lower fat loss than sweet cream; but according to Mc Dowell, the reverse is true.
4. **Physical properties of fat:** The softer the fat, the more the fat loss and vice versa.
5. **Condition of cooling and ageing:** Insufficient cooling and ageing i.e. improper crystallization causes more fat loss and vice versa
6. **Conditions of churn:** Overloading, gross under loading and under churning all a cause a greater fat loss in butter milk.

(j) Washing

1. Draining the butter milk: When the cream has been churned the churn is stopped in the proper position, a drain-plug fixed and the buttermilk removed thorough sieve.

2. The purpose of washing:

1. Remove all loose buttermilk adhering to butter grains so as to reduce the curd content of butter, thereby improving its keeping quality.
2. To correct defects in the firmness of butter by proper adjustment of wash water temperatures, and
3. To decease the intensity of certain off flavours

3. Procedure of washing: After buttermilk has been drained chilled water is added to the butter grains in the churn. The temperature of water is usually1-2°C lower than the churning temperature of cream and an amount equal to the quantity of buttermilk removed. Normally one wash is enough for good quality butter. The quality of water should

be physically clean and bacteriologically and chemically safe. It is better to use freshly pasteurized and cooled water.

(k) Salting: This refers to addition of salt to butter.

(i) Purpose

1. To improve keeping quality
2. To enhance the taste.
3. To increase overrun.

(ii) Amount and quality: Salt is usually added at the rate of 2.0-2.5 % of the butterfat. Excessive salt damages the quality of butter.

(iii) Methods of Salting:

1. Dry salting: The calculated amount of salt may be added to butter either by sprinkling the powder salt over the butter surface during working
2. Wet salting: Calculated amount of salt is wetted in the least amount of potable water and then sprinkled over the butter during working.
3. Brine salting: The salt is added in the form of a saturated solution of brine. Practicable when only light salting is required.

(iv) Specifications of butter salt: The salt should be a coarse grained and free from lump. It should pass completely through and IS Sieve-85 (aperture 842 microns). Purity is 99.5–99.85% sodium chloride on dry matter. Bacterial counts less than 10/kg. Completely soluble, high rate of solution and negligible sediment.

(l) Working: This refers to the kneading of butter,

(i) Objectives

1. To completely dissolve, uniformly distribute and properly incorporate the salt.
2. To expel buttermilk and to control the moisture content of butter,
3. To fully incorporate the added makeup water in butter.
4. To bring the butter grains together into a compact mass for convenient handling and packaging.

(ii) Procedure of working: The working should be continued until the butter has a compact body, closely-knit grain, a tough waxy texture, and an even distribution of salt and moisture. Indicator paper develops a coloured spot if free moisture is present. Both over working and under working should be avoided; the overworking damages the body and

texture of butter and under working produces leaky butter. Working increases the air content of butter. Normally worked butter has an air-content of 0.5-10 ml/100 g. The air content of butter is important because it affects:

1. The density of butter
2. Its microbial spoilage; and
3. Its oxidative spoilage.

(m) Packaging of Butter

1. Wood or timber: Firkin Teak wood etc.
2. Parchment paper/substitutes: Vegetable parchment paper/ butter paper; cellophane; pliofilm; polythene. etc.
3. Aluminum foil /laminates: moisture and greaseproof
4. Tin-plates cans: Tin plate cans are advantageous in tropical countries not only in preventing melted butter from escaping in hot weather but also in preventing absorption of foreign flavours.
5. The packaging may be hand moulding and wrapping; or it may be mechanical moulding, patting, and wrapping.

(n) Storage: The temperature of cold storage of butter ranges from –23 to–29°C. There is invariably some flavour deterioration of butter during commercial cold storage, thus a fishy off flavour develops in salted acid butter. Bacteriological deterioration plays a negligible part, while chemical degradation plays a leading one.

(o) Overrun in butter: Overrun may be defined as the increase in the amount of butter made from a given amount of fat. It is usually expressed as a percentage. Overrun is caused by the presence of moisture, curd, salt etc., in butter. It is a source of profit to the butter maker and helps to check the efficiency of factory operations

(i) Types of Overrun

Theoretical overrun: Maximum obtainable, viz. 25%. Since the minimum legal fat content of butter is 80%, the maximum amount of butter that can be made from 100 kg of fat are 100/80x100=125kg. This gives an overrun of 25 %, which is not obtainable in actual practice.

Actual overrun: On the basis of fat actually bought and butter made there from.

(ii) Factors Influencing Over Run

1. Inaccuracy in weight, fat test of milk, cream or butter:
2. Fat losses in skim milk or buttermilk:

3. Mechanical fat losses
4. Unavoidable fluctuations in the fat content of butter
5. Weight allowances in butter packs or cream or butter
6. Handling losses etc.

Formula for calculation of overrun

$$\% \text{ OR} = \frac{B - F}{F} \times 100$$

Where

OR = Overrun in butter

B = Butter made (kg)

F = Fat in churn (kg)

Theories of Churning

They are three main theories on the churning of cream in to butter. Viz.

1. Fisher and Hooker's Phase Reversal Theory
2. Rahn's foam theory
3. King's Modern Theory.

1. Fisher and Hookers Phase Reversal Theory: According to this theory, churning is a process of phase reversal, i.e. changing an oil-in-water type emulsion (Cream) to a water-in-oil type emulsion (Butter). Agitation of cream in the churning process causes coalescence and clumping of fat globules until eventually the ratio of the surface area to the volume of fat units becomes so small that it can no longer contain all the buttermilk in stable form. The fat-in-water emulsion then suddenly breaks, yielding butter grains and free buttermilk.

Drawback: Butter is not true water in fat emulsion. Microscopic studies reveal that a proportion of fat globules in butter is still intact in the worked butter.

2. Rahn's Foam Theory: According to this theory, the presence of foam/froth is essential for churning. It also postulates that there is a foam producing substance present in cream, which gradually solidifies as the cream, or milk is agitated. Foam is created during the churning period. The fat globules due to surface tension effect tend to concentrate and clump on the foam bubbles. The foam producing substance assumes a solid character and the foam collapses. The fat globules then coalesce and butter is formed.

Drawbacks: Foam formation is not required in some continuous butter making processes.

3. King's Modern Theory: According to this theory.

1. In cooled cream at churning temperature, the fat is present as clusters of fat globules; and within each globule it is present partly in liquid and partly in solid form.
2. Churning breaks up the cluster and cause foam/ froth formation. The globules become concentrated to some extent in the film around the air bubbles in the foam and are thus brought into close contact with each other.
3. The movement of the globules over one another in the foam film and the direct concussion between them causes a gradual wearing away of the emulsion protecting surface layer of the phospholipid protein complex. The globules then adhere together to form larger and larger particles. Eventually these particles become visible as butter grains. As the granules form, they enclose some of the air from the foam. The fat in the granules is still mainly in globular form.
4. The working of butter grains cause the globules to move over one another, under the effect of friction and pressure, some of them yield up a portion of liquid fat. Others are broken up during working. Finally there is enough free liquid fat present to enclose all the water droplets, air bubbles and intact fat globules.

Continuous Butter Making (CBM): A number of types of continuous butter making machines have been developed for commercial use.

Advantages of CBM

1. More economical due to lower capital cost, lower running cost (reduced power, labour, refrigeration, steam, detergent etc.,).
2. Reduced floor space, no expensive foundations to prepare, no time loss for fat crystallization, less in butter wastage, etc.,
3. More hygienic due to it's being a closed system, free from air borne contamination.

Disadvantages

1. Lack of uniformity on the quality.
2. Difficulty in grading and analysis.

Basic principles: The continuous methods of butter production developed since 1935 may be divided into three main groups.

Groups I: Fritz process or churning process: This involves the use of high-speed beaters to destabilize the fat emulsion in the chilled cream, and thus cause the formation of grains of butter in matter of seconds.

The buttermilk is drained away and the resulting grains worked in a kneading section prior to extrusion.

Group II: Alfa-Laval process or the concentration and phase reversal process. This involves a system whereby cream of 30-40 per cent fat is concentrated in a special cream separator to 80-82% fat. After standardization, the concentrated cream or butter mix is subjected to combined cooling and mechanical action, which causes phase reversal and the formation of butter, followed by its expulsion from the machine.

Group III: Cherry Burrell process which again involves the concentration of 30-40 % cream. During concentration, the emulsion is broken and the fat, water, and salt content are standardized. This is followed by re-emulsification, cooling working, and finally extrusion.

Uses of Butter

1. Direct consumption.
2. In the preparation of sauces.
3. As a cooking medium.
4. In the baking and confectionery industries.
5. In the manufacture of Ice cream, butter oil and ghee.
6. In the production of reconstituted milk.

Ghee

Even today, the country or desi method of ghee making contributes more than 90 per cent of Indians ghee production, the remaining quantity begins manufactured by the organized sector of the industries which employ modern methods.

Definition: Ghee may be defined as clarified butter fat prepared chiefly from cow or buffalo milk. (To clarify means 'to make clear' a liquid or something liquefied, by removing unwanted solid matter or impurities).

According to the PFA (1976), ghee is the pure clarified fat derived solely from milk or from desi (cooking) butter or from cream to which no colouring matter is added.

Composition: The general chemical composition of ghee is following.

Chemical Composition of Ghee

Characteristics	Requirements	
	Cow	Buffalo
Milk fat	99 to 99.5 per cent	
Moisture	Not more than 0.5 per cent	

Unsaponifiable matter		
(a) Carotene (μg./g.)	3.2-7.4	-
(b) Vit. A (I.U./g)	19-34	17-38
(c) Tocopherol (μg./g.)	26-48	18-37
Free fatty acids (% oleic)	Max. 2.8 (Agmark)	
Charred casein, salts of copper and iron, etc.,	Traces	

Sources: Milk production of India by Srinivasan and Anantakrishnan (1964).

Physico-chemical constants: Ghee is characterized by certain physico-chemical properties, which show some natural variations depending on such factors as, method of manufacture, age and condition of the sample, species, breed, individuality and animals stage of lactation, season of the year, region of the country, feed of the animal etc., Some of the important analytical constants or standards of mixed ghee produced under standard conditions are given below:

1. **Melting and solidifying points:** The melting point varies from 28°C to 44°C, while the solidifying point varies from 28°C to 15°C. (As ghee fat is of a mixture of glycerides, it does not have a sharp melting or solidifying point).
2. **Specific gravity:** This varies from 0.93 – 0.94.
3. **Refractive index:** The Butyro-Refractometer (B.R), reading (at 40°C) varies from 40-45.
4. **Reichert-Meissl (RM) Value:** This is also known as Reichert value, and this should be normally not less than 28. However, ghee from cottonseed feeding areas, the limit is 20.
5. **Polenske value (P.V):** This should be normally not more than 2 (except for cotton-seed feeding areas, where the limit is 1.5)
6. **Saponification value:** This should be normally not less than 220.
7. **Iodine value:** This should normally vary from 26 to 38.

Process of Ghee

- More than 90 percent of ghee is produced by the traditional method from desi butter or makkhan and then converting it into ghee.

- Makkhan, which is produced by churning the curd, is heated in a metallic vessel and stirred over a low fire to evaporate the moisture.
- When practically all the moisture is removed, further heating is stopped and the vessel is removed from the fire. After the residue has settled down on cooling, the clear fat is decanted into suitable containers a ghee.

Methods of preparation: Ghee can be prepared by various methods, which are as follows:

1. Country/Desi method
2. Creamery butter method
3. Pre-stratification method
4. Direct cream method
5. Continuous method

1. Country/Desi Method: Fresh or accumulated over a few days, makkhan (butter) is taken in a suitable open mud-pot or metallic vessel, and heated and stirred on a low fire to drive out the moisture. When practically all the moisture has been removed, a stage judged by experience, further heating is stopped and the vessel removed from the fire. On cooling, when the residue has settled down, the clear fat is decanted into suitable containers.

Merits

1. Desirable flavour, body and texture

Demerits

1. Extremely small scale in operation, problems in collecting and marketing
2. Low keeping quality and vitamin content.

2. Creamery butter method: A standard method, where unsalted creamery butter is used. Butter is heated in an ghee boiler consisting of a steel jacketed pan with a manual stirrer, with steam supply and control valves. The butter mass, is cut into small pieces and heated on low heat and carefully stirred. Later after complete melting, the steam pressure is raised to bring the liquid mass to boil at a temperature of 90°C. The temperature remain contents as long as the moisture begins driven out. The contents are constantly agitated throughout the process of conversion of butter into ghee, to prevent scorching. The scum, gathering on the top of the boiling is periodically removed, until there is profuse effervescence, followed by crackling sound, initially. When all the moisture has been driven out, the temperature of the liquid mass

suddenly shoots up and end point is indicated by the appearance of effervescence for the second time together with browning of the curd particles. At this stage, characteristic ghee flavour emanates and the temperature now will be around 110-120°C. Heating is then stopped, and after cooling and sedimentation, the ghee is filtered through a muslin cloth and then goes for granulation and packaging.

Merits

Saving in labour, physical exertion and exposure to uncomfortably high temperatures and humidity during actual ghee making compared to the direct-cream heating method.

3. Pre-stratification method: The two above mentioned practices of ghee making has the disadvantage of poor quality of the finished product characterized by an over-heated and smoky flavour, along with high acidity, making it greasy and with reduced shelf life. Hence research work has led to the evolving of a newer method/technique, which yields a higher grade product at lower cost, known as the pre-stratification process.

The basic principle of this method is that, when butter is left undisturbed 80-85°C for 15 to 30 minutes, it stratifies, i.e., separates into 3 distinct layers, viz., a top layer of floating denatured particles of curd, a middle layer of fat, and a bottom layer of buttermilk. This separation is called pre-stratification. The bottom layer of buttermilk contains 60-70 percent of solids-not-fat and also over 80 per cent of the moisture originally present in butter. The buttermilk is removed mechanically without disturbing the top and middle layers. After wards, the temperature of the remaining two upper layers (of denatured curd and fat) is raised to the usual clarifying temperature of 110-120°C.

Merits

1. Economy in fuel consumption as compared to direct clarification
2. Production of ghee with lower acidity and longer shelf life.

4. Direct-cream method: In this method the cream obtained by normal separation of milk is heated in the same ghee boiler described for the creamery-butter method and the procedure for heating and moisture removal, final temperature of clarification, cooling and sedimentation, granulation and packaging also remain the same.

Merits

1. Overall economy in labor compared to the creamery method

Demerits

1. Low fat recovery due to greater fat loss in ghee residue.
2. Slightly greasy texture of ghee.
3. Ghee has flat flavour.

5. Continuous Method

Objects

1. To manufacture ghee on an industrial scale as an continuous process
2. To reduce human labor

Advantages

1. Large scale handling
2. Utilization of machines for a large number of operations
3. Uniform demand on services
4. High fat recovery
5. Possibility of in-place-cleaning (IPC)
6. No stirring, no scrapping and no laborious effort on the part of the ghee operators required.

Final temperature of clarification: The final temperature (110°C to 120°C) to which ghee is heated during the manufacture depends up on the region of the country, normally it is around 110°C in north India and 120°C in south India.

Cooling and granulation (crystallization): Granularity in ghee is considered as an important criterion of quality and even purity. Granular form of ghee is assumed primarily to certain content of glycerides of high-melting saturated fatty acids, especially palmitic and stearic acid. In this regard the buffalo ghee, which is more saturated, crystallizes more effectively than cow ghee. It has been observed that heating ghee to 60-100°C, followed by rapid cooling, yields small grains in ghee; however, if the ghee is held for crystallization at a temperature about 1°C above the melting point of ghee (cow ghee 29°C and buffalo 31°C), a large number of big grains result. Cold storage of ghee should be avoided, as it leads to a loss of granularity and the product develops a waxy consistency in the stored products.

Packaging: Non-toxic, non-tainting material with easy availability, economical, resistance to rough handling, suited for printing and capacity to hold desirable volume are preferred.

Storage: Under the existing condition of handling, the storage temperature of ghee may vary from 5 to 38°C throughout the country depending on the session of year.

Renovation of Ghee: This refers to the market practice of attempting to improve the old and rancid ghee, so as to make it marketable as a product of secondary quality.

Methods Employed for the Renovation of Ghee Include

1. Re-heating inferior ghee with curd, betel or curry leaves, etc., and subsequently filtering it;
2. Adding a yellow substance, such saffron, annatto, turmeric juice, etc to make it as cow ghee
3. Blending an inferior ghee with a superior quality product.

Neutralizing high-acid ghee: Market ghee sometimes develops large quantities of free fatty acid (oleic), which results from faulty methods of preparation and storage. This produces harmful effects in the body system. This is tackled by the process of neutralization. In one, the neutralizer used is sodium hydroxide, and other is lime. Lime is more preferable than sodium hydroxide. First the high acid ghee to be refined, is heated to 60-70°C, and finely ground, good quality lime(shell-lime), powdered to 60 mesh is then sprinkled on the surface @ 3 % of ghee. The temperature is quickly raised to108°C with gentle stirring and mass is cooled and filtered at 60°C.

Keeping quality of Ghee: Keeping quality of Ghee can be extended by the addition of antioxidants. Under the PFA Rules (1976), BHA at not exceeding 0.02 per cent can be added. Naturally, betel leaves and curry leaves are added while ghee making at 1 per cent level, which gives good flavour to ghee as well as it acts as antioxidant.

The AGMARK ghee-grading scheme: Literally Agmark is an insignia-AG for "agricultural" and MARK for 'marketing ' with a view to develop the orderly marketing of agricultural produce on all India basis, the Indian legislature had passed the agricultural produce (grading and marketing) act 1937, this act provides for the grading of ghee on a voluntary basis.

Objectives

1. To assure the consumer a produce of pre-tested quality and purity.
2. To enable manufacturers of a high-grade product to obtain better returns.
3. To develop an orderly marketing of the commodities by eliminating malpractices when transferring them from the producer to the consumer.

AGMARK ghee specifications: The original specification introduced in 1938 have been modified and amended from time to time.

Agmark Standard for Ghee

Sr. No	Tests	All-India	Regional	
			Winter	Summer
1.	Baudouin	Negative		
2.	Phytosterol test	Negative		
3.	B.R. reading(40°C)	40.0 - 43.0	41.5 - 44.0	42.5 - 45.0
4.	R.M. value	Not less than 28.0	Not less than 23.0	Not less than 21.0
5.	Polenske value	1.0 - 2.0	0.5 –1.2	0.5 -1.0
6.	Moisture	Not more than 0.3		
7.	Free fatty acids(%oleic acid) (a) Special grade (Agmark Red Label) (b) General grade (Agmark Green Label)	Not more than 1.4 Not more than 2.5		

Uses of Ghee

(a) Major Uses

1. As a cooking or frying medium
2. In confectionery
3. For direct consumption

(b) Minor Use

1. In indigenous pharmaceutical preparations (mainly cow ghee).

10 Preparation of Ice-Cream

The name, Ice Cream has derived from the phrase "Iced Cream". The name was later abbreviated to "ice cream" the name we know today. Ice cream is a frozen dessert made from dairy products, such as milk and cream, and often combined with fruits or other ingredients and flavours.

(a) Definition: Ice cream may be defined as a frozen dairy product made by suitable blending and processing of cream and other dairy products together with sugar and flavour, with or without stabilizers or colour, and with the incorporation of air during the freezing process.

According to the PFA rules (1976) ice cream is the frozen product obtained from the cow or buffalo milk or a combination thereof or from cream and or on the milk products, with or without the addition of cane sugar, eggs, fruits, fruit juices, preserved fruits, nuts, chocolate, edible flavours and permitted colours. It may contain permitted stabilizers and emulsifiers not exceeding 0.5 per cent by weight. The mixture must be suitably heated before freezing. The product should contain not less than 10%milk fat, 3.5% protein, and 36% total solids. However, when any of the aforesaid preparations contains fruits or nuts or both, the content of milk fat may be proportionately reduced but not less than 8% by weight, starch may be added to a maximum extent of 5%, with a declaration to that effect on the label.

(b) Classification: Some of the frozen desserts can be classified as follows.

1. **Plain:** Ice cream in which the colour and flavouring ingredients together amounts to less than 5 per cent of the volume of the unfrozen ice cream. Example: vanilla and coffee ice creams.
2. **Chocolate:** Ice cream flavoured with cocoa or chocolate.
3. **Fruit:** Ice cream containing fruits with or without additional fruit flavouring or colour. Fruits such as strawberry, apricot, pineapple, mango, banana, etc., maybe fresh, frozen, frozen packed, canned or preserved.
4. **Nut:** Ice-cream containing nuts, such as almonds, pistachio, walnuts, cashew nut, etc., with or without additional flavoring or color.

5. **Milk ices or lollies:** According to the PFA rules (1976) these refers to the frozen product obtained from the milk or skim milk or milk products with or without the addition of cane sugar, eggs, fruits, nuts, chocolate, edible flavours, and permitted food colours. It may contain permitted stabilizers not exceeding 0.5 % of the product. The mixture should be suitably heat treated before freezing. The product should contain not more than 2.0% milk fat, less than 3.5% proteins and not less than 20.0% total solids.
6. **Ices:** Made of fruit juices sugar and stabilizers with or without additional fruit acid, colour, flavouring, or water, and frozen to the consistency of ice cream. Usually contain 28 to 30 per cent sugar, 20 – 25 per cent over run and no dairy products.
7. **Sherbet:** Made of juices, sugar, stabilizers and milk products. It is similar to an ice except that milk, either whole, skim, condensed or powdered, or ice cream mix, are used in place of all or, part of the water in an ice.
8. **Fancy moulded:** Moulded in fancy shapes and composed either of one colour and flavour of ice cream or a combination of colours and flavours, or especially decorated. Examples: Rick Ice Cream, Cakes, Cake Roll, Moulds Representing Fruits, etc.
9. **Novelties:** Novelty ice cream or frozen is an especially shaped and usually low priced package containing an individual serving whose main appeal consists in its shape, size, colour or convenience for eating.
10. **Soft ice cream:** Sold as drawn from the freezer without hardening.

(c) Composition: The composition of ice cream is usually expressed as a percentage of its constituents, i.e. a percentage of milk fat, milk solids not fat, sugar, stabilizers, total solids etc. Its composition varies in different localities and in different markets. In ice cream, the percentage of milk fat varies more than any other constituents. As the fat content is increased, the milk solids not fat must be decreased so as to avoid sandiness.

Composition of Commercial Ice Cream (%)

Milk Fat	Milk solids not fat/ serum solids	Sugar	Stabilizer and emulsifier	Total solids
Economy Ice Cream 10	10 to 11	13 to 15	0.30 to 0.50	35 to 37
Good Average Ice Cream 12	11	15	0.30	37.5 to 39

Sources: Ice Cream by Arbuckle (1968).

The ISI specifications for ice cream (IS: 2802, 1964) are given below.

Sr. No.	Characteristics	Requirements
1	Weight (g./litre) (min.)	525
2	Total solids(%wt) (min.)	36.0
3	Milk fat (% wt) (min.)	10.0 (Tentative)
4	Acidity (% lactic acid) (max.)	0.25
5	Sucrose (%wt.) (Max.)	15.0
6	Stabilizers/Emulsifiers (%wt.) (Max)	0.5
7	Standard plate counts (per g.)	Not more than 2,50,000
8	Coliform count (per g.)	Not more than 90
9	Phosphatase test.	Negative.

(d) Food and nutritive value: Ice cream contains two to three times as much fat and slightly more protein than does milk. In addition it may contain other food products such as fruits, nuts, eggs, and sugar which enhance its food value. However, like milk, it lacks iron, vitamin C, and some of the trace minerals. Ice cream is a rich source of calcium, phosphorus and other minerals of vital importance in building good bones and teeth. Being rich in lactose, ice cream favours greater assimilation of the calcium content on the diet.

The protein content of ice cream rates high both in quantity and quality. The milk and egg proteins are complete, that is, they contain all the amino acids essential to animal life and are especially important sources of tryptophan and lysine which are lacking in many plant proteins. Ice cream provides these valuable proteins in a very palatable form. In fact, ice cream is the most palatable source of milk proteins to vegetarians.

Ice cream is an excellent source of food energy. Having twice to three times the fat content of milk and more than half of its total solids being sugar the energy value of ice cream is very high. It is therefore, a very desirable food item for growing children and persons who need to put on weight. Ice Cream, is a rich source of essential vitamins, without which normal health and growth cannot be maintained. Thus it is an excellent source of vitamin A, a good source of vitamin B (Thiamine) and G (riboflavin) and a fairly good source of Niacin, vitamin E, and in fruit ice cream, of vitamin C. The digestibility and palatability of ice cream is also very high.

(e) Role of the Constituents in Ice Cream

1. **Milk fat:** This is high in food value, but expensive. It enriches and mellows the ice cream, giving it a full, rich, creamy flavour. If the milk fat is even slightly off-flavoured, the defect will be noticeable. The fat also contributes to the body and melting resistance of ice cream while producing a smoothness of texture. Fat gives stability to the ice cream but impairs whipping ability.
2. **Milk solids not fat (SNF):** They add very little to the smell, but improve its body and texture, however milk sugar adds to the sweet taste. The milk proteins help to make ice cream more compact and smooth. Milk SNF should be added in as large quantity as possible without risking the danger of sandiness. They are high in food value and also inexpensive.
3. **Sugar:** The main function of sugar is to increase the acceptability of ice cream. The desired sweetening effect is only produced by sucrose. Sugars are usually the cheapest source of total solids in the mix.
4. **Stabilizers:** These are used to prevent the formation of objectionable large ice crystals in ice cream, especially during storage. Since they are added in very small quantities, they have a negligible influence on food value and flavor.
5. **Emulsifiers:** These are used mainly to improve upon and provide a uniform whipping quality of the mixture, and to produce a drier ice cream with smoother body and texture.
6. **Flavor and colours:** Flavor increases the acceptability of ice cream, and colours its aesthetic appeal.

(f) Properties of mix: The properties of practical importance on the mix are

1. **Viscosity:** This is defined as the resistance offered by liquids to flow. Viscosity is considered an important property of the ice cream mix, and a certain amount of it seems essential for proper whipping and the retention of air. Two types of viscosity exists in ice cream mixes: Apparent viscosity which is a thickened condition that disappears with agitation; and Basic viscosity: which remains after the apparent viscosity disappears. The viscosity of ice cream mix is influenced by composition, kind and quality of ingredients, processing and handling of the mix, total solids concentration and temperature.
2. **Acidity and pH:** The normal acidity of ice cream mixes is dependent upon the serum solids content. The normal acidity

of ice cream mix is 0.15% and the pH of ice cream mix should be 6.3. If the mix acidity is more, it may be neutralized with suitable neutralizers Eg. Sodium bicarbonate. It should be remembered that good ice cream couldn't be made from a highly acidic cream.

3. **Mix stability:** This refers to stability or resistance to separation by the milk proteins in an ice cream mix. Instability results in separation of milk proteins as coagulated or precipitated material in the mix, and the resulting ice cream has a curdled appearance on melting. Mix stability is affected by high mix acidity, low citrate and phosphate content, high calcium and magnesium continent, high homogenizing pressure, high heat treatment, low ageing time, destabilizing effect of freezing etc.
4. **Specific gravity:** The specific gravity or density of a ice cream mix varies with its composition and may range from 1.05 to 1.12.
5. **Surface tension:** This refers to the force of attraction between the molecules of a liquid at its surface. The greater the attraction between the molecules, the higher the surface tension and vice versa. The unit of measurement of surface tension is dyne.
6. **Freezing point:** The freezing point of ice cream is dependent on the soluble constituents and varies with its composition. The mix constituents, which affect the freezing point directly, are sugar, milk sugar, milk salts, and any other substances that may have been added and are in true solution. Freezing point is indirectly affected by fat, protein and any other constituents not in true solution by replacing water.

   Glucose, sucrose and corn sugar depress the freezing point in the descending order. In fruit ice-cream the freezing point will depend on the type of sugar used in fruit preparations. An average mix has a freezing point of 27.5°F. Mixes with high sugar and milk solids not fat contents may range to 26.5°F, while high fat, low MSNF or low sugar content mixes may range to 29.5°F.
7. **Whipping rate:** A high whipping rate means the ability to whip rapidly to a high over run. The present hypothesis is that whipping ability is based on the tensile strength and the strength of the lamella (i.e. walls around the air cells). Whipping ability is improved by a high processing temperature, proper homogenization and ageing the mix for 2-4 hours. Smaller fat globules and less clumping of fat globules increase the whipping ability.

Mixes made from butter, butter oil, or frozen cream have poor whipping ability. Egg yolk solids, fresh cream, buttermilk solids improve whipping ability. Sugar decrease the whipping ability except when added after homogenization, in which case it increases it. Finally, the construction and operation of freezer itself determine whether the maximum whipping ability of a given mix can be obtained.

The rate of whipping is measured by calculation the overrun at one minute intervals while the mix is being frozen in a batch freezer, normally within 3 to 5 minutes after the freezing process starts, the mix is frozen and within 7 minutes an overrun of 90 percent is obtained. In mixes, which have a rapid whipping rate, 90 per cent overrun may be reached in 5 minutes or less. Mixes requiring 8 minutes of more to reach 90 per cent overrun are considered to have slow whipping rate.

Method of Manufacture, Packaging, Hardening and Storage

Selection of Ingredients
↓
Figuring the mix
↓
Mixing Ice cream mix
↓
(SMP, Cream, Sugar, Stabilizer, Emulsifier)
↓
Homogenization of mix (1st stage-2500 psi & 2nd stage 500 psi)
↓
Pasteurization the mix (68°C for 30 min)
↓
Cooling (15°C)
↓
Ageing of mix (4°C/3-4 hrs.)
↓
Freezing the mix (–4 to –5°C)
↓
Packaging of ice-cream
↓
Hardening and storage of ice-cream (–23 to –29°C)

(aa) Selection of ingredients: The selection of ingredients is without doubt the most important factor in successful ice cream manufacture. A

clean fresh creamy flavour of the ice cream can be secured only by the use of products, which have been carefully selected and handled. The selection of ingredients depend on

- Availability of milk products
- Perishability of the products
- Convenience in handling
- Effect on flavour body and texture of ice cream
- Cost
- Equipment available.

Ice cream ingredients may be grouped into dairy and nondairy products.

I. Dairy Products

Source of fat

1. Sweet cream: This is the most desirable concentrated source of fat for use in a mix.
2. Frozen cream.
3. Plastic cream.
4. Unsalted butter.
5. Butter oil.

Source of Milk Solids not Fat

1. Skim milk.
2. Skim milk powder this is most frequently used in the spray dried or flaked form.
3. Condensed skim milk.
4. Sweet cream buttermilk.

Sources of Both Fat and Solids not Fat

1. Whole milk.
2. Whole milk powder
3. Condensed whole milk.
4. Evaporated milk.

II. Non-Dairy Products

(a) Sweetening Agents

1. Cane sugar or beet sugar.
2. Corn sugar (dextrose).
3. Corn syrup solids (dextrose + maltose).

4. Corn syrup.
5. Invert sugar (glucose + fructose).
6. Saccharin.

(b) Stabilizers: These may be defined as substances which help to preserve emulsions. Although added in small amount, they perform a very important role. Stabilizers are added in ice cream to produce smoothness in body and texture, retard or reduces ice crystal growth during storage and provide uniformity in the product and resistance to melting. Stabilizers function through their ability to form gel structure in water or combine with water of hydration. Among the stabilizers which are permitted and used in making ice cream are below.

1. Gelatin (Animal origin): It was one of the first of the commercial stabilizers and still used, its advantages lies in its ability to form a gel in the mix during the aging periods as well as during the freezing process, and even after the frozen product is placed in the hardening room its peculiar gel stricture and its great affinity for waste prevent the formation of large ice crystals in ice cream and contribute to the smoothness on texture and firmness in body of the frozen product.

The amount of gelatin used depends on several factors such as the source of gelatin whether from calf, pork skin or bone materials. Its gel structure is measured by bloom test. Commonly gelatin is used at the rate of 0.25 to 0.5 % for a 250 bloom gelatin. The ice-cream mix stabilized with gelatin usually requires about 4 hours of ageing to develop complete stabilizing properties, while other stabilizing materials do not require an ageing period.

2. Sodium alginate (Vegetable origin): The basic stabilizing principle 'algin' is extracted from ocean kelp (seaweed) growing on the shores of California and in Japan. This product improves whipping ability and leaves a slightly cleaner flavour in the mouth. It dissolves properly only when added to the mix at about 68-71°C. A slightly smaller amount is needed to produce the same stabilizing effect as gelatin.

3. Guar gum (Indian origin): This is a carbohydrate obtained from Indian legume. Often used in combination with carageenan. It is readily soluble in cold solutions and used as stabilizers for mixes to be pasteurized by HTST.

4. Carageenan: This is extracted from Carageen, a seaweed growing on the coast of Ireland, etc. It is used together with gum stabilizers.

5. Agar-agar: This is a products extracted from red algae (Seaweed) growing on the pacific coast. More useful for sherbets and ices.

6. Carboxy methyl cellulose (CMC): This has a high water holding capacity and can be easily dissolved in the mix. Also acts as an emulsifier. Slightly less of it is to be used than gelatin.

7. Pectin: This is a carbohydrate obtained mainly from citrus fruits. Not satisfactory as a stabilizers for ice cream.

(c) Emulsifiers: They are substances which help to form emulsions. The value of emulsifying agents in the manufacture of ice cream lies mainly in the improved whipping quality of the mix, the production of a drier ice cream with smoother body and texture. There are two kinds of emulsifiers used in the manufacture of ice cream, viz., mono and di-glycerides derived by the chemical reaction of naturally occurring glycerides and polyoxythylene derivatives of hexahydric alcohols, glycol and glycol esters. The mono glycerides improve fat dispersion and whipping ability and have a moderate effect on stiffness and the melting rate. The poly-derivatives are effective in producing dryness, stiffness and increasing the melting time.

(d) Flavours: The most popular flavours, viz., vanilla and chocolate.

(1) Vanilla: This is the most popular flavour all over the world. Vanilla flavouring is obtained from the perennial climbing plant Vanilla Planifolia Andrews, a member of the orchid family, vanillin is the principal flavouring material. The typical vanilla flavours is developed by fermenting the bean.

(2) Chocolate: Chocolate and cocoa rank second only to vanilla as flavoring of ice cream. They are obtained from cocoa beans, the fruit of the perennial tree. Theobroma cocoa, growing in such tropical regions as Mexico, Ghana, Brazil, Venezuela, Nigeria, Ecuador, The West Indies, African Gold Coast and The East Indies etc. Other flavours are Strawberry, Pineapple, Lemon, Banana, Mango and Orange.

(e) Colour: Only harmless, edible, permitted food colours should be used. Generally, colours are matched with the flavours added ex. Yellow, Green and Pink.

(f) Egg solids: Frozen and powdered egg yolks are used by many ice cream manufactures. Usually not more than 0.5 percent of egg yolk solids are added to the mix. Egg yolk solids improve the whipping ability of the mix.

(g) Fruits and nuts: The fruit in ice cream may be fresh, frozen or canned. Fresh fruits must be considered the best source of flavour when available at low price. In India, pieces of chopped mango, banana, etc. are added towords the end of the freezing process @ 7-8% of the mix. Ex. Apple, Banana, Mango, Pineapple, Grape. Nuts should be sound, clean and free from rancid flavours. All nuts should be chopped into

very small pieces before they are added to the mix at 1-3%. Ex. Almond, Pistachio, Cashew nut, Walnut and Groundnut.

(bb) Figuring the mix: Knowledge of calculation of ice cream mix is helpful in properly balancing the mix, in establishing and maintaining a uniform quality and in producing ice cream that conforms to legal standards. Ice cream mix may be divided into two groups, namely simple and complex. Simple mixes require the least calculation and are made of ingredients, each of which supplies one constituent. Complex mixes are more difficult to calculate they include mixes where at least, one constituent is obtained from two or more products. Complex mixes require the use of the Pearson's square, algebraic methods.

The calculations are simplified by first finding the weight of ingredients needed to make 100kg of the desired mix. With the algebraic method, symbols such as X,Y, Z is used to represent the weights of dairy ingredients required for a 100 kg batch of mix. These symbols are then used in writing the three equations that express the weight of fat serum solids (SNF) and the total weight of dairy ingredients needed for 100 kg of mix.

(cc) Making the mix: The selection of good, wholesome ingredients and calculation of a satisfactory composition precede the mixing of the ingredients in a vat, where they can be heated to facilitate dissolving, blending and pasteurization. The order in which ingredients are added is as follows; all liquid ingredients are placed in a jacketed vat provided with a power stirrer, and the agitation and heating started at once, the dry ingredients, including skim milk powder, sugar, and stabilizers are added while the liquid material is agitated before the temperature reaches 49°C. Proper suspension to avoid lumpiness of the dry ingredients may be obtained by either mixing the dry materials thoroughly with part of the sugar before slowly adding it to the liquid, or by sifting / slowly adding these substances to the liquid.

If gelatin is the stabilizer used, it is best added after it is thoroughly mixed with equal volume of sugar, and before the liquid material reaches 49°C. Alternatively, it can be sprinkled on the surface of the cold liquid and allowed to soak before the mixture is heated or soaked in water and the mixture heated to completely dissolve the gelatin, which is usually added to warm (38-49°C) mix.

If sodium alginate is used, it should not be added until the temperature of the liquid material has reached at least 66°C. The dry Dariloid is not allowed to soak but is stirred up with cold water and immediately dumped into the hot mix.

If butter, plastic cream, frozen cream, or other products are used, they should be cut into small pieces and added after time has been given

to allow for complete melting before the pasteurizing temperature is reached. With a few exceptions, colouring and flavouring materials are added when the mix is frozen.

(dd) Pasteurization of mix: The Purpose of Pasteurization are

- It renders the mix completely free of pathogenic bacteria,
- It dissolves and helps to blend the ingredients of the mix.
- It improves flavour,
- It improves keeping quality, and it produces a more uniform products

Proper pasteurization consists in rapidly heating the mix to a definite temperature, holding it at that temperature for a definite minimum period of time and then rapidly cooling it to below 5°C. The ISI specification for pasteurization temperature – time combination for ice cream mix are as follows:

- Batch system - 68.5°C for not less than 30min.
- HTST method - 80 °C for not less than 25 sec.

(ee) Homogenization of mix: The main purpose of homogenization is to make a permanent and uniform suspension of the fat by reducing the size of the fat globules to a very small diameter, preferably not more than 2 microns. The advantages of homogenization are:

- It prevents fat separation during ageing.
- Produces more uniform ice cream with a smoother texture.
- Improves whipping ability
- Shortens ageing period.
- Decreases the risk of churning occurring in the freezer and
- Leads to the use of slightly less stabilizer.

The mix is usually homogenized at temperature from 63 to 77°C at pressure of 2500 to 3000 psi with one valve, or 2,500 to 3000 psi at the first stage and 500 psi at the second stage will usually give good results for an average mix (with 3 to 12% fat).

(ff) Cooling and ageing of mix: Cooling the mix immediately after homogenization to 0 - 5°C is essential, after which it should be held in ageing tanks until used. Ageing refers to holding the mix at a low temperature for a definite time before freezing. The ageing temperature should not exceed 5°C. The ageing time under commercial conditions may range from 3 to 4 hours, except for sodium alginate which requires no ageing. Ageing produce following results:

- It improves the body and texture of ice cream
- It improves the whipping capacity
- Increases maximum overrun
- Increases melting resistance.

(gg) Freezing the mix: Freezing is one of the most important operations in the making of ice cream for upon it depends the quality, palatability and yield of the finished product. The freezing process may be divided into two parts. The mix with the proper amount of colour and flavouring materials generally added at the freezer, is quickly frozen while being agitated to incorporate air in such away as to produce and control the formation of small ice crystals which is necessary to give smoothness in body and texture, palatability and satisfactory overrun in the finished product. When ice cream is partially frozen to a certain consistency it is drawn from the freezer into packages and quickly transferred to cold storage rooms where the freezing and hardening is completed without agitation.

(a) Classification of freezers: Freezers may be classified as follows:

1. **Batch freezer:** Horizontal position and direct expansion type.
2. **Continuous freezer:** Horizontal position and direct expansion type.
3. **Soft serve freezer:** Batch and automatic continuous freezer of the direct expansion type.

(b) Importance of rapid freezing: Rapid freezing is essential for a smooth product because ice crystals that are formed quickly are smaller than those formed slowly. Therefore it is desirable to freeze and draw the mix from the freezer in as short a time as possible. A continuous freezer accomplishes this in a few seconds, while batch freezer takes 6 to 10 minutes.

Since the freezing continues after the ice cream is placed in the hardening rooms, the ice crystal formed during the hardening period are larger because they are formed more slowly than in the freeze. For this reason it is desirable to freeze the ice cream as stiffly as possible and yet have it liquid enough to be able to draw it out of the freezer.

(c) Factors Influencing the Freezing Time

I. Mechanical:

1. Type of freezer
2. Condition of freezer walls and blades
3. Speed of dasher.

4. Temperature of refrigerant.
5. Velocity of refrigerant passing around the freezing chamber.
6. Overrun desired.
7. Temperature at which ice cream is frozen.
8. Rate at which freezer is unloaded.

(ii) Characteristics of Mix

1. Composition of mix.
2. Freezing point of mix.
3. Acidity of the mix.
4. Kind of ingredients, particularly those carrying fat.
5. Methods by which the mix is processed.
6. Kind and amount of flavouring materials added.

(d) Changes occurring during freezing of ice cream mix: The function of freezing process is to freeze a portion of the water of the mix and to incorporate air into the mix. This involves: Lowering of temperature of the mix from the ageing temperature to the freezing point, freezing a portion of the water in the mix, Incorporating air into the mix and cooling the ice cream from the temperature at which it is drawn from the freezer to hardening room temperature.

The temperature of mix, which is put into the freezer drops very rapidly, this process, takes less than a minute or two. Mean while the rapid agitation reduces the viscosity by partially destroying the gel structure and by breaking up the fat globule cluster. The gel structure may restore itself partially during the hardening process in the hardening room. The rapid agitation hastens incorporation of air into the mix.

When freezing point is reached the liquid water changes to ice crystals, which appear on the mix. These ice crystals are practically pure water in a solid form, and thus the sugar as well as the other solutes becomes more concentrated on the remaining liquid water. Increasing the concentration of these solutes slightly depresses the freezing point of the liquid portion, so that the temperature must be lowered before anymore ice crystals will form.

Thus, in freezing ice cream, the freezing point is continually being lowered by the formation of ice crystals and while the temperature drops, more ice crystals are formed, increasing the concentration of sugar and other solutes in the remaining liquid water until the concentration is so great that further freezing will not occur, thus all the water dose not freeze even after long periods in the hardening room.

The first phase of freezing process accounts for the freezing of 3 to 6 per cent the water, depending on the drawing temperature. The second phase i.e. hardening process, this account for the freezing another 23 to 57 per cent, depending on the drawing temperature.

(e) Batch freezing: The batch freezer consists mainly of a freezing chamber and a dasher. The dasher consists of two parts viz., scraper blades and the beater, the dasher performs the following functions: Aids in transmission of refrigeration by keeping the mix in continuous contact with the freezer walls, scrape freezer walls free of ice crystals, beats in air, continually pushes mix forward, which is essential for unloading the freezer. The freezing procedure in batch freezer consists of:

i. **Preparation of the freezer:** The freezer parts should be inspected to ensure that they are clean and dry. Then they should be assembled in accordance with the instructions from the manufacturer. The freezer is sanitized by running hot water at 82°C or chemical (chlorine) sanitizers.

ii. **Adding the mix:** It is always desirable that the temperature of the mix be below 5°C when it goes into the freezer. Colour and flavouring should be added only when some ice crystals have been formed. The total volume of the mix, flavour, and colour should be about half the total capacity of the freezer chamber. The flavour and colour must be so added as to be uniformly distributed.

iii. **Freezing and incorporation of air:** The mix should now run into the freezer, the dasher should be started and then the refrigerant turned into the freezing chamber. This sequence must be maintained, the dasher must never be operated when there is no mix in the freezer; neither should the refrigerant be turned on unless the dasher is in motion.

iv. **Drawing of ice cream:** When ice cream is drawn from the freezer, it should be sufficiently stiff to 'ribbon' or almost hold its shape, and yet soft enough to 'settle' or lose its shape within a minute or two. The container or package into which the ice cream is drawn should be cooled sufficiently to prevent the ice cream melting; this melting may cause a coarse and icy texture around the edge of the package and also loss of overrun. Formation of air pockets within the container should be avoided, the freezer should emptied rapidly to prevent wide fluctuations in overrun in the package, Therefore, avoid filling small packages directly from the batch freezer.

(f) Continuous freezing: Continuous freezers are of two types in the market today:

1. One of these has two pumps just behind the freezer barrel. One pump pulls the mix from the supply tank and pumps it to the

second one. The second pump operated at a speed approximately twice as great as that of the first pump. This has the effect of creating a partial vacuum between the two pumps. The valve device in the piping between the pumps allows for air to be sucked in. The amount of air incorporated can be regulated very accurately. The second pump pumps the mix and air into the freezer barrel. As the mix freezes, semi-frozen ice cream is forced out from the front of the machine.

2. The second type operated in a very similar fashion as the first, however here the two pumps are at the front. One pump pushes the mix into rear end of the freezer and the other helps to pull out the semi frozen ice cream form the freezer. There is separate air pump that forces air directly into the freezer barrel. This second kind of continuous freezer is just as versatile as first one. Some important advantages of the continuous freezing method are:
   i. Less stabilizer is needed.
   ii. A shorter ageing period.
   iii. Less flavouring material is needed.
   iv. Smoother ice cream is obtained.
   v. Tendency towards development of sandiness decreases
   vi. A more uniform yield is obtained with less variation between packages.

Some disadvantages are:

1. Great care must be taken in handling the parts of the machine that fit with very small clearance. Operators and mechanics must have special experience and training in order to avoid operational difficulties and possible damage to the equipment.
2. It is difficult to prevent excessive overrun.
3. There is greater tendency for ice cream to shrink on volume after hardening.
4. Initial cost of the equipment is high.

(hh) Overrun in ice cream: Overrun is usually defined as the volume of ice cream obtained in excess of the volume of the mix. It is usually expressed as a percentage. This increased volume is composed mainly of the air incorporated during the freezing process. The amount of air incorporated depends on the composition of the mix and the way it is processed; and is regulated so as to give that percentage overrun or yield which will give the proper body, texture and palatability necessary

to a good quality product. Too much air will produce a snowy, fluffy, unpalatable ice cream while too little, a soggy, heavy product. The desirable overrun percentage in different ice creams is given below.

Percentage overrun in ice creams

Sr. No.	Products	% Overrun
1	Ice cream packaged	70-80
2	Ice-cream bulk	90-100
3	Softy ice cream	30-50

There are two basic or fundamental methods of calculating percentage overrun, viz., by volume and by weight.

(a) $$\% \text{ OR} = \frac{(\text{volume of ice cream}) - (\text{volume of mix})}{\text{volume of mix}} \times 100$$

(b) $$\% \text{OR} = \frac{\left(\begin{matrix}\text{weight of unit volume} \\ \text{of mix}\end{matrix}\right) - \left(\begin{matrix}\text{weight of unit volume} \\ \text{of ice cream}\end{matrix}\right)}{\text{weight of unit volume of ice cream}} \times 100$$

(ii) Packaging of ice cream: When ice cream is drawn from the freezer it is usually collected in containers, which give it the desired shape or size for convenient handling during the hardening and marketing processes. Ice cream packages are of two types

i. **Multi service:** This type of container is seldom used in modern operations; a tinned steel can is an example. They are cleaned, sanitized, and used repeatedly.

ii. **Single service:** This type of package has met with increasing favour, especially during recent years. The quantity filled may be for bulk or retail sale. Most bulk ice cream is packaged in fibre board cartons coated with wax or polythene wax blends for protection against moisture and oxygen. The most recent trends for such factory filled carry home packages are the all plastic cylindrical container with a reclosable lid. Retail ice cream may be packaged in cups, stick, or bars. Cups may be of paper or cardboard treated as above to make them impervious to moisture. The recent trends is towards multipacks.

(jj) Hardening and storage of ice-cream: The freezing process is continued without agitation in the hardening rooms until the temperature of the ice cream is reaches –18°C or below. Here quick hardening is desirable, since slow hardening favours large ice crystals and coarseness. Most operators allow hardening time of 12 hours. The factors affecting hardening time are: speed of air circulation, size and shape of the ice cream package, temperature of cooling air, section of

hardening room, temperature of ice cream drawn from the freezer, composition of mix and percentage overrun in the ice cream being hardened. The different hardening systems are:

1. Hardening rooms –12 to –45°C.
2. Hardening tunnels –34 to –45°C.
3. Hardening cabinets –23 to –26°C.

Ice cream can be stored in the room where the temperature is maintained uniformly at a point between –23 to –18°C and the packages should be piled very close to delay change in ice-cream temperature.

Distribution of ice cream: When ice cream is marketed the manufacturer usually transports it to retailer under refrigeration at the same temperature as is maintained in the retailers cabinet. The various means of refrigerating the distribution vehicle are

i. **Dry ice refrigeration:** Dry ice is solid carbon dioxide with a freezing point of –78°C. It is used extensively for package deliveries.
ii. **Refrigerated truck:** This is most commonly used means of transport for ice cream in developed countries, the truck is refrigerated overnight, and loaded with hardened ice cream the following morning.
iii. **Frozen brine:** Jacketed metal containers, which contain calcium chloride solution of specific gravity of 1.1 are refrigerated to –21°C and are then placed around the packaged ice cream, this arrangement is used by a few ice cream factories for air lifting of ice cream in this country.

(kk) Softy ice cream: There is a marked demand for the ice cream, which has generally come to be known as 'Softy'. This term has been applied largely because this product is marketed in a soft condition and is ready for consumption shortly after it is drawn for the freezer. The problem involved in the preparation of softy ice cream are some what different from those encountered in the manufacture of regular ice cream.

Composition of softy ice cream

Sr. No.	Composition	Percentage
1	Fat	3-6
2	Milk Solids Not Fat	11-14
3	Sugar	12-15
4	Stabilizers and emulsifiers	0.4 –0.6

Softy ice cream is usually drawn from the freezer at around –8 to –7°C. The overrun may range from 30 to 50 %.

11 Preparation of Cheese

Cheese is one of the important and popular products of the dairy world. Cheese making has been an age old practice as it served as a means of preserving milk solids through moisture removal by coagulation and had a longer shelf life besides high nutritive value. The manufacture of cheese did not develop in India during the past mainly because animal rennet could not be used by Indians (mostly Hindus), who considered cow slaughter sinful.

In India, cheddar cheese, processed cheese and cottage and other allied types (panner and chhana) are the common verities like by the consumers. Today there are at least 2000 names assigned to various types of cheese, although several of them differ very little. However, there are approximately 400 established verities and 18 distinct types of cheese.

Davis (1965) has defined cheese as "a product made from the curd obtained from milk by coagulating the casein with the help of rennet or similar enzymes in the presence of lactic acid produced by added or adventitious micro-organisms, from which part of the moisture has been removed by cutting, cooking and or pressing, which has been shaped in a mould, and then ripened by holding it for sometimes at suitable temperatures and humidities".

According to the description of hard cheese in PFA Rules (1976), cheese (hard) means the product obtained by draining after coagulation of milk with a harmless milk coagulating agent under the influence of harmless bacterial cultures. It shall not contain any ingredients not found in milk, except coagulating agents, sodium chloride, calcium chloride not exceeding 0.2% by weight, annatto or carotene colour and may contain certain emulsifiers and or stabilizers, viz., citric acid, sodium citrate or sodium salts of orthophosphoric acid and polyphosphoric acid, not exceeds 0.2% by weight; wax used for covering the outer surface should not contain any thing harmful to the health.

In case the wax is coloured, only permitted food colours should be used. Hard cheese should contain not more than 43% moisture and not less than 42% milk fat of the dry matter. Hard cheese may contain 0.1% of sorbic acid or its sodium, potassium or calcium salts; or 0.1% of nisin.

Types of Cheeses: The classification of cheese is based on a number of factores like raw material, types of consistency, appearance (interior and exterior), fat content, moisture content and ripening methods. However, the most commonly used criteria are the moisture content of the finished production and mode of ripening.

(i) Types Based on Moisture Content

(a) Very hard (maximum 34%moisture)
(b) Hard (maximum 39% moisture)
(c) Semi-hard/semi-soft (39-50% moisture)
(d) Soft (50-80% moisture)

(ii) Types Based on Mode of Ripening

(a) Bacterial ripening: Ripening is brought about by different bacteria like *lactic streptococci, lactobacilli, pediococci, leuconostoes, propionibacteria, brevibacterium etc.*
(b) Mold ripening: Ripening is brought about by mold species like penicillium spp.
(c) Unripened: Ripening not involved.

Classification of Cheese

Country of origin	Soft	Semi-hard	Hard	External Mould	Internal Mould
U.K	Cream	Lancashire	Cheddar cheshire		Stilton
France	Neufchatel		Gruyere(P)	Camembert	Roquefort/ Blue (G)
Belgium		Limburger			
Italy			Parmesan(VH) Romano Provolone(W)		Gorgonzola
German	Romadur	Munster			
Holland			Edam Gouda		
USA	Cottage Cream(C)	Brick	Cheddar Swiss (P)		Blue(G)
Sweden			Herrgard(P)		
Switzerland			Swiss/ Emmental		
Norway					Gammelost
Hungary	Liptau				

C = High fat content, G = A general name, P = Propionic fermentation leading to holes (eye), R = Ripened, S = Surface Slim, U = Unripened, W = Washed (plastic) curd, VH = Very hard.

Composition of Different Types of Cheese

Types of cheese	Moisture %	Fat %	Protein %	Calcium %	Vitamins			Energy Content (KCal/100g)
					Vita. A (µg/100g)	Thiamin (µg/100g)	Riboflavin (mg/100g)	
Hard (Cheddar)	35	33	26	0.83	380	50	0.5	400
Semi hard (Edam)	43	24	26	0.76	250	60	0.35	320
Blue-veined (Roquefort)	40	31	21	0.32	300	30	0.7	360
Soft (Camembert)	51	23	19	0.38	240	50	0.45	280
Unripened (Cottage)	79	0.4	16.9	0.09	3	30	0.28	82

Food and Nutritive Value

Cheese has high food and nutritive value

1. It is an excellent source of milk proteins.
2. A rich source of calcium and phosphorus.
3. An excellent source for several fat-soluble vitamins, such as A, D, E, and K.
4. A concentrate form of energy; cheddar cheese gives about 400 calories/100g.
5. Cheese is both palatable and digestible.

Cheddar Cheese: Cheddar cheese originated in the town of cheddar located in the county of Somerset in south-western England. Cheddar is probably the best known cheese in the world.

Manufacture of Cheddar Cheese

1. Receiving of milk: It is well know that only high grade milk can yield high-grade cheese. The quality of finished cheese depends upon the initial quality of the milk which it is made. Successful cheese factories follow a system of daily efficient grading of all milk received. This consists of:

1. Determining the odour of the milk in each can/tanker. No off flavour should be accepted.
2. Inspecting the appearance of the milk, which should be free from all extraneous matter.
3. Determining sediment, either once a week or every ten days, in each can of milk.

4. Performing MBR, Resazurin and rennet –curd test on the milk once a fortnight or so for each producer/supplier and more frequently (weekly or even daily) on milk of doubtful quality.
5. Determining the percentage of acidity.
6. Examining milk for bacteriophage, antibiotics and inhibitory substances.

After the milk has been examined for quality and accepted, it is weighed than a representative samples is taken for determination of fat and casein contents.

Flow Diagram of Manufacture

Receiving milk
⇩
Preheating (35-40°C)
⇩
Filtration/Clarification
⇩
Standardization (Casein/ fat ratio is 0.68-0.7)
⇩
Pasteurization (63°C/30 min)
⇩
Adding Starter (Ripening) 31°C (Str. Cremoris/lactis)
⇩
After 30 minutes
⇩
Adding Rennet (31°C) 1-2gm/100 lit.
⇩
Coagulation/setting (for 45 min)
⇩
Cutting (5 mm cubes)
⇩
Cooking (upto 37-39°C for 45 min.)
⇩
Drainage of whey
⇩
Cheddaring (2 hrs.)
⇩
Milling (Acidity -0.6%)
⇩
Salting (3%)
⇩
Hooping

⇩
Dressing
⇩
Pressing 75-80 psi (15 to 18 hrs) 2kg/cm2
⇩
Drying (15°C for 5 days)
⇩
Paraffining (125°C)
⇩
Vacuum packaging
⇩
Ripening (5-10°C, 80-92% RH for 3-12 months)

2. Filtration/ Clarification: The chief object of this step is to remove any visible dirt in milk so as to improve the aesthetic quality of the cheese made. The milk is usally preheated to 35-40°C for efficient filtration/clarification.

3. Standardization: In cheese making standardization referes to adjustment of the casein/fat ratio in cheese milk to 0.68-070. The objectives are.

1. To regulate the fat in the dry matter of cheese
2. To produce the maximum amount of cheese per kg of fat in cheese milk.

4. Pasteurization: The usual temperature time employed for pasteurization of cheese milk is.

1. Holder- 63°C for 30 min.
2. HTST- 71°C for 15 sec.

Limitations:

1. It destroys the typical flavour and body of cheese.
2. It entails a longer ripening period.
3. It encourages the use of low quality milk
4. It increase the overall cost of cheese making.

5. Homogenization: Cheese milk is normally not homogenized because a soft curd is formed.

6. Addition of calcium chloride: Excessive heat treatment couses the precipitation of a part of the calcium salts in milk. This results in slower renneting action and a weaker curd, which can be corrected by addition of 0.01-0.03 percent calcium chloride to milk.

7. Adding starter (ripening): Ripening or souring of milk refers to the development of acidity in milk from the time it is received in the cheese

vat until renneting. In cheese milk, ripening is done by the addition of starter. The starter is the heart of cheese. A bad starter is almost certain to give low quality cheese. A cheddar cheese starter usually contains Str. Lactis and or Str. Cremoris.

The usual time to add the starter is before all the milk has been received in the vat. The amount of starter added is to the extent of 0.5 to 1 per cent of the milk, and the temperature of addition is 30 to 31°C. Before being added to the milk, the starter should be examined for its quality; it should then be stirred until smooth and creamy in consistency; then strained and added in the required quantity and mixed thoroughly and uniformly into the milk. Ripening in milk is measured by titration, rennet test and pH meter.

8. Adding colour: Colour is added just before renneting. The colour of cheese is usually an alkaline solution of annatto. The colour is usually diluted with approximately 20 times its volume of potable water for even distribution. The usual amount of colour is 30 to 200 ml or more for 1000 kg of milk.

9. Renneting: Adding rennet to milk in cheese making is commonly known as renneting or setting.

(a) What is rennet? Rennet is the crude preparation or extract from the abomasum, rennet contains two principal enzymes viz., rennin, and pepsin. Rennin is an extremely powerful clotting enzyme, which causes rapid clotting without much proteolysis. On the other hand pepsin induces proteolysis is leading to bitterness in cheese. Rennet is available as liquid or powder or as tablet. Commercial rennet should be stored in a closed vessel, in a dark room at below 10°C.

(b) How is it prepared? Rennet is the preparation obtained commercially from the fourth of true stomach (Abomasum) of the young calf, known as the vell. Milk fade calves yield the purest rennet. The lining of the stomach is washed, dried, cut into small pieces and macerated into water containing about 4 percent boric acid at 30°C for 5 days

(c) Properties of Rennet: It is a sulphur containing protein. One part can clot about 5 million parts of milk. In cheese making one part of liquid rennet (2%) is used for about 5000 parts of milk. It is very sensitive to alkali. Heating to 70°C at pH 6.8 –7.0 will destroy it in 14 minutes.

(d) Factors Affecting Rennin Action

i. **Temperature:** Below 20°C, rennin is almost inactive; from 30 to 48°C it is about equally active, the optimum temperature being 41°C. Above 50°C the activity falls off rapidly.

ii. **Acidity:** The rate of clotting increases rapidly with small increase in acidity. Alkalis retard the clotting of by rennet.

iii. **Calcium ions:** Coagulation of milk is very sensitive to changes in concentration of calcium ions, it is common practice to add calcium chloride to which has been severely pasteurized. E.g. at 80°C for 30 seconds. This acts in three ways by lowering the pH value, increasing the calcium ion concentration and raising the colloidal calcium phosphate content.

iv. **Inhibitory substances:** Many colloidal substances interfere with rennin coagulation e.g. albumin, serum peptones, etc. Albumin and globulin retard coagulation (mastitis milk clots slowly with rennet; the alkalinity of such milk also contribute to this effect.). Boiling of milk resulting in denaturation of the protein removes the inhibitory effect. Five per cent peptone almost prevents clotting.

v. **Homogenization:** This has an accelerating effect on rennet clotting, but decreases the curd tension.

vi. **Heating the milk:** Heat not only destroys rennin but also makes clotting of the milk by the enzyme less easy. The major reason for this is the removal or precipitation of calcium ion.

(e) Rennet preparations other than calf rennet: These include goat and lamb rennets, plant enzymes such as withania coagulans, ficus,papain etc

(f) Difference in behavior between animal and vegetable rennet: Although vegetable rennet clots cow and vegetable milks, animal rennet do not clot vegetable milks even if the calcium ion concentration is raised to that of cow's milk.

(g) Bacterial rennet: The use of enzymes from microorganisms, particularly aerobic spore formers (Bacillus subtilis) and some of the gram-negative rods *(Serratia Marcescens)* is a fairly new development in cheese technology.

(h) Adding rennet: Rennet is added when it has been determined that the acid is developing at the desired rate. Thus, when making cheese from ripened milk rennet is added when the acidity has increased from initial level by 0.02 %. The ideal temperature for setting raw milk under normal conditions is 30°C and for pasteurized milk , 31°C. The amount of rennet extract used should be such as to form curd that is firmer enough to be cut in 25-30 minutes after the addition of rennet.

The amount of rennet, which should be added, depends on: i) strength of the rennet; ii) temperature of milk; iii) acidity of the milk; iv) composition of milk. Usually, liquid rennet is diluted with 20-40 times its volume of (potable) water before it is added, to ensure proper distribution for uniform coagulation.

The milk is thoroughly stirred during the addition of the rennet and also for 3 to 5 minutes afterwards. The vat is covered as soon as possible

when the stirring is over, to keep the surface warm and protect it from contaminating dust particles.

(i) Coagulation: This refers to liquid milk changing to a semi-solid junket. The first signs of coagulation are that bubbles of air stirred into the milk surface take longer to break and a spatula dipped into the milk and withdrawn shows small flakes of curd.

10. Cutting: This refers to cutting of the 'firm' coagulum into cubes of a specific size.

(a) When to cut the curd: When a glass rod inserted at a 45° angle and lifted straight up makes a clean break in the curd, it is ready for cutting. If the curd is cut too soon, there will be a lower yield of cheese, if cut too late, cutting will be difficult and moisture expulsion delayed.

(b) Curd Knives: These consist of stainless steel wires or strips, 6 or 9 mm apart, one horizontal and the vertical.

(c) Method of cutting: The curd is usually first cut with the horizontal knife lengthwise, then with the vertical knife lengthwise and widthwise.

(d) Syneresis of cheese curd: This refers to the expulsion of whey and contraction of the curd. From the cheese-making point of view, the factors controlling the loss of whey and contraction of the coagulum are: cutting, temperature, acidity, agitation, time and salt.

(e) Behavior of curd after cutting: After the curd is cut, whey begins to appear between the cubes and a film begins to form on the outer surface of each cube. This film should not harden, i.e., become firm, too rapidly. Care has also to be taken to avoid breaking this film.

(f) Acidity of the curd after cutting: Decreases by 0.05 % to 0.08 % from that of milk at renneting.

(g) Stirring curd after cutting: During the first 2-3 minutes after cutting, the curd is not stirred. Then gentle stirring starts. The speed of stirring increases with the gradual firming of curd cubes. Matting is indicative of inadequate stirring.

11. Cooking: This refers to the heating of curd cubes; it begins within 15 minutes of cutting.

(a) Regulation of heating: The heat is applied slowly to begin with. If the temperature is raised too rapidly, a condition similar to 'case-hardening' will result, and the curd cubes will be hard on the outside but soft and 'water-logged' inside. The rate of heating is such that the temperature rises to 32°C in about 15 minutes and thereafter to a maximum cooking temperature (37 to 39°C) at the rate of 1° C every 4 minutes.

(b) Amount of heat required: This depends chiefly on the type of cheese required. For cheddar, a maximum of 37 to 39°C is normal. Too high a temperature can reduce the souring rate and activity of the starter organisms.

12. Drainage of whey (dipping): This refers to the removal of whey from the curd. When the cubes have been reduced to about one-half of their size at cutting, the acidity approaches a desirable limit and the cube attains a desirable consistency (elastic feel when squeezed), stirring is stopped and the cubes 'pitched'. (Pitching refers to the curd cubes dropped to the bottom of the vat and piling them up together). The curd cubes are pushed away from the gate of the vat, a strainer is inserted in the gate, a curd-pail is hung on the curd out let, and the whey is drawn from the vat.

13. Cheddaring: This refers to the combined operations of packing, turning, piling and repiling the curd cubes.

(a) Packing: After the bulk drainage of whey, the curd cubes are kept closely together in two heaps with a channel in between. This is known as packing and takes 5 to 15 minutes after dipping. It results in the formation of two long slabs of curd. These are cut with a cheese knife into blocks or strips 15 to 20 cm wide.

(b) Turning: As soon as the blocks (strips) of curd can be handled without breaking, they are rolled bottom-side in the vat. This is called turning and is carried out every 15 minutes till the curd is ready for milling and salting.

(c) Piling and repiling: Within 30 to 45 minutes of packing, blocks of curd are turned and laid one over another in twos or threes. This is called piling. Then the position of the curd block is altered and this is known as repiling.

(d) The cheddaring operation: usually lasts for two hours or more and is very important not only for moisture control but also for improving body and texture. After cheddaring, the curd becomes drier, more mellow and silky and changes from a sorbo rubber-like material to one resembling chicken breast-meat. In the latter stages, it tends to tear apart in fibrous shreds and develops a characteristic nutty and buttery aroma.

14. Milling: This refers to the mechanical operation of cutting the blocks of cheddared curd into small pieces with the help of a cheese mill,

(a) Objectives

1. To promote the further removal of whey
2. To enable quick distribution of salt in the curd
3. To prepare curd for pressing into final form
4. And also for de-odourization, cooling of the curd etc.

Type of mill: The curd mill may be hand or power driven. Milling is done in such a way that the cut curd falls into the vat itself.

15. Salting: This refers to the addition of common salt to the curd pieces. Salt in cheese affects flavour, body and texture, and keeping quality. Cheese without salt is often soft, ripen quickly and rapidly develop unpleasant flavors.

(a) Objectives

1. For further removal of whey
2. Hardening and shrinking of curd
3. Retarding further formation of lactic acid
4. Checking the undesirable fermentation
5. Produce desirable quality characteristics

(b) Holding before salting: Salting may be delayed (by more than 15 minutes if needed) when it seems desirable to develop more acid in the cheese curd to encourage further drainage of whey, or to aerate the curd to improve its flavour.

(c) When done: Cheese curd is normally salted:

1. About 15 minutes after milling
2. When a hot iron test shows threads 1 to 2 cm long
3. When acidity is 0.4 to 0.5 % and
4. When pH is 5.4 to 5.0

(d) Amount of salt added: This depends on the amount of curd in the vat and salt desired, and generally varies from 1 to 2 % (average 1.5 %).

16. Hooping: This refers to the curd being placed in hoops or moulds in which the cheeses curd is pressed into its final shape. For hooping operation all the added salt should have completely dissolved and the temperature at hooping should be 30-32°C. Hooping and pressing at too high a temperature causes an excessive loss of fat, decreased yield, development of abnormal flavours and exaggeration of bacterial defects. On the other hand, hooping and pressing at a too a low temperature result in an open texture, imperfect rind formation and lack of whey drainage.

17. Dressing: Refers to arrangement of the cheese cloth before and after pressing.

(a) Before pressing: Large cheese hoops are lined with cloth before they are filled with cheese curd for pressing. Small hoops are filled with curd and pressed for 30-60 minutes without any cloth.

(b) After (first) pressing: To remove all wrinkles formed during pressing.

18. Pressing: This refers to the operation of forcing the particles of milled and salted curd in the hoops into the smallest possible space

to give cheese its final shapes. Cheese pressing is done with the help of presses which may be, Screw or Pneumatic or Hydraulic or spring types.

19. Drying: This is done for rind formation in cheese. It involves first taking the cheese out of the hoop and then stamping of date, batch, variety name etc., and keeping in a drying room at 12 to 16 ° C for a few days. The cheese is turned at 24-hour intervals so that both ends and sides of the cheese can dry and form the desired rind.

20. Paraffining: This refers to the operation of dipping the cheese for a few seconds in a bath of melted paraffin, whereby a thin coating of the paraffin is applied to the surface of the cheese.

(a) Objectives

1. To reduce the loss of moisture during curing
2. To prevent extensive mould growth
3. To protect it against insects

(b) Kinds of paraffin:

1. Paraffin wax, this is a product of the petroleum industry.
2. Micro-crystal wax. More water and vapour resistant.
3. Flexible waxes.

These are combination of paraffin and micro-crystal waxes.

21. Curing

(a) Definition: The curing/ripening/souring/maturing of cheese refers to the storage of cheese for at least 2 to 3 months at a given low temperature (0-16°C) during which its physical, chemical and bacteriological properties are profoundly changed, resulting in the development of a characteristic flavour, body and texture. The term 'green cheese' is usually applied to hard-pressed cheese in the early stages of ripening before the characteristic flavour, body and texture of ripened cheese developed.

(b) Ripening agents: These influence the rate, extent and nature of ripening in cheese and include micro-organisms (chiefly bacteria and moulds) and enzymes, including pepsin.

(c) Changes During Curing

A. Physico Chemical Changes

1. Flavour: From a mildly acid taste and aroma in green cheese to the development of characteristic flavour of ripened cheese which is a blend of several odours and tastes of diacetyl, traces of butyric and caproic acid, esters of alcohol, salts of propionic and acetic acids in well aged cheeses.

2. Body: The cheese becomes slightly harder, due to loss of moisture. There is a gradual change from the rubbery body in the green cheese to a mellow and waxy body in the ripened cheese.

3. Texture: Curd cheese tends to acquire a fairly close to close texture.

4. Chemical: The chief chemical changes which occur during the curing of cheddar cheese are: fermentation of lactose to lactic acid and small amounts of acetic acid, propionic acid and carbon dioxide; proteolysis; and a slight fat break down. The most obvious chemical changes are the breakdown of the proteins, the newly created solubility of about 25 % of total proteins in the cured cheddar cheese. In addition to fat breakdown, the ammonia produced by moulds and certain bacteria may have considerable effect on the bacteria and so assist in the growth of other types of bacteria.

An increase in acidity and decrease in pH takes place for the first few days. The pH is lowest in cheese on about the third or fourth day after pressing, and is normally 5.10 to 5.05. It then decreases slowly and steadily during the curing period.

B. Microbiological changes: In cheddar cheese type, which are low in moisture and close in texture sustain a steady changeover from Streptococci to Lactobacilli, some of which contribute to the flavour. Other types are of course present and the higher the proportion of miscellaneous types, the quicker is the curing and greater the possibility of off-flavours.

C. Shrinkage in Cheese: This refers to the loss in weight of cheese during curing/storage. Although a slight shrinkage is natural, excessive shrinkage should be prevented. Shrinkage is caused mainly by 'loss of moisture'.

Factors Causing Shrinkage in Cheese

1. Temperature of curing: Higher the temperature, higher the shrinkage, and vice-versa
2. Relative humidity of curing: Higher the humidity lower the shrinkage.
3. Size (and shape) of cheese: Larger the size, lower the shrinkage and vice-versa
4. Moisture content of cheese: Higher the moisture content, higher the shrinkage
5. Paraffining of cheese: Paraffined cheese undergoes less shrinkage, than non-paraffined cheese.

22. Freezing: Freezing of cheese may damage its texture. On average the freezing point of cheddar may be - 4.5°C in fresh cheese and -14.5°C in cheese more than a year old.

23. Yield: This is affected by the quality of milk, nature of manufacturing operations, the skill of the manufacturer, and curing procedures.

Cottage Cheese

This is a soft, unripened cheese usually made from skim milk. It has mildly acid flavour and consists of small particles or flakes or curd, which have a meaty consistency. Creamed cottage cheese has cream mixed to it upto 4 % level. Both varieties are salted always.

Cottage cheese can be prepared by both, direct acidification and by starter culture methods. In direct acidification method, any food grade acid, phosphoric, lactic acid etc., are used.

Method of Manufacture:

A. Flow Diagram of Manufacture

Receiving (pasteurized) skim milk
↓
Adding calcium chloride
↓
Adding starter culture (lactic cultures)
↓
Adding rennet (0.01 %)
↓
Setting (30°C)
↓
Cutting
↓
Cooking (upto 46°C, over 2 hrs)
↓
Drainage of whey
↓
Washing and draining the curd
↓
Salting
↓
Creaming
↓
Packaging and storage

B. Details of Manufacture

1. Receiving skim milk: The skim milk should be fresh, sweet, low in fat and bacterial count and clean in flavour. It should be pasteurized

immediately after separation preferably by LTLT method as it produces softer curd.

2. Adding calcium chloride: Calcium chloride is generally added to the skim milk at the rate of 1 ml saturated solution per 100 liters of milk, with the objective of increase the concentration of calcium for better setting

3. Adding starter culture: A high quality starter consisting of either *S.lactis, S.cremoris, S.diacetylactis, L.dextranicum,* etc., singly or in combination is then added at 1 to 2.5 % levels and thoroughly mixed into it.

4. Adding rennet: Rennet is added at 2 - 2.5 ml per 1000 lit. of milk. It is diluted with water upto 40 times with water before adding.

5. Setting: The temperature is set around 29 - 32°C for 4 - 6 hrs.

6. Cutting: The most desirable acidity of whey at cutting is approximately 0.5 % (pH 4.6 - 4.7).The whey should be clear and free from curd particles.

7. Cooking: This begins soon after cutting and continues for an hour or two until the temperature reaches 45°C or until the curd becomes hard enough for removal of whey. Stirring during cooking is done very gently and at a minimum rate..

8. Drainage of whey: Whey is removed when the curd cubes no longer have a 'soft center' and when a handful of them squeezed gently show slight elasticity. The whey is removed from the curd approximately two hours after cutting.

9. Washing and draining: The curd is washed after removing all the whey to produce desired firmness and mildness in flavour. The wash water is at the temperature of 21°C and after soaking for 15 minutes the water is drained. Second washing is done with water at 16°C. Draining should be thorough; the cubes are placed in draining rack with perforations at the bottom, which can be wheeled into cold store rooms.

10. Salting: This is done when the free moisture has been drained from the curd. Salt can be added to the curd in the vat, or it can be dissolved in the cream for creamed cottage cheese. Salt is added @ 1 % of curd weight.

11. Creaming: This is done immediately after draining, if the product is to packed at once. Holding the curd overnight in a cold room before it is creamed makes it more firm when creaming. The amount of 20 % cream added is @ 4 % level of the curd weight.

12. Packaging and storage: Cottage cheese, creamed or uncreamed, may be packed in waxed / polythene-coated paper cups or in polythene bags. Storage is at 5°C or less.

13. Yield: The yield of cottage cheese before creaming depends essentially on:

1. The composition of the milk
2. Manufacturing losses
3. Moisture content of the cheese
4. Approximately yield of cottage cheese is 15 %.
5. Keeping quality of cottage cheese
6. The keeping quality is short even under refrigerated conditions (5-10°C).

14. Keeping quality: The keeping quality of cottage cheese, weather uncreamed or creamed is short even under refrigeration storage (5-10°C). It is 90 days or longer by freezing or by brine storage.

12 Preparation of Indigenous Milk Products

Dahi

Curd (Dahi): In India curd is also called Dahi. As per PFA rules, dahi is a product obtained from pasteurized or boiled milk by souring using previously cultured milk or otherwise by using selective lactic cultures.

(a) Starter Required

1. Sweet Dahi: *Streptococcus diacetilactis, Streptococcus cremoris, Streptococcus Lactis and Leuconostoc citrovorum or Leuconostoc dextranicum.*
2. Sour Dahi: *Lactobacillus Bulgaricus or Streptococcus thermophillus.*

(b) Food and Nutritive Value

1. Dahi is more palatable and those who usually do not like drinking milk would consume it readily.
2. Dahi is more easily digested and assimilated than milk.
3. Dahi seems to exert a possible therapeutic value in the stomach and during intestinal disorder.

(c) Composition: An average composition of whole milk dahi is given.

Composition of Whole Milk Dahi (%)

Water	Fat	Protein	Lactose	Ash	Lactic acid
85-88	5-8	3.2-3.4	4.6-5.2	0.70-0.72	0.5-0.11

(d) Requirements for Dahi

Characteristics	Requirement for sweet dahi	Requirement for sour dahi
Acidity, lactic (% weight)	0.70	1.0
Yeast and mould count/gm.	100	100
Coliform count/gm.	10	10
Phosphatase test	Negative	Negative

(e) Method of Production

i. **Traditional method:** In the house hold the milk is boiled, cooled to body temperature, inculcated with 0.5-1 percent starter (Previous days dahi or butter milk) and then allowed to set undisturbed overnight. In cooler weather, the dahi setting vessel is usually wrapped up with Woollen cloth to maintain warmth.

ii. **Standardized Method**

- Pasteurize milk at 85°C for 10 min and cool the milk immediately (20-25°C) by keeping the container in running water, by simultaneous agitation. Add 0.5-1.0% of starter culture at this temperature and mix thoroughly.
- Distribute this cultured milk into a small sterile steel or glass container taking aseptic precautions, and fix a sterile aluminium cap.
- Permit the cultured milk to remain undisturbed for 16-18 hours at 22-25°C, or to reach the acidity of 0.8-0.98.

(f) Packaging and storage: The traditional container for dahi is an earthenware cup. However, modern packaging includes glass bottle and plastic/plastic coated cup. The recommended storage temperature is around 5-10 °C

(g) Keeping quality: Under refrigerated storage (5-10°C), it usually keeps well for a maximum period of one week.

(h) Uses of Dahi:

1. For direct consumption.
2. For preparation of chakka and shrikhand.
3. For preparation of makkhan.

Lassi

(a) Definition: Lassi is also called as chhas or matha, refers to desi butter milk, which is the by-product of butter (Makkhan).

(b) Composition: The composition of lassi varies considerably, depending upon the composition of dahi, the extent to which the curd is diluted when churning and the efficiently of fat removal.

Composition of Lassi (%)

Water	Total solids	Solid not fat	Fat	Protein	Lactose	Ash	Lactic acid
96.2	3.8	3.0	0.8	1.3	1.2	0.4	0.44

(c) Method of Production

1. The method involves standardization, heating & cooling of milk to inoculation temperature, addition of starter culture and incubation at a suitable temperature. To the set curd desired level of sugar is added in the form of syrup which has already been pasteurized and cooled separately. Smooth consistency of lassi is obtained which is then passed through a homogenizer. The product is filled in pouches and stored at refrigerated temperature.
2. To the butter milk desired level of sugar is added in the form of syrup. Smooth consistency of lassi is obtained which is then passed through a homogenizer. The product is filled in pouches and stored at refrigerated temperature.
3. Same technology can be used to prepare spiced lassi. In this case sugar is replaced with the spices. The shelf life of lassi is about one week at refrigeration.

Advantages of the Technology

- Provides an opening for the disposal of good quality returned surplus milk.
- Increase the profitability of dairy plants.
- Requires minimum additional facilities.
- Amenable to large-scale production.
- Simple technology and does not require sophisticated equipments or machinery.
- Excellent quality milk based beverage
- Has high nutritive and therapeutic value

(d) Food and nutritive value: Lassi contains appreciable amount of milk protein and phospholipids, and is an excellent beverage for quenching thirsts.

(e) Uses

1. As a beverage, usually after the addition of ice, sugar or salt, with or without additional flavours.
2. As a starter culture.
3. As a coagulants.

Khoa

(a) Definition: Khoa/ Mawa / Khava refers to the partially dehydrated (heat coagulated) whole milk product prepared by continuous heating of milk in a karahi over a direct fire, while also constantly stirring-cum-scraping by using a khunti till it reaches a semi solid consistency.

Thereafter the pan contents are removed from the fire and worked up in to a solid mass known as khoa pat.

According to PFA Rules (1976), khoa is the product obtained from cow or buffalo (or goat or sheep) milk, or a combination thereof, by rapid drying. The milk fat content should not be less than 20 percent of the finished product.

(b) Classification: Three main types of khoa, viz., Pindi, Dhap and Danedar are prepared in the country. The three main varieties are "pindi" for burfi, "dhap" for gulabjamun, pantooa etc., and "danedar" used for kalakand. Khoa making has been the easiest way of preserving rurally produced milk in the flush season.

(c) Composition: Chemical composition of Khoa

Composition of Khoa %

Type of milk	Moisture	Fat	Protein	Lactose	Ash	Iron (ppm)
Cow	25.6	25.7	19.2	25.5	3.8	103
Buffalo	19.2	37.1	17.8	22.1	3.6	101

(d) Uses: Khoa forms the base material for a variety of Indian sweets like gulabjamun and dishes.

(e) Method of Preparation of Khoa: Three methods have been known in the production of khoa. They are

1. Existing practice
2. Improved method
3. Continuous method

1. Existing practice/trade practice: Required quantity of milk is taken as per batch and boiled in a karahi (vessel) of different shape and size over a brisk non-smoky fire. The milk is stirred vigorously and constantly with a circular motion by a ladle or khunti. During this operation, all parts of the pan with which the milk comes in contact are lightly scraped to prevent the milk from scorching.

Constant evaporation of moisture takes place and the milk thickens slowly. However, no sugar is added and milk-dehydration continues until heat-coagulation of milk proteins begins and the concentrate becomes insoluble in water. There is change of colour at this stage, and heating is continued with greater control with increased stirring-cum scraping speed. Soon the viscous mass reaches a semi-solid/pasty consistency and begins to dry up. The final product is ready when it shows signs of leaving the bottom and sides of the karahi and sticking together, which is known as the khoa-pat. This is invariably made after removing the pan from the fire and working the contents up and down into a single compact mass.

2. Improved method: This is followed in organized firms, where, the equipment, conditions of dehydration, and the quality of the milk used are given importance. The karahi and open fire substituted with stainless steel jacketed-pan or kettle, which is heated by water or steam. Milk is boiled till it assumes pasty consistency and then held at 85°C and stirring at 100 rpm.

Regarding the quality of the milk used, buffalo milk is preferred over cow milk as the latter produces soft, loose body and gives smooth a granular texture which is not relished. The milk should contain 4% and 5% fat respectively for cow and buffalo milk. Neutralization of acid milk improves the texture but does not improve the flavour of khoa. Starch adulterated milk gives hard khoa. Homogenization of milk produces softer body and fat leakage.

3. Continuous method: Milk is continuously heated in the steam jacketed drum heater, where it is partially concentrated. This is followed by another heating and concentration of the milk in open pans till a viscous semi-solid product is obtained and is removed mechanically. The equipment basically consists of

- A steam-jacket drum heater with a rotary scrapper and milk outlet
- Two open steam-jacketed pans with outlet valves
- Two sets of scrapers for the pans and
- A power drive for the scrapers.
- A cover is provided in the pan to prevent any dust or dirt falling into it.

(f) Physicochemical changes in milk on conversion into khoa: The following are the changes that are encountered while milk is being converted into khoa.

- **Change of state:** From liquid milk to solid khoa (due to considerable dehydration)
- **Change in intensity of colour:** From 'light' to a more intense shade of colour with a tinge of brown.
- **Homogenization of milk fat:** The fat globules are appreciably subdivided due to vigorous agitation of the milk at a high temperature.
- **Free-fat formation:** Considerable free fat is produced due to rupturing of the fat globule membrane by the scraping action of the stirrer.
- **Heat coagulation of milk proteins:** The serum proteins are coagulated by the action of heat and concentration.

- **Super-saturated solution of lactose:** From a dilute solution in milk, lactose is present in khoa as a super-saturated solution.
- **Partial precipitation of milk salts:** A portion of the milk salts are precipitated by the action of heat.
- **Increase in Iron content:** From 2 to 4 ppm in milk, the iron content in khoa exceeds 100 ppm due to scrapping of the pan surfaces during the manufacture.

(g) Yield: The type of milk, cow or Buffalo, influences the yield by virtue of its total solids content. Buffalo milk with higher total solids give higher yield than cow milk. Normally the yield of khoa ranges from 17 to 19 % from cow milk and 21 to 23 % for buffalo milk.

(h) Over-run: The over-run in khoa refers to the 'the excess weight of khoa obtained over the amount of total (milk) solids used. It is influenced by the moisture chiefly. The formula for calculating overrun (OR) in khoa is:

$$\%\,OR = \frac{K - TS}{TS} \times 100$$

Where, K = Weight of khoa (in kg)

TS = Weight of total solids in milk (in kg).

Chhana

(a) Definition: Chhana is a product obtained by acid coagulation of milk near its boiling point followed by removal of whey. The acids commonly used are lactic or citric acid in both natural and chemical form.

As per PFA rules, chhana may be defined as a product obtained from cow or buffalo milk or a combination thereof by precipitating with sour milk, lactic acid or citric acid. It should not contain more than 70 per cent moisture and milk fat should not be less than 50 per cent of the dry matter.

Chhana differs from paneer in the method of preparation that no pressure is applied to remove the whey. The coagulum is collected in a cloth and hung on a peg without applying pressure to drain off the whey.

(b) Composition: The chemical composition of chhana depends mainly on the initial composition of milk.

Chemical Composition of Chhana (%)

Type of milk	Moisture	Fat	Protein	Lactose	Ash
Cow	53.4	24.8	17.4	2.1	2.1
Buffalo	51.6	29.6	14.4	2.3	2.0

(c) Food and nutritive value: As chhana has a fairly fat and protein content and also contains some minerals, especially calcium and phosphorus. It is also good source of fat soluble vitamins A and D. Chhana is highly recommended for diabetic patents.

(d) Method of Preparation

1. Flow Diagram of Chhana:

Milk
↓
Standardization (4.0%fat)
↓
Heating (to boil)
↓
Cooling (80°C)
↓
Addition of coagulant (1-2% citric acid solution)
↓
Coagulation (80°C)
↓
Straining
↓
Draining
↓
Cooling
↓
Packaging
↓
Storage (4-6°C)

2. Traditional Method: Milk for chhana making is brought to boil by heating it directly in a large iron karahi over an open fire with continuous stirring to prevent burning. This hot milk is ladled out in batches of about 1-1.5 litre in a separate coagulation vessel and adequate amount of coagulant solution normally cleansed sour chhana-whey (1-2 % citric acid) is added to it while gentle stirring the contents.

After complete coagulation, the content is poured over a piece of clean muslin cloth stretched over another vessel. The process is repeated till all the milk is used up. The cloth containing the coagulated mass is then removed, tied into a bundle and hung up to drain out the whey completely without applying any external pressure.

3. Bulk Method: In this method, the same steps are followed as mentioned above except instead of small lots of 1-1.5 litres, all the milk (normally 5-20 litr.) is coagulated in one bulk. Coagulant solution is

added slowly with uniform constant stirring till the clear whey separate out. The coagulated solid mass is collected by straining it through a muslin cloth.

4. Improved Method: The method of production of chhana remains the same as mentioned above with some improvements. These involve consideration of equipment, quality of milk, conditions of coagulation and method of straining etc. In this process boiling, cooling, coagulation of milk are accomplished in the same steam jacketed stainless steel kettle by employing steam or tap water as and when required, coagulation of milk is carried out at 80°C by adding 1-2 per cent citric acid solution (pH of coagulation 5.4).

The coagulant solution is added within 30-60 seconds and while addition of coagulant a slow stirring is preferred so as to avoid foam formation and also breakage of curd into fines. After attaining complete coagulation of milk the coagulated mass is strained as mentioned above.

5. Continuous Method: In order to overcome problems of small scale, attempts have been made to mechanize chhana production. Aprototype machine with a capacity of 40 kg chhana per hour has been developed at NDRI, Karnal. The major components of the equipment are a balance tank, injection chamber, tubular heat exchanger, cooling chamber, mechanized strainer, whey tank etc. In this mechanized process, standardized cow milk is pumped from a balance tank at the rate of 250-litres/ hour to an injection chamber where culinary live steam is directly injected into the milk.

Steam gets completely condensed in milk and raises its temperature to 90-95°C.There after, milk is brought in contact with coagulant solution, the quantity of which is regulated manually in proportion to the rate of flow of milk. The mixture of milk and coagulant is circulated through a holding coil to facilitate complete coagulation of milk. The coagulated product, along with the whey is then pumped to a double-jacketed cooling tank, where it is cooled down to room temperature. Finally, the coagulated mass is directed to a mechanical strainer, a double jacketed inclined sieve, where it is drained thoroughly.

Chhana with 55-65 per cent moisture is discharged through the outlet and collected in the basket. Drained whey is transferred to a separate tank for subsequent use. Finally the chhana is packaged in suitable packaging material and stored in cold room.

(e) Physical Appearance

1. Cow milk channa is light yellow in colour has a moist surface, soft body and smooth texture.
2. Buffalo milk is whitish in colour.
3. Both buffalo and cow milk have a pleasant, sweetish mildly acid flavour.

(f) Uses

1. As a base and filler for the preparation of a large number of indigenous milk sweets such as rassogolla, sandesh, chhana-kheer and pantooa
2. For direct consumption with addition of sugar or salt.
3. For preparation of cooked vegetables

Paneer

(a) Definition: Paneer is also acid coagulated product of hot milk. It resembles chhana is its manufacturing procedures, but differs in that it is lightly pressed in blocks. The pure foods act of India requires paneer to have not more than 70% moisture and not less than 50% fat on a dry matter basis.

(b) Composition of paneer: The typical composition is given below

Chemical Composition of Paneer (%)

Moisture	Fat	Protein	Lactose	Minerals
53-55	23-26	17-18	2.0-2.5	1.5-2.0

(c) Method of Preparation

1. Flow Diagram of Paneer

Buffalo milk
↓
Standardization (6% Fat)
↓
Heating (90°C)
↓
Cooling (70°C)
↓
Addition of coagulant (1 to 2% eitric acid)
↓
Coagulation (70°C)
↓
Draining
↓
Hooping
↓
Pressing
↓
Packaging
↓
Storage

2. Process of Paneer: Buffalo milk is an ideal raw material for manufacture of good quality paneer. Buffalo milk is standardized to a fat level of 5.8 to 6.0 per cent using fresh buffalo skim milk. The standardized milk is heated to 90°C without holding. Thereafter, the temperature of milk is brought down to 70°C and coagulated at this temperature using 1 per cent citric acid solution. The temperature of citric acid solution is also maintained at 70°C. Citric acid solution is added with continuous stirring till clear whey separate out. After complete coagulation, the stirring is stopped and the coagulated mass (curd) is allowed to settle down for about 5 minutes. The whey is then drained through stainless steel strainer. The temperature of the content is not allowed to drop below 63°C until this stage. The curd is collected and filled in hoops (with holes on all its sides to facilitates the expulsion of whey) lined with clean fine cloth (Paneer press). The hoops containing curd is pressed for about 10-20 minutes. Thereafter, the pressed block of curd is removed, cut into pieces and immersed in chilled water of 5-6°C for about 2 hours. Dipping of paneer pieces facilitates cooling of product and also it absorbs moisture and improves the body and texture of paneer. The chilled pieces of paneer are then removed from water and placed on wooden planks for 10-15 minutes to allow loose water to drain. The paneer is cut into desirable size and packaged in suitable packaging material. Finally it is stored under refrigeration till marketing and consumption.

(d) Paneer Dishes

1. Mattar paneer (paneer with peas).
2. Paneer majestic (paneer fried in a spicy batter).
3. Saag paneer or Palak paneer (paneer along with spinach).
4. Shahi paneer (paneer cooked in a rice, Mughlai curry).
5. Shahi tukda (a dessert from frying paneer).
6. Paneer tikka (a vegetarian version of chicken tikka, paneer placed on skewers and roasted)

(e) Keeping quality

Keeping quality of paneer under refrigerated storage (5-10°C) is 6 days.

13 Preparation of Milk By-Products

1. Introduction: A dairy by-product may be defined as a product of commercial value produced during the manufacture of a main product. Because of their great and unique nutritional value, the most rational method of utilization from the stand point of general welfare is in the form of food products; usually this is the most profitable method as well.

It has always been realized that economic disposal of by products is an essential prerequisite to profitable dairying. The important dairy by-products obtained from Indian dairy industry are skim milk, butter milk, lassi, ghee residue, whey, etc.

2. Classification: The by-products of dairy industry are given below

Main Products and By-Products

Sr. No.	Main products	By-products
1	Cream	Skim Milk
2	Butter	Butter milk
3	Ghee	Ghee Residue
4	Chhana/paneer/Cheese/Casein	Whey

3. Composition: The average composition of these by-products is given

Composition of Some Important Dairy By-Products

Sr. No.	Name of by-product (from cow milk)	Composition %				
		Water	Fat	Pro-tein	Lac-tose	Ash
1	Skim milk	90.6	0.1	3.6	5.0	0.7
2	Butter milk	91.0	0.4	3.4	4.5	0.7
3	Lassi	96.2	0.8	1.4	1.2	0.4
4	Ghee residue	9.7	61.4	24.8	-	4.1
5	Chhana whey	93.6	0.5	0.4	5.1	0.4
6	Cheese whey	93.1	0.3	0.9	4.9	0.6
7	Acid casein whey	93.1	0.1	1.0	5.1	0.7

4. Principles of Utilization

The broad principles for the utilization of the various dairy by-products, both foreign as well as indigenous together with the names of commonly made food products are given below.

By-Products, their Principles of Utilization and Names of Food Products Made

Sr. No.	By-products	Principle of utilization	Food products made
1	Skim milk	Pasteurization	Flavoured milks
		Sterilization	Sterilized flavoured milk
		Fermentation	Cultured butter milk. Acidophilus milk. Bulgarian butter milk
		Fermentation & concentration	Concentrated sour skim milk
		Concentration	Plain condensed skim milk. Low lactose condensed skim milk, Sweetened condensed skim milk
		Drying	Dried skim milk
		Coagulation	Cottage cheese, Bakers cheese, Quarg, Gammelost Sapsago, casein (edible)
2	Butter milk	Fermentation & concentration	Condensed butter milk
		Drying	Dried butter milk
		Coagulation	Soft cheese
3	Whey	Fermentation	Whey beverage, Yeast whey
		Concentration	Plain condensed whey Sweetened condensed whey, Whey protein concentrate, Whey paste, Lactose.
		Drying	Dried whey

		Coagulation	Ricotta cheese, Mysost, Gjetost, Primost Cheese.
4	Lassi	Icing and seasoning	Beverage
5	Ghee - residue	Processing	Toffee or paste or indigenous sweetmeat

5. Methods of Utilization

A. Skim Milk

i. **Flavoured milk:** Flavoured milk are milks to which some flavour have been added. Different flavours used for flavoured milk are chocolate, fruit etc. and sugar is added 5 to 7.5% with stabilizers 0.2%.

ii. **Sterilized flavoured milk:** Sterilized flavour milk may be defined as(homogenized) milk which has been heated to a temperature of 100°C or above for such length of time that it remains fit for human consumption for at least 7 days at room temperature.

iii. **Cultured butter milk:** This is obtained by inoculation and incubation of pasteurized skim milk with lactic starter *(Streptococcus diacetilactis, Streptococcus cremoris, Streptococcus Lactis and Leuconostoc citrovorum or Leuconostoc dextranicum).*

iv. **Acidophilus milk:** This type of fermented milk is produced by development in milk of a culture of *lactobacillus acidophilus.*

v. **Bulgarian milk:** *Lactobacillus bulgaricus* starter strain is used for Bulgarian milk.

vi. **Skim Milk Powder:** According to the PFA rules (1976), skim milk powder is the products obtained from (the skim milk of) cow or buffalo milk or a combination thereof by the removal of water.

B. Butter Milk

(a) Condensed butter milk: Semi-solid butter milk, this is usually sweet cream butter milk which has been ripened and condensed in such the same way as concentrated sour skim milk. Alternatively, sour-cream butter milk may be used without ripening, but the acidity of the buttermilk prior to its being condensed should be greater than 1.6 per cent. This will ensure in the finished product a 5-6% acidity, which is necessary for long keeping quality.

Either sour whey or an organic acid may be used to increase the acidity of butter milk. While the older packages were air-tight wooden barrels, a modem package, might consist of an acid-resistant tight plastic film with a suitable fibre board container. Condensed buttermilk is also used for feeding pigs and poultry.

C. Whey

The whey obtained as a by-product of cheese industry has long been utilized in the production of fermented beverages, both alcoholic and non-alcoholic (acidic). The production technique of one such beverage of the acidic type is described below.

i. **Whevit:** Utilization of whey in the production of a soft drink called whevit may be outlined as follows. Fresh whey obtained from chhana (paneer) making it passed through a cream separator and the separated whey then steamed for 30 minutes in a double jacketed vat. It is cooled to room temperature and kept overnight to allow the precipitated proteins to settle down, and then filtered through a muslin cloth.

Now 50 percent sugar syrup @ 22 -23 percent of whey, 10 percent citric acid @ 2 -2.1 percent of whey and sugar mixture, and selected colour and flavour in requisite amounts are added. The above mixture is inoculated with a culture of Saccharomyces cerevisae @1 % of yeast whey broth.Whey broth is prepared by adding 1% sugar in filtered whey and sterilizing it at 15 lb. pressure for 15 minutes, the inoculum is prepared by inoculating it with the yeast culture and incubating it for 48 hours at 25°C. After mixing the inoculum in the whey, it is incubated at 22°C for 14 -16 hours. The bottles with their contents are then crown-corked to retain the carbon dioxide and pasteurized at 75°C for 30 minutes in hot water, cooled and stored at a low temperature (5 -10°C).

ii. **Carbonated whevit:** Carbonated whevit may be prepared, by using the same method as described for the whevit. The process consists of receiving the whey, separating and steaming it, cooling and precipitating proteins, filtering, adding sugar syrup, citric acid, colour and flavour. The mixture is then filled at 5 -10°C into the carbonation -cum -bottling plant for carbonation and crown -corking. The bottles and their contents are subjected to pasteurization at 75°C for 10 minutes in hot water, then slowly cooled and stored at a low temperature (5 –10 °C) till marketed. Whevit bottles should be gently inverted 2 to 3 times, if required, to make the product homogeneous.

iii. **Yeast-whey:** This is a newly developed product. A world-wide protein deficiency makes the production of yeast from whey particularly attractive, as a much needed food supplement rich in proteins and vitamins can be made from a substrate which is often used in efficiently or wasted.

iv. **Plain condensed whey:** This is made by pasteurizing the separated whey and draining the hot liquid into the vacuum pan. The whey may be condensed to any degree up to 70% total solids.

Condensed whey is highly supersaturated with lactose, which may crystallize in the pan itself and prevent the viscous product from flavouring. If crystallization in the pan can be avoided, a smooth bodied paste will flow from the pan as a clear syrup. This must then be rapidly cooled to about 32°C (90°F), seeded with lactose and run into containers. It may be used in human and animal food products, it is also valuable in confectionery, baked foods and cheese foods and make excellent chicken feed.

v. **Sweetened condensed whey:** Separated, pasteurized, sweet, cheese whey with a quantity of sugar equal to the weight of the solids in the whey, is drawn into a vacuum pan and condensed to at least 75% total solids. The concentrate is cooled to 35°C (95°F), seeded and stirred slowly for 1 to 3 hours to crystallize the lactose. It may be packed in barrels or cans. It has a salty taste and a characteristic whey flavour. It has good whipping properties, the whipped, sweetened and condensed whey may be used in fruit whips, certain candies and frozen dessert preparations.

vi. **Whey Paste:** This is manufactured in Russia by pre-concentrating a mixture of whey and skim milk in a vacuum evaporator, and adding sufficient sugar syrup, butter and cream so as to obtain 15 percent sugar, 15% fat and 65% total solids in the finished product. After slowly cooling it to 48°C, vanilla is added and the product cooled rapidly to 18 –20°C before packaging.

vii. **Whey protein concentrate:** These include soluble and coagulated whey proteins. More recently, using the modern tools of elctrodialysis, gel filtration, and reverse osmosis, commercial process for concentrating undenatured whey proteins have been developed.

viii. **Lactose:** The lactose is separated from the whey (rennet or acid) by the simple process of concentration and crystallization. The common grade of lactose are crude, edible and USP.

D. Lassi

Also known as desi (indigenous) butter milk, this is largely consumed by the producers house hold after seasoning it with salt or sugar.

E. Ghee –residue

This is a nutritious food containing fat, denatured protein, burnt lactose and minerals. Although usually consumed with sugar in the producers household it is generally wasted in the large ghee manufacturing blending concerns. Indian research so far has shown that it can be utilized in the preparation of candy and chocolate, and can be used as a paste, as sandwich spread and with Indian foods such as idli-dosa and samosa.

i. **Candy:** Fifty percent sugar syrup is first prepared. The cooked residue is thoroughly mixed into the syrup in a karahi by means of a Khunti. The mixture is heated on a low fire until a good deal of water has evaporated. To this sticky mass, powdered dry coconut may be added and the candy evenly spread on a plate (5–10°C) for an hour or so to allow it to cool. It is then cut into small cubes and wrapped in parchment paper for storage at room temperature till marketed.

(Note: For every 1 kg of cooled residue, the sugar required is 500 to 625 gm and dry coconut powder 125 to 250 gm).

ii. **Chocolate:** The cooked residue is mixed with 50% sugar syrup as above. To the sticky mass, cocoa powder and skim milk powder are added and stirred vigorously till a pat is formed. The finished producer is spread on a plate and allowed to cool (5 –10 °C) overnight. It is then cut into slabs or cubes and wrapped in parchment paper for storage at room temperature before being marketed as above. (Note: For every 1 kg of cooked residue, the sugar required is 500 to 625 gm, cocoa powder 60 to 90 gm, and skim milk powder 250 g).

14 Cleaning and Sanitation of Milk Plant

All dairy equipment should be properly cleaned and sanitize because milk provide excellent medium for the growth of microorganisms at the same decrease the keeping quality of milk. Detergent and sanitizers are used for cleaning and sanitization. They should be select properly. Cleaning & sanitization are complementary process either of them will not achieve the desired result.

(a) Definitions

Cleaning: Cleaning or washing of dairy equipment is removal of soil from the surface of equipment soil consist milk or milk product residue or foreign matter.

Sanitization: Sanitization is defined as destruction of all pathogenic and almost all non pathogenic microorganisms from surface of equipment.

Detergent: Detergent is the substances capable of assisting cleaning. It is also known as cleaning/washing compound.

Sanitizer: Sanitizers are substances capable of destroying all pathogenic & almost all non-pathogenic microorganisms.

Antiseptic: An agent used against sepsis or putrefaction in connection with human beings or animals.

Disinfectant: An agent that is applied to inanimate objects; it does not necessarily kill all organisms.

Germicide: An agent that destroys microorganisms.

Bactericide: An agent that causes the death of a specific group of microorganisms.

Bacteriostat: An agent that prevents the growth of a specific group of microorganisms but does not necessarily kill them.

Sterilization: The process of destroying all microorganisms.

(b) Detergent

(i) Characters of Detergent

Following are the character of good detergent.

1. Wetting & penetrating power. Due to this property the surface tension of substance reduces and it is become moisten.

2. It should have emulsifying power i.e. disintegrating the fat to small globule.
3. It should have saponification power in which fatty acid react with alkali form soap reduce surface tension.
4. It should have deflocculating (disintegrating of protein) power disintegration protein which can be later on easily dissolved.
5. It should dissolve very easy.
6. It should have ability to make water soft.
7. It should be germicidal in action which related to destruction of microorganisms.
8. It should be non-corrosive.
9. It should be non toxic, safe for human being, it should be economical.
10. It should be stored for long period.
11. It should be mild on hands.
12. It should be sequestering (separate) chelating (help to separate).

(ii) Classification of detergent: Detergents are classified into four groups.

1. Alkali Detergent

- NaOH: It is use for hand cleaning, generally for stainless steel.
- Na_2CO_3: It is generally use for aluminum cans & tined copper.
- Sesquicarbonate: (NaOH + Na_2CO_3) it is use generally for aluminum, tined surface. It is considered as mild alkali.
- Sodium meta silicate: It is generally use as emulsifier & deflocculating power.

2. Acid Detergent

There are two types:

- Strong acid: In the strong acid nitric acid is generally use it should not be more than 1% conc.
- Mild acid: In the mild acid phosphoric, tartaric, citric, gluconic acid use generally. It is use up to 7%.

3. Polyphosphates: These are use with alkali or acid detergent. It is generally use when hard water is use for cleaning e.g. Tertameta phosphate, hexameta phosphate tripolyphosphate

4. Surface active & wetting agent: These detergents are having property of wetting and surface active substances. These substances are sulphonated alcohol with end of molecules is hydrophobic and

hydrophilic. Normally they contain carbon chain length of 8-20 carbon. E.g. teepol, Idet-10, common soaps lisopol, picino-N

(c) Sanitizer

(i) Properties of sanitizer: Sanitizer should have following desirable properties;

1. It should be non toxic.
2. Relatively non-corrosive to hand and equipment.
3. Quick acting, easily applied.
4. Economical, relatively inexpensive.

The common dairy sanitizers are steam, hot water and chemicals (chlorine compounds, ionospheres & quaternary ammonium compounds.)

(ii) Common Sanitizers

1. Chlorine compounds like chlorine gas, chloramines-T, hypochlorite's, hexachlorophenol, dichloroisocyanic acid and trichloroisocyanic acid.
2. Iodophors like iodine.
3. Quaternary ammonium compounds like acetyl trimethyl ammonium bromide

(d) Principles of Cleaning and Sanitization

1. In the selection of any particular detergent consideration should be given to type of soil, quality of water supply, material of surface and the equipment to be cleaned and method of cleaning viz., soaking, brushing, spraying and or re circulation
2. Heat is most reliable sanitizer especially when both temperature and time are controlled. Thus effective sanitization can be done by steam (15psi / 5 minutes or 0 psi / 15 minutes) or scalding water (90 – 95°C/10 minutes)
3. Drainage: To remove any residual loose milk and any other matter
4. Pre rinsing: With cold or hot tap water to remove as much milk residue and other matters as possible
5. Warm to hot detergent washing: With detergent solution of 0.15 to 0.60% alkalinity to remove the remaining milk solids.
6. Hot water rinsing: To remove traces of detergents
7. Sanitization: To destroy all the pathogenic and almost all the non pathogenic micro organisume.
8. Drainage and drying: To prevent bacterial growth and corrosion. Drying readily accomplished by heat and ventilation.

(e) Method of cleaning & sanitization: Following are the method of cleaning:

1. Hand cleaning.
2. Mechanical cleaning.
3. CIP (Clean In Place).

1. Hand cleaning: Can and bottles are cleaned manually. In this method prepare the detergent solution. Rinse the equipment with cold water applies the detergent solution wash the equipment with cold water allows it for draining.

2. Mechanical: In this method machines are use for cleaning and sanitization. Generally can & bottle washing & sanitization is done by this method.

(a) Can washing: Following procedure should be follow for cleaning & sanitization of can.

- Draining of milk residuals.
- Prerinsing with cold water
- Drainage
- Application of detergent solution at not less than 70°C.
- Application of steam & water not less than 88°C.
- Fresh water rinsing with steam at the temperature of 88°C to 93°C.
- Application of steam.
- Hot air drying 95°C to 115°C.

Type of can washer: There are two types of can washer rotary can washer and straight through can washer. In the both types the inverted can are moved by mechanical action over washing jets for the cleaning of the interior and pass under washing jets for the cleaning of exterior. The can is then sterilized with jets of steam or boiling water and finally dried with a blast of hot air.

Rotary can washer	Straight through/tunnel can washer
It is used in small scale plant	It is used on large scale/big plant
Loading and unloading at the same space	Loading at one end and unloading at another end
Single person can operate	At least two persons are required
2-6 can per minute wash and sanitize	4-12 can wash and sanitize
Can moves in circle	Can moves in straingth.

(b) Bottle washing

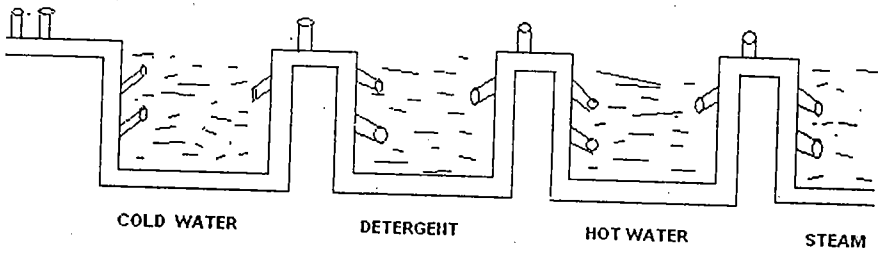
Following stage are adopted for bottle washing

- Prerinse with water at 32°C to 38°C
- Detergent wash at 62°C to 75°C
- Sanitize the bottle.
- Cold water rinsing at 25°C to 30°C
- Cold water rinsing with chlorine solution.
- Draining of bottle.

Types of bottle washing: Bottle washers are either the jet type or soaker type. Some of these washers discharge the cleaned bottle at end at which the dirty bottles enter, other at the opposite end. Following is method of bottle washer.

(i) Jet type bottle washer: In this type a series of jets spray of water and washing solution both inside and outside the surface of bottle.

(ii) Soaker type bottle washers: The washers are provided with pockets to hold the bottle that moves in the machine for stage of treatment. Clean bottle discharge at a level above that at which the dirty bottles are fed to washer.



Soaker Type Bottle Washers

3. CIP/ICP [Clean In Place/In Place Cleaning]: This refers to that system of cleaning and sanitization of equipment does not require the daily dismantling of dairy equipment.

(a) Process

1. Pre rinse with cold water till discharge water runs clear.
2. Acid rinse with phosphoric acid solution of 0.15 to 0.60% acidity, re-circulated at 65 to 71°C for 20 to 30 minutes (wetting agent may be added to increase cleaning ability).
3. Drain out acid solution.
4. Hot water rinse with water at 65 to 71°C for 5 to 7 minutes. Rinse water should drain out.

5. Alkali rinse with alkali detergent solution of 0.15 to 0.60% alkalinity, re-circulated at 65 to 71°C for 20 to 30 minutes (wetting agent may be added to increase cleaning ability)
6. Drain out alkali solution.
7. Final hot water rinse with water at 71 to 82°C till the whole system has been sanitized. Rinse hot water should be drained out.

(b) Advantages of CIP

- Uniform treatment day after day to all equipment.
- Less damage to equipment.
- Fewer (less) requirements of labour.
- Saving the time & manpower.
- Reduce the possibility of contamination.
- Increase the efficiency of equipment machinery.
- Less noise.

(c) The Success Factor of CIP

- Proper selection of pipe and fitting.
- Proper temperature of cleaning solution.
- Adequate velocity of detergent solution.
- Proper concentration of detergent solution.
- Sufficient cleaning time & drainage.

(d) Type of CIP

- Manual control
- Automation control
- Low level
- Medium level
- High level

15 Regulation in Milk Industry

A. Hazard Analysis Critical Control Point

(a) Introduction: The first section of this document sets out the principles of the Hazard Analysis and Critical Control Point (HACCP) system adopted by the Codex Alimentarius Commission. The second section provides general guidance for the application of the system while recognizing that the details of application may vary depending on the circumstances of the food operation. The HACCP system, which is science based and systematic, identifies specific hazards and measures for their control to ensure the safety of food. HACCP can be applied throughout the food chain from primary production to final consumption and its implementation should be guided by scientific evidence of risks to human health.

As well as enhancing food safety, implementation of HACCP can provide other significant benefits. The successful application of HACCP requires the full commitment and involvement of management and the work force. The application of HACCP is compatible with the implementation of quality management systems, such as the ISO 9000 series, and is the system of choice in the management of food safety within such systems.

(b) Definitions

1. **Control measure:** Any action and activity that can be used to prevent or eliminate a food safety hazard or reduce it to an acceptable level.
2. **Corrective action:** Any action to be taken when the results of monitoring at the CCP indicate a loss of control.
3. **Critical Control Point (CCP):** A step at which control can be applied and is essential to prevent or eliminate a food safety hazard or reduce it to an acceptable level.
4. **Critical limit:** A criterion which separates acceptability from unacceptability.
5. **Deviation:** Failure to meet a critical limit.

6. **Flow diagram:** A systematic representation of the sequence of steps or operations used in the production or manufacture of a particular food item.
7. **HACCP:** A system which identifies, evaluates, and controls hazards which are significant for food safety.
8. **HACCP plan:** A document prepared in accordance with the principles of HACCP to ensure control of hazards which are significant for food safety in the segment of the food chain under consideration.
9. **Hazard:** A biological, chemical or physical agent in, or condition of food with the potential to cause an adverse health effect.
10. **Hazard analysis:** The process of collecting and evaluating information on hazards and conditions leading to their presence to decide which are significant for food safety and therefore should be addressed in the HACCP plan.
11. **Monitor:** The act of conducting a planned sequence of observations or measurements of control parameters to assess whether a CCP is under control.
12. **Verification:** The application of methods, procedures, tests and other evaluations, in addition to monitoring to determine compliance with the HACCP plan.

(c) Principles of the Haccp System: The HACCP system consists of the following seven principles.

1. Conduct a hazard analysis.
2. Determine the Critical Control Points (CCPs).
3. Establish critical limit(s).
4. Establish a system to monitor control of the CCP.
5. Establish the corrective action to be taken when monitoring indicates that a particular CCP is not under control.
6. Establish procedures for verification to confirm that the HACCP system is working effectively.
7. Establish documentation concerning all procedures and records appropriate to these principles and their application.

(d) Guidelines for the Application: HACCP management commitment is necessary for implementation of an effective HACCP system. During hazard identification, evaluation, and subsequent operations in designing and applying HACCP systems, consideration must be given to the impact of raw materials, ingredients, food manufacturing practices, role of manufacturing processes to control hazards, likely end-use of the product, categories of consumers of concern, and epidemiological evidence relative to food safety.

HACCP should be applied to each specific operation separately. CCPs identified in any given example in any Codex Code of Hygienic Practice might not be the only ones identified for a specific application or might be of a different nature. The HACCP application should be reviewed and necessary changes made when any modification is made in the product, process, or any step.

Application: The application of HACCP principles consists of the following tasks as identified in the Logic Sequence for Application of HACCP

1. Assemble HACCP team: Assembling a multidisciplinary team. Where such expertise is not available on site, expert advice should be obtained from other sources. The scope of the HACCP plan should be identified. The scope should describe which segment of the food chain is involved and the general classes of hazards to be addressed.

2. Describe product: A full description of the product should be drawn up, including relevant safety information such as: composition, physical/chemical structure (including aW, pH, etc.), microbial/static treatments (heat-treatment, freezing, smoking, etc.), packaging, durability and storage conditions and method of distribution.

3. Identify intended use: The intended use should be based on the expected uses of the product by the end user or consumer. In specific cases, vulnerable groups of the population, e.g. institutional feeding, may have to be considered.

4. Construct flow diagram: The flow diagram should be constructed by the HACCP team. The flow diagram should cover all steps in the operation. When applying HACCP to a given operation, consideration should be given to steps preceding and following the specified operation.

(a) The HACCP team should confirm the processing operation against the flow diagram during all stages and hours of operation and amend the flow diagram where appropriate.

(b) List all potential hazards associated with each step, conduct a hazard analysis, and consider any measures to control identified hazards

(c) The HACCP team should list all of the hazards that may be reasonably expected to occur at each step from primary production, processing, manufacture, and distribution until the point of consumption.

(d) The HACCP team should next conduct a hazard analysis to identify for the HACCP plan which hazards are of such a nature that their elimination or reduction to acceptable levels is essential to the production of a safe food.

(e) In conducting the hazard analysis, wherever possible the following should be included:

- Occurrence of hazards and severity of their adverse health effects.
- Qualitative and/or quantitative evaluation of the presence of hazards.
- Survival or multiplication of microorganisms of concern.
- Production or persistence in foods of toxins, chemicals or physical agents and conditions leading to the above.
- The HACCP team must then consider what control measures, if any, exist which can be applied for each hazard.
- More than one control measure may be required to control a specific hazard(s) and more than one hazard may be controlled by a specified control measure.

5. Determine Critical Control Points: There may be more than one CCP at which control is applied to address the same hazard. The determination of a CCP in the HACCP system can be facilitated by the application of a decision tree, which indicates a logic reasoning approach. Application of a decision tree should be flexible, given whether the operation is for production, slaughter, processing, storage, distribution or other. It should be used for guidance when determining CCPs. If a hazard has been identified at a step where control is necessary for safety, and no control measure exists at that step, or any other, then the product or process should be modified at that step, or at any earlier or later stage, to include a control measure.

6. Establish critical limits for each CCP: Critical limits must be specified and validated if possible for each Critical Control Point. In some cases more than one critical limit will be elaborated at a particular step. Criteria often used include measurements of temperature, time, moisture level, pH, aW, available chlorine, and sensory parameters such as visual appearance and texture.

7. Establish a monitoring system for each CCP: Monitoring is the scheduled measurement or observation of a CCP relative to its critical limits. The monitoring procedures must be able to detect loss of control at the CCP. If monitoring is not continuous, then the amount or frequency of monitoring must be sufficient to guarantee the CCP is in control. Most monitoring procedures for CCPs will need to be done rapidly because they relate to on-line processes and there will not be time for lengthy analytical testing. Physical and chemical measurements are often preferred to microbiological testing because they may be done rapidly and can often indicate the microbiological control of the product.

All records and documents associated with monitoring CCPs must be signed by the person(s) doing the monitoring and by a responsible reviewing official(s) of the company.

8. **Establish corrective actions:** Specific corrective actions must be developed for each CCP in the HACCP system in order to deal with deviations when they occur. The actions must ensure that the CCP has been brought under control. Actions taken must also include proper disposition of the affected product. Deviation and product disposition procedures must be documented in the HACCP record keeping.

9. **Establish verification procedures:** Establish procedures for verification. Verification and auditing methods, procedures and tests, including random sampling and analysis, can be used to determine if the HACCP system is working correctly. Examples of verification activities include:

- Review of the HACCP system and its records.
- Review of deviations and product dispositions.
- Confirmation that CCPs are kept under control.
- Where possible, validation activities should include actions to confirm the efficacy of all elements of the HACCP plan.

10. **Establish Documentation and Record Keeping:** Efficient and accurate record keeping is essential to the application of a HACCP system. HACCP procedures should be documented. Documentation and record keeping should be appropriate to the nature and size of the operation.

- Documentation examples are
  1. Hazard analysis.
  2. CCP determination.
  3. Critical limit determination.
- Record examples are
  1. CCP monitoring activities.
  2. Deviations and associated corrective actions.
  3. Modifications to the HACCP system.

B. Milk and Milk Products Order (MMPO)

(a) Introduction: The Milk and Milk Products Order (MMPO), 1992 is a regulatory order of the Government of India, regulated by the Ministry of Agriculture through Department of Animal Husbandry and Dairying. It is exercised under Essential Commodities Act, 1995 vide notification number S.O.405 (E) dated June 9, 1992, published in the notification part-II, section 3(ii): 226-227. According to this order,

seeking registration is compulsory for all dairy plants processing more than 10,000 lit of milk/day or handling more than 500 tones of milk solids per annum. Plants processing up to 75,000 lit per day or equivalent quantity of total milk solids per annum are to be registered with the director of Animal Husbandry, of the respective State while those handling more than 75,000 lit./ milk/day or equivalent quantity of milk solids per annum have to register with related authority of the Central Government. The plants registered are free to procure milk locally only if they process liquid milk. In case they process milk products, milk has to be procured only through cooperative Unions/Federation.

(b) Main Provisions of the MMPO

1. The order defines various terminologies related to dairy industry like milk, milk sheds etc.
2. Milk and Milk Product Advisory Board is constituted to assist and advice to the Government. It also establishes norms and conditions for license like milk collection area, marketing area, capacity, balancing of uneven supply, standards etc.
3. Registration in specified from with all information regarding trade is compulsory for above said capacity.
4. No person, shall handle, produce of deal with any item of milk in excess of capacities specified without prior permission from the competent authorities.
5. Certificate holder shall collect milk only within the specified milk sheds. The certificate may be canceled or suspended if holder fails to comply any norms and conditions of certificate or of order.
6. The milk controller may put temporary restriction of milk product manufacturing to maintain or increase liquid milk supply for not more than 90 days.
7. The levy on SMP or milk may be imposed by the controller to ensure liquid milk supply to general public in any region.
8. The order specifies stringent sanitary and hygienic requirements for premises in which milk and or milk products are being handled, processed, manufactured, stored of distributed. The organization or person shall conform to the specified requirements before license is given.
9. It specifies criteria for packing, marking and labeling.
10. The product shall get certified to conform to national standards of Central Government e.g. PFA Acts and Rules, Standards of Weights and Measures Act and Rules etc.

11. The milk controllers' implements and control the order and exercises all the powers and functions under the order.
12. The registering authority will deal with application and all functions related to registration.
13. The order provides prosecution and penalty for the organization that fails to meet the requirements of the order.
14. The procedure for sampling, analysis and prohibition for sale processing, manufacturing etc, will remain as per PFA Rules.
15. Samples shall be analyzed only in the BIS recommended laboratories.

The MMPO will help in quality improvement through stringent sanitary and hygienic control. This order will assist in evolving certain norms with respect to processing machinery, essential services and other ancillary requirements of dairy plants. It will promote orderly growth capacity wise of classified organizations and permit free flow of surplus in areas of deficit, as and when need arises. It has several benefits; however, it may be detrimental to liberalization of Indian economy.

C. Prvention of Food Adulteration Act (PFA)

(a) Background and Need: Laws existed in a number of states in India for the prevention of adulteration of food stuffs but they lacked uniformity having been passed at different times without mutual consultation between the stated. The need of central legislation for the whole country in this matter has been felt since 1937 when a committee appointed by the Central Advisory Board of Health recommended this step.

Adulteration of foodstuffs and other goods is now included in the concurrent list (III) in the constitution of India. It has, therefore, become possible for Central Government to enact all India legislation on this subject.

(b) Introduction: The standards of quality of any food item have been fixed by the Government under PFA Act after wide circulation among public and after consulting a committee of competent persons. The PFA (The Prevention of Food Adulteration) rules were framed in 1954 and PFA act came into force in 1955 and has been amended several times. It extends to the whole of India. The standards are formulated and revised whenever required by an expert committee called Central Committee for Food Standards (CCFS).

These standards prescribe the minimum and maximum requirements for all types and categories of food. Any food that does not conform to the minimum standards laid down by the PFA rules is said

to be adulterated in wide perspective, irrespective of whether any thing has been added to or removed from the original food. It establishes the consumer-supplier relations and protects the former against outright fraud.

(c) Main Provisions of PFA Act and Rules

- It is operative in whole of India.
- PFA, 1954 and Fruit Products Order (FPO), 1955 are supplementary and cumulative. The Essential Commodities Act may supersede the PFA Act.
- The adulteration has been defined in wide perspective in terms of purity, nutrition, wholesomeness, safety against contaminants, toxicants, bacterial toxins, preservatives, colorants, insecticides and pest infection, misbranded or mislabeled and anything else which may be injurious to health.
- The Central Government under this Act has constituted a committee called the Central Committee for Food Standards (CCFS) to advise the Central Government and the State Governments on matters arising out of the administration of this Act and to carry out the other functions assigned to it. The committee consists of the following members for three year term and subject to re-nomination.

1. Director-General, Health Services, Ex-officio chairman.
2. Director(s) of Central Food Laboratory(ies)
3. Two experts nominated by the Central Government
4. The representative each of the departments of Food and Agriculture in the Central Ministry of Food and Agriculture and one representative each of the Central Ministry of Commerce, Defense, Industry and Supply and Railways, nominated by the Central Government
5. One representative each nominated by the Government of each state
6. Two representative nominated by the Central Government from Union Territories
7. One representative each, nominated by the Central Government to represent the agricultural, commercial and industrial interests.
8. Five representatives nominated by the Central Government to represent the consumers interests, one of whom shall be from the hotel industry

The committee works under the direction of Central Government and makes bye-laws for the purpose of regulating its own procedure and the transaction of its business subject to the previous approval of the Central Government.

(d) Miscellaneous Provisions

- Manufacturers, distributors and dealers to give warranty.
- Vendor to disclose the name etc. of the person from whom the article of food was purchased.
- Notification of food poisoning.
- Penalties for different types of offenses, penalty varies from a sentence of imprisonment for a term not less than 3 months to 1 year which may also be extended further or a fine which shall not be less than Rs. 200-500.
- Power to court to try cases summarily.
- Offenses by the companies.
- Forfeiture of property.
- Defenses which may or may not be allowed in prosecutions under this Act.
- Cognizance and trial of offenses.
- Power of court to implied manufacturer etc.
- Application of the probation of the offenders Act and the Code of Criminal Procedure.
- Magistrate's power to impose enhanced penalties.
- Protection of action taken in good faith.
- Power of Central Government to give direction.
- Power of Central Government to make rules.
- Power of State Government to make rules.
- Repeal and saving.

(e) Limitation of PFA

- Method of sampling for food products not given
- Random sampling formula as suggested by BIS not followed
- In penal clause, consideration not given to the difference between the adulterated and substandard quality and the extent of adulteration
- The PFA Rules rely mostly on chemical tests. There is almost no importance given to microbiological standards

- It does not make a distinction between the cases coming under it on the basis of degree of adulteration
- It does not provide for aggregation of offense based on the extent of contamination
- The offense and punishment are the same whether the adulteration is great or small
- It is also difficult to coin a suitable definition for a substandard product which could be generally be agreed to
- No provisions exist for defining a standard set of conditions required to manufacture/process, package and preserve the food as obtainable in other countries
- Number of Government testing laboratories, staff and facilities are inadequate
- No provision for methods of analysis of foods
- No modern and sophisticated instruments available in the local laboratories
- Delay in legal proceedings

D. International Organization for Standardization (ISO 9000)

All organization including milk and dairy product manufacturing units are striving to adopt and get ISO 9000 certification to enhance their image for boosting their domestic sale and to gain easy entry in international market. Bureau of Indian Standards is the agency which is authorized by Govt. of India under Bureau of Indian Standards Act 1986 and Rules and Regulations framed their under to operate Quality Systems certification in India which has been launched from 1991.

The ISO 9000 series of standards is adopted as IS-14000 series by BIS is a series of standard which seeks to accomplish following three objectives with regard to quality (Sohrab,1997)

1. The enterprise should achieve and sustain the quality of the products produced so as to meet customers stated or implied needs.
2. The enterprise should provide confidence to its own management that the intended quality is being achieved and maintained.
3. The enterprise should provide confidence to the customer that the intended quality is being achieved in the developed products i.e. brand name is synonym with quality.

(A) ISO 9001, Quality systems – Model for quality assurance in design, development, production, installation and servicing: For use when conformance to specified requirements is to be assured by the supplier during design, development, production, installation and servicing.

(B) ISO 9002, Quality systems – Model for quality assurance in production, installation and servicing: For use when conformance to specified requirements is to be assured by the supplier during production, installation and servicing.

(C) ISO 9003, Quality systems – Model for quality assurance in final inspection and test: For use when conformance to specified requirements is to be assured by the supplier solely at final inspection and test.

E. Bureau of Indian Standards (BIS)

The role of standards in supporting and facilitating the smooth functioning of the chain of processes and also embedding quality in the system cannot be understimated. Standards provide the needed guidelines to the entrepreneurs in ensuring the right practices to be followed during various stages, such as storage, transportation, selection of raw material, processing and packing and also pinpoint the specifications which should be adhered to, in order to promote quality and safety of the end product. So for integrated speedy development of dairy sector, standardization has a vital role to play.

Bureau of Indian Standards (earlier ISI) had started the work with respect to formulation of standards for various food products since its inception in 1947. However, the work of food legislation and standardization got a fillip from 1956 onwards when the food and agriculture division councils of BIS started functioning. Other legislation to bring out uniformity in food quality at mandatory level in the country resulted in enactment of Prevention of Food Adulteration Act, 1954.

Indian Standards are formulated by the technical committees comprising experts from the Industry, Govt., departments testing laboratories, users and consumers. National Standards are evolved through a consensus opinion of the members comprising of a technical committee. In this context, the Bureau of Indian Standards which is the national organization for standardization in our country has so far formulated over 16000 Indian Standards.

F. Good Manufacturing Practices (GMP)

Good manufacturing practice (GMP) is that part of quality assurance which ensures that products are consistently produced and controlled to the quality standards appropriate to their intended use and as required by the marketing authorization.

GMP is aimed primarily at diminishing the risks inherent in any pharmaceutical production, which may broadly be categorized in two groups: cross contamination/mix-ups and false labeling. Above all, manufacturers must not place patients at risk due to inadequate safety,

quality or efficacy; for this reason, risk assessment has come to play an important role in WHO quality assurance guidelines.

GMP describes the methods, equipment, facilities, and controls for producing processed food. As the minimum sanitary and processing requirements for producing safe and wholesome food, they are an important part of regulatory control over the safety of the nation's food supply.

Guidelines for Good Manufacturing Practices (GMP)

1. Employees working in a food plant are required to wash their hands with a sanitizing soap prior to beginning or returning to handling of food.
2. Loose hair poses serious sanitation problems in the plant. Therefore, all persons working or visiting the production area must wear authorized head covering to avoid contact of loose hair with food product.
3. Sanitary precautions are required to be taken by employees when sneezing or coughing. An employee with infectious skin eruption, communicable disease or other infected conditions must have plant management clearance before allowed to handle food.
4. Employees in production areas should wear clean uniforms. The uniforms should be changed daily or sooner if soiled for any reason. Shirts are required to be buttoned and tucked into trousers.
5. Workers in the production area will not wear rings, and neck/ear jewellery. This practice will preclude mix-up of the foreign materials in food. Watches, pens, pencils and loose materials should be removed prior to entry to production areas.
6. Smoking, spitting or chewing of tobacco is prohibited in the production as well as storage areas.
7. Consumption of beverage or food is allowed exclusively in the designated area.
8. Nail polish and/or perfume is not allowed in production or storage areas.
9. Containers and equipment made of glass, including glass thermometers, should not be permitted in the production area.
10. Good housekeeping in the production area is necessary for work efficiency and workers' safety.

16 Packaging, Transportation, Storage and Distribution of Milk and Milk Products

Packaging of Milk

(a) Introduction: Packing is the technique of using the most appropriate containers and components to protect, carry, identify and merchandise any product. It constitutes a vital link between the manufacturer and eventual consumers for the safe delivery of the product through the various stages of manufacture, storage, transport, distribution and marketing. The importance of packaging as a means of promoting exports has been well recognized in recent times. Packing is today an essential part of any industrial activity and an integral part of the modern marketing complex.

(b) Definition: Packing means placing a commodity into a protective wrapper or container for transport or storage.

(c) Functions: A packing must perform the three-fold functions of containing, protecting and merchandising:

(1) To contain the product: The package container should be adequately large to hold the product. It should have proper constructional features so as not to allow leakage and spoillage. It should have enough strength to withstand handling, transportation and storage hazard. Finally, it should also be as compatible as possible with the product.

(2) To protect the product: The package should safeguard the product against contamination or loss and damage or degradation due to microbial action, exposure to heat, light, moisture and oxygen, accidental spillage, evaporation, pilferage, etc.

(3) To help in selling the product: The package should be favorable to dispensation and reclosure, and to its disposal or use.

(d) Packaging Forms:

1. Bottle: The glass bottle still continues to be used for packaging of milk in some parts of the world. However in several developed countries and some of the developing countries it has lost ground to single service containers for packing milk.

2. Carton: The cartons are the preferred medium or form for packaging milk. They are also used for packaging liquid, frozen and coagulated milk products. Cartons are commonly made of food grade paper coated

on the inside with wax or plastics; or lined with paper, plastic films or aluminium foil; or made of laminates.

3. Sachet/Bag/Pouch: Flexible waterproof plastic bags are commonly used for packaging milk and liquid dairy products. Since it is difficult to pour from these, a jug is usually also provided. The popular laminate for such bag is black or dark brown (to exclude UV light) or white.

4. Can: This is commonly used for packaging all types of solid, semi solid and powdered dairy products. Cans are traditionally made of soldered tin plate steel, generally lacquered on the inner surface to prevent corrosion. Aluminium cans have now become famous. Cans are the most convenient for gas packing.

5. Box/Tub: It may be made up of wood or paper board. White wooden boxes/tubs are used for bulk packing of butter and butter oil with butter paper/plastic liners, paperboard boxes are generally used as overwraps.

6. Barrel/Cask: Commonly made up of wood and coated with wax on the inner surface. Used for bulk packaging of sweetened condensed milk, semi solid butter milk / whey, butter oil etc.

7. Cup: They are generally made up of paper with wax or plastic coating on the inside. Used for packing frozen dairy products such as ice cream and coagulated milk products.

8. Collapsible tube: They are made up of aluminium and lacquered on the inside. Low cost, lightweight, ease of handling and dispensing, product protection are its advantages. Used for packing semi fluid products such as sweetened condensed milk, processed cheese spread etc.

(e) Packing Materials: The packaging materials include paper and paper based products (coated or lined), glass, tin plate, aluminium foil, timber (wood), plastics and laminates (Tetra packs).

1. Paper and paper based products: The paper and paper based products form an excellent packaging material for milk and milk products. They may be kraft paper, grease proof paper, vegetable parchment paper, glassine paper, wax coated paper, plastic coated paper, paper boards, solid fibre boards, liner boards, box boards etc. The advantage of using paper is that it is weightless, capability for printing on the surface, low cost and easy disposability. The disadvantages include low wet and tear strength.

2. Glass: The glass may be transparent or opaque. Glass is used in the form of bottles, tumblers, jars, jugs etc. The advantages cited for glass as a packaging material include its strength, rigidity, ability to have a barrier for water and gas and inertness to chemical substances. The disadvantage is its heavy weight, and fragility.

3. Tin plate: It may be made up of a thin sheet (0.025 mm thick) of mild steel coated on both sides with a layer of pure tin. It is desirable to have an internally lacquered can, which provides better resistance to corrosion. The advantages cited for tin containers as a packaging material are their good strength and excellent barrier properties. The disadvantages are their high cost, heavy weight, difficulty in closing the lid of the container, and disposal. The containers are mostly used in the form of can.

4. Aluminium foil: The common thickness of the foil used is 0.012 – 0.015 mm. To increase corrosion resistance, it may be lacquered (coated with lacquer) or a thin film of plastic can be applied for packing dairy products. The advantages of these containers are good barrier properties, grease proof, non-absorption, shrink proof, odourless, tasteless, hygienic, non toxic, opaque to light, bright in appearance etc. The demerits are its low tear strength, susceptibility to strong acids and alkalis. It is mostly used in the form of wrapper, carton and box.

5. Timber: The required qualities for the timber to act as a packaging material are it should be free from odour, have an attractive appearance, and required mechanical strength. It may be treated with casein formalin, or sprayed with paraffin wax or plastics or to make it more water resistant and to avoid the passage of timber taint to butter. It is generally used in the form of a box, tub, cask or barrel.

6. Plastics: The use of plastics in packaging has made tremendous progress in recent years all over the world. A wide variety of plastics can be used as thermoformed, injection moulded or blow moulded containers, such as bottles, cartons, cups, boxes etc. The merits of rigid plastic containers are its low cost and ease of fabrication. The demerits cited are lack of product compatibility, low barrier properties, plastic deterioration, lack of resistance to high heat and fragility at lower temperatures. Flexible plastic packaging films are used as wrappers or sachets or bags or pouches for packaging milk and dairy products. The flexible plastics can be classified in to two types.

i. Low polymers: They include cellophane (coated with plain or nitrocellulose / saran / polyethylene), treated with cellulose etc.

ii. High polymers: Polyethylene, polypropylene, polystyrene, poly vinyl chloride, poly vinyledene chloride (cryovac), rubber hydrochloride (pliofilm), polyester, polyamide (nylon), saran (a mixed polymer), etc. form good packaging materials. The merits cited for flexible packaging films are they can be easily applied and the packaging process can be readily mechanized; loss of moisture from the dairy product is practically nil; it confers protection to dairy products against attack by microorganisms, insects etc. The demerits are: not all technical problems in film packaging have been solved; failure to obtain a perfect

seal and removal of all air before packaging may lead to spoilage; the most careful attention to detail is necessary, else faulty production will result; etc. Care has to be exercised in selecting food grade plastics for packaging of milk and dairy products; otherwise toxicity, if any, from the package will be transferred to the products.

7. Laminates: They are formed by combining the complete surfaces of 2 or more webs of different films with the primary object of overcoming the defects of single films. Usually laminations are made to strengthen the film material, to improve barrier properties, to improve grease resistance, to provide a surface that will heat seal, etc. Some of the typical laminates available for packaging are paper-polythene, cellophane-polythene, aluminium foil-polythene, paper aluminium foil-polythene, polyester-polythene, etc.

8. Tetra pack: It has Internal polyethylene layer seals in the liquid, Polyethylene layer needed for lamination process and aluminium foil provides a barrier to oxygen, flavours and light Polyethylene adhesion layer needed for lamination process. Paperboard gives stability and strength to Polyethylene layer protects food from external moisture. Advantages of Aspetic Tetra Pak System: a) Long keeping quality. b) Better convenience. c) Pilfer-proof. d) Saving in transportation. e) Better milk marketing. f) Milk can be made available even in remote areas. g) For filling other products than liquid milk e.g. Fruit juices condensed milk etc.

Transportation of Milk and Milk Products

The transportation is necessary where milk is collected from scattered area or remote areas. Under Indian conditions milk has to be regularly collected and transported twice a day (morning & evening). Milk is collected at centers and each to be transported to city milk plants. There are various modes of transportation based on various conditions. These depend upon carrying load, the distance of collections and local conditions. The modes of transportations are

1. Road transportation
2. Rail transportation
3. Boat transportation

1. Road transportation: In this transportation loading & unloading possible directly at godown of seller and buyer. It is cheaper than rail transportation over short distances. Road transportation mode includes head load, shoulder-sling, pack animal, bullock-cart, tonga, bicycle, cycle-rickshaw, auto-rickshaw, motor-truck, road tankers, etc.

(a) Head load: This method is follows where as roads are poor. The producers will carry the milk to city supply scheme in cans or any other

containers available with him on the head. Nearly 15 to 25 kg of milk can be transported to a distance of 3 to 8 km. It is important in hilly areas.

(b) Shoulder-sling: This method is followed in rural areas. It is meant for heavier loads but for shorter distances than head load. Nearly 40 kg of milk can be transported to a distance of 2 to 6 km.

(c) Pack animal: An animal like donkeys, horses, mules, camels, are used for milk transportation where road facility is not available or hilly areas. 70 to 80 kg of milk is transported to optimum distance of 6 to 8 km by these animals.

(d) Bullock-cart: It is followed in south India. The milk is filled in can and transported through bullock-cart. 300 to 400 kg of milk is transported to an optimum distance of 10 to 12 km.

(e) Tonga: It is used in U.P. Large quantity of milk is transported by this way. It is faster than head load, shoulder-sling and pack animal. 250 to 300 kg of milk is transported up to optimum distance of 12 or more than 12 km.

(f) Bicycle: Bicycle can be used if roads are poor. Transportation of milk is quick. It is easily accessible to milk producer's home. 40 or more than 40 kg of milk can be transported to optimum distance of 15 or more km.

(g) Cycle rickshaw: More quantity of milk is transported than bicycle. Nearly 150 to 200 kg. of milk is transported up to distance of 10 or more km.

(h) Auto rickshaw: Carrying capacity of auto rickshaw is more and faster than cycle rickshaw. 250 to 500 kg of milk is transported up to a distance of 15 or more km.

(i) Motor-truck: As the automobile industry in India has shown remarkable progress, use of motor trucks as a means of transporting milk in truck has become very popular. Almost all the dairies make use of motor trucks for the transportation of milk particularly when the milk is to be transported in cans. They carry approximately 0.5 to 3.5 ton load for more than 100 km. Due to improvement in road facilities and construction of all season roads, motor trucks have been found most effective means of transportation.

(j) Road Tankers: Road tankers were firstly used in 1922. Insulated stainless steel tanks are mounted either on the road truck. They are definitely meant for bulk handling and for long distance transportation. Recently a good number of road tankers are in use. These tankers are outcome of technological advancement in the field of design and development of equipment within the country. Contribution made by the NDDB and IDC in providing these facilities is remarkable. 5 to more than 5 tons of milk is transported up to a distance of 80 to more km. Road tankers has great scope in future.

(k) Refrigerated Van: Refrigerated van is using for distribution of milk products Butter, Paneer, Ice cream and other dairy products supply to wholesale dealers.

2. Rail Transportation: In this method larger quantity of milk can be handled at a time. It is cheaper than road over long distances. In this method individual producer will carry milk with him; there is provision of separate compartment for milk cans. In rail transportation milk is transported by railway wagon or railway tankers. Rail tankers were used for transportation in 1903. The rail tankers firstly used by AMUL for transporting milk from ANAND to Bombay.

The national milk grid program has started milk train from ANAND to Kolcatta of 5 liquid milk tankers with capacity of 40,000 thousand liters each. Insulated stainless steel tanks are mounted either on the railway. They are definitely meant for bulk handling and for long distance transportation. Rail tankers are outcome of technological advancement in the field of design and development of equipment within the country. Contribution made by the NDDB and IDC in providing these facilities is remarkable.

3. Boat Transportation: It is followed where milk produced across the river i.e. in Assam, Bihar, and U.P. & West Bengal. 40 to 200 kg of milk is transported up to 2 to 8 km.

Problems in relation to collection and transportation of milk

1. Milk is liquid, perishable and bulky.
2. Small and scattered production of milk.
3. Tropical climate.
4. Lack of transport facilities.
5. Lack of country wide organizations for milk collection & transport.
6. Vested interests among local milk merchants.

Storage of Milk

(a) Introduction: Modern milk plants hold both raw and pasteurized milks for a much longer period than before. Normally the milk storage capacity is equal to one day's intake. This allows a more nearly uniform work-day for processing and bottling operations with less dependence on the time for receiving raw milk. Storage tanks are used in Milk Plants for the storage of raw, pasteurized, or processed products, often in very large volumes.

Because of the longer periods of holding, storage tanks are among the most important items of equipment. They must be designed for ease in sanitization, preferably by the circulation-cleaning method. In

addition, the tanks should be insulated or refrigerated, so that they can maintain the required temperature throughout the holding period. Agitation should be adequate for homogeneous mixing, but gentle enough to prevent churning and incorporation of air.

(b) Objectives

1. To maintain milk at a low temperature so as to prevent any deterioration in quality prior to processing/product manufacture.
2. To facilitate bulking of the raw milk supply, this will ensure uniform composition.
3. To allow for uninterrupted operation during processing and bottling.
4. To facilitate standardization of the milk.

(c) Types of Storage

1. Insulated or Refrigerated: In the form, there are 5 to 7.5 cm. of insulating material between the inner and outer linings; in the latter, the space between the two linings is used for circulation of the cooling medium. Another variation of the refrigerated type is the Cole-wall tank.

2. Horizontal or Vertical: Both horizontal and vertical types of refrigerated storage tanks are used for storage of milk at lower temeperature.

3. Built for gravity flow, air pressure or vacuum operation: The first is the most common. However, air pressure is sometimes used to evacuate the product. This requires special construction of the storage tank for greater strength than necessary for normal operations under gravity flow (i.e. atmospheric pressure).

4. Cold Storage for Dairy Products: Different quality and efficient cold storage for dairy products are:

- Ice Cream Freezing Tunnels (–40° C)
- Ice Cream Cold Stores (–25° C)
- Cheese & Butter Cold Stores (–2°C/+15° C)

Distribution of Milk and Milk Products

Milk spoils so easily, it should, ideally, be distributed as quickly as possible. In many countries milk used to be delivered to households daily, but economic pressure has made milk delivery much less popular, and in many areas daily delivery is no longer available. People buy it chilled at grocery or convenience stores or similar retail outlets. Prior to the widespread use of plastics, milk was often distributed to

consumers in glass bottles, and before that in bulk that was ladled into the customer's container.

In India, milk is delivered daily by:

1. Local milkman carrying bulk quantities in a metal container, usually on a bicycle
2. Metropolitan city milk is usually bought or delivered in a plastic bags or cartons via-shops or supermarkets.
3. UHT process milk is sold in cartons often called a "brick" that lack the peak of the traditional milk carton.
4. Glass milk containers are now rare.

Most people purchase milk in bags, plastic jugs or waxed-paper cartons. Ultraviolet light from fluorescent lighting can destroy some of the proteins in milk so many companies that once distributed milk in transparent or highly translucent containers are now using thicker materials that block the UV light. In India milk sold Commonly in 500 ml plastic bags. It is still customary to serve the milk boiled, despite pasteurization. Flavored milk is sold in most convenience stores in waxed cardboard containers. Convenience stores also sell many varieties of milk (such as flavored and ultra-pasteurized) in different sizes, usually in Tetra-Pak cartons.

Factor which help in deciding the mode of distribution of milk are as follows:

1. Keeping quality of milk
2. Perishable nature
3. Possible contamination.
4. Distance to which milk is transported.
5. Supervision and control in distribution.
6. Costs of delivery of milk in distribution.

17 Toxic/Pesticide Residues in Milk and Milk Products

Milk is considered to be the most balanced food ever found in nature containing most of the nutrients. It is consumed by people of all ages and nationalities, although the amount may vary according to food habit and availability. However, it is mostly required for infants, children, the aged and the sick peoples.

Milk is the only source of animal protein in the diet of many vegetarians, whose numbers are increasing across the globe. Milk is also regarded as a sacred food for people with particular religious beliefs. Besides raw milk, various products made from it, such as butter, ghee, cheese, paneer, yoghurt, sweetmeats, ice creams etc., are produced and consumed by people all over the world.

However, milk of animals is exposed to and gets contaminated with diverse chemical substances at various stages during its production and storage. Milk is produced inside the body by a physiological process incorporating biochemical conversion of nutrients derived from various dietary constituents. During the process it is exposed to many chemical contaminants carried by the feed, fodder and other dietary constituents including water.

The various types of chemical contaminants which may be found in milk are:

1. Pesticide residues
2. Heavy metals
3. Radionuclides
4. Veterinary drugs and antibiotics
5. Aflatoxins and mycotoxins
6. Nitrites and nitrates
7. Detergents and disinfectants.

1. Pesticide Residues

A dairy cow can be exposed to pesticide through the air it breathes, the water it drinks, and the forage it consumes. Depending on the type and properties of the pesticide, the residue is excreted through urine, adsorbed in the adipose tissue or excreted into the milk. Pesticide may

be applied directly to livestock or to structure (ex. Barns and housing units) or to control pests.

The potential pollutants from pesticide include the active and inert ingredients, diluents and persistent degradation products. Contamination of milk with pesticides may arises from the practice of dipping cattle in pesticides to prevent tick borne diseases, insect control in stables, feeding stuffs and contamination from the environment.

Feeding stuffs might be the most important source of contamination, especially in areas where large amounts of concentration are fed. Cutaneous contact with organochlorine pesticide and intake of pesticides contained in fodder and soil give rise to residues in the body fat of the animal and in the milk. Milk and milk products can be contaminated with residues in the body fat of the animal and in the milk.

Milk and milk products can be contaminated with residues of organophosphate and carbamates from various sources like animal feed treated with insecticides, animal feed manufactured from plant material which has been treated during the growing season, use of insecticides in animal dwellings, direct contact of animals with treated areas. These compounds are readily metabolized in the alimentary tract and in the organs associated with digestive processes.

(a) Organochlorines (OCs)

Dairy milk and its products have been contaminated with residues of pesticides, particularly organochlorines (OCs), which has been a matter of great concern because of the special significance of these food items in the diet of infants and children. The reason for the existence of pesticide residues in milk or dairy products is not due to any direct application but is due mainly to transfer from contaminated feed and fodder offered to animals for consumption.

Organochlorines as a class are regarded as having low acute toxicity but possess a greater potential for chronic toxicity when compared to other classes, i.e., organophosphates, carbamates, synthetic pyrethroids and others. Organochlorinated pesticides (OCPs), particularly DDT and HCH along with cyclodienes like aldrin, dieldrin, chlordane and heptachlor, have been encountered in milk and its products over the past few decades.

OC pesticides, designated as the most persistent, have been extensively used throughout the world, particularly in the tropics, in order to boost agricultural production by controlling insect pests and in human and animal health applications to control the vectors causing diseases like malaria and others. During the global malaria control programme thousands of tonnes of DDT were used for vector control

from the 1950s to the 1970s. After that their use was discontinued in developed countries but still continued in developing countries of Asia, Africa, Central and South America.

The unusual and high persistence of the OCPs is due to their chemical inertness, stability and lipophilic character. Among the different OCs, DDTs and HCBs are very highly lipophilic while HCHs are comparatively less lipophilic. Their very high lipid solubility allows the OCPs to accumulate in the tissues of the body.

After entering the body they enter a steady state and bioconcentrate in tissue lipids, according to equilibrium patterns of internal transport and lipid tissue content. The lipid-rich tissues act as depots or reservoirs of persistent OCs by virtue of their physico-chemical interactions with cellular components, and their concentrations decline at a very slow rate, even after sources of contamination are eliminated.

OCPs display various types and degrees of toxicity. Residues accumulate in fatty tissues, thus building up in the vital organs such as thyroid, heart, kidney, liver, mammary gland and testes. Several health effects ranging from systemic cardiovascular and respiratory effects to genotoxicity have been reported (Kalpana, 1999).

These residues can also be transferred from the umbilical cord to the foetus and through breast-feeding to babies. Researchers have linked exposure to them with increased risk of cancer in humans and some of these pesticides are carcinogenic in animal test systems. Many of the OCPs are now recognised as potential endocrine disruptors in humans, even at low levels of exposure (Colborn et al., 1993; Kalpana, 1999).

(b) DDT Residues (Dichoro diphenyl trichlooe thane)

Surveys conducted since 1965 have shown residues of DDT in bovine milk often exceeding the MRL (Maximum Residue Limit) of 1.25 mg/kg (fat basis) or 0.05 mg/kg (whole milk basis) as prescribed by FAO/WHO.

Extent of contamination of dairy milk samples with DDT residues in different states of India

State	No. of samples analyzed	No. of samples above detection Limit	No. of samples with residue above MRL	Mean conc. (mg/kg, whole milk basis)
Himachal Pradesh	120	120 (100%)	67 (55.8%)	0.091
Haryana	120	116 (96.7%)	6 (5.0%)	0.022
Uttar Pradesh	240	137 (57.1%)	26 (10.8%)	0.030
Madhya Pradesh	240	230 (95.8%)	52 (21.7%)	0.042
Maharashtra	299	299 (100%)	222 (74.2%)	0.080
Gujarat	120	120 (100%)	84 (70.0%)	0.091
West Bengal	120	43 (35.8%)	15 (12.5%)	0.021
Bihar	120	115 (95.8%)	23 (19.2%)	0.041
Andhra Pradesh	240	208 (86.7%)	137 (57.1%)	0.207
Karnataka	203	45 (22.2%)	36 (17.7%)	0.047
Kerala	120	115 (95.8%)	11 (9.2%)	0.030
Combined	2205	1805 (81.9%)	812 (36.8%)	0.075

Source: Kalra et al. (1999).

(c) HCH residues

HCH (hexachlorocyclohexane) is also known as BHC (benzene hexachloride), although this is a misnomer. The residues of HCH in α, β, γ any product or substrate are estimated on the basis of its four major isomers, viz., α, β, and δ. Total γ-HCH refers to the summation of all these four isomers present in a single sample. Technical or commercial HCH consists of 55-70% α-HCH, 5-14% β-HCH, 10- 18% γ-HCH, 6-18% δ-HCH and 3-4% Ɒ and Ɛ -HCH. of these the β and γ isomers are inactive, the α, δ and Ɒ isomers are slightly to moderately active, and the isomer is the most active. The acceptable daily intakes (ADI) of α-HCH, β-HCH and γ-HCH are 0.005, 0.001 and 0.005 mg/kg/day, respectively. HCH is reported to be a carcinogen.

(d) Endosulfan Residues

Reports of the occurrence of endosulfan residues in milk are very rare, unlike the DDTs, HCHs and some other OCs. There are a few reasons for this. Firstly, endosulfan came onto the market after the DDTs and HCHs. Secondly, the use of endosulfan is very much restricted only to the field of agriculture, unlike DDTs and HCHs which, apart from agricultural use, have been and still are being used widely as part of public health programmes to combat malaria and some other diseases and also in veterinary healthcare to control animal ectoparasites.

Thirdly, but most importantly, endosulfan is not as persistent as the other OCs because it metabolises at a much faster rate to water-soluble metabolites and has a lower partition coefficient. Fourthly, endosulfan has a very much lower transfer coefficient compared to DDTs, HCHs and other OCs. The transfer coefficient of endosulfan has been calculated to be 0.23-0.33% in goat milk (Nag et al., 2007). It has even been reported that endosulfan does not pass into the milk of cattle when ingested in feed for a prolonged period of time (Li et al., 1970; Surendra Nath et al., 2000).

(e) Other OCPs and POPs (Persistent Organic Pollutants)

Apart from DDTs, HCHs and endosulfans, other OCPs such as aldrin, dieldrin, chlordane, endrin, heptachlor, etc., have also been detected in milk on many occasions. Milks from Lucknow were found to be contaminated with aldrin at an average level of 0.02 mg/kg in buffalo and 0.005 mg/kg in goat milk (Saxena and Siddiqui, 1982).

Khandekar et al. (1981) found 12 out of 23 milk samples collected from local vendors in Bombay had alarming levels of dieldrin which varied from 38 to 126 mg/kg on a fat basis. This level was more than 500 times higher than the MRL of 0.05 mg/kg as recommended by FAO/WHO (1979). Aldrin, dieldrin and heptachlor were detected in significant amounts in the samples of milk collected from Andhra Pradesh, India (Kannan et al., 1992).

(f) Organophosphate Pesticide Residue (OPPR)

The highly persistent organochlorinated pesticides, after their banning or restriction in use, have been replaced mostly with organophosphate (OP) compounds which have wide application, mainly as insecticides. Organophosphates are characterised as less persistent and are easily metabolised in the system, although they are highly toxic to mammals. These substances are mainly acetylcholine esterase inhibitors with associated neurological and neuro-muscular effects. They are also carcinogens, teratogens and mutagens.

(g) Pesticide Residues in Butter and Ghee

Butter has often been found to be contaminated with OCPs in India. All the eight samples of butter from Delhi were excessively contaminated with DDT (Agnihotri et al., 1974). The residues in branded butter varied from 1.1 to 8.0 mg/kg while the unbranded local butter revealed a contamination level of 2.8 to 3.8 mg/kg. Duggan and Duggan (1973) reported that only five out of 1141 samples of the dairy products analysed in the USA contained DDT residues above 2 mg/kg, while 80 and 48 of the 145 butter samples contained more than 2 mg/kg DDT and HCH, respectively, in India (Kalra and Chawla, 1985).

(h) Baby Milk Powder

The presence of residues of HCH and DDT in infant food is of particular concern, since newborn children often depend on them for a substantial period of time. Different brands of infant formula from Gujarat, Maharashtra and Punjab in India were found to be contaminated with DDT above the MRL (Dhaliwal and Kalra, 1978).

Infant food collected from Ludhiana was found to be 100% contaminated with DDT, 90% of which were above the tolerance level (Shastry, 1983). The presence of DDT and HCH residues in different brands of baby milk powder was also reported by Kathpal et al.(1992).

2. Heavy Metal Pollution in Milk

Bioaccumulation of toxic heavy metals, e.g. lead (Pb), cadmium (Cd), arsenic (As), mercury (Hg), etc., in milk has evoked great concern in recent years, just like the presence of any other toxic matter in milk. Heavy metal contaminants enter animal systems through pollution of air, water, soil and feeds. Once in the animal, these metals can persist for several weeks even after discontinuation of the exposure. The accumulation of heavy metals in dairy animals adversely affects health and milk production (Dey et al., 1996; Dogra et al., 1996). It has been found through experiment that the cardiovascular and immune systems are affected before the appearance of clinical symptoms caused by heavy metal poisoning (Dey et al., 1993; Haneef et al., 1995).

The toxic metals mercury, lead and cadmium have reproductive and endocrine system disrupting effects. Elevated levels of lead could be detected in animal tissues up to 12-19 months after exposure has ceased (Hatch, 1988).

Normally, very low concentrations of these metals are secreted into milk because the carryover rate of metals from forage to milk is approximately 1:500 (Bluthgen et al., 1997). The body has no homeostatic mechanism for regulating these metals in tissues. Some of the heavy metals, e.g. As, Cd, Hg and Pb, are cumulative poisons and are toxic even at a low concentration. A higher Pb burden in blood and milk from animals reared in urban localities and around polluting industrial units in various parts of India (Swarup et al., 2004) and elsewhere in the world (Baars et al., 1990) has been documented.

It is expected that animals exposed to industrial Pb will excrete higher amount of Pb in milk. Because of its toxicity and passage into milk, Codex International has established specifications for Pb in milk and milk products (0.02 ppm) and in butter (0.05 ppm). Higher concentrations of Pb in milk and blood were reported in some Indian cities such as Delhi (Dey et al., 1996), Varanasi (Bhatia and Chowdhury, 1996) and Kanpur (Swarup et al., 1997).

3. Radionuclides

Radioactive elements may be present in soil naturally or by accumulation from industrial wastes. Radionuclides are radioactive isotopes of certain elements. Such isotopes are always present in milk but in minute quantities. Their presence in milk is attributed mainly due to passage

from grasses and fodder to milk. Plants get contaminated either by uptake from contaminated soil or through deposition on a leaf surface. Radionuclides enter the milk and milk products through naturally occurring radioisotopes (Yousef and Marth, 1985).

The potentially hazardous radionuclides in milk are 90Sr, 89Sr, 131I, 133I, 134Cs, 140Ba and 40K. Among the naturally occurring radionuclides 40K is the most important and is present at a higher level.

Among the artificial radioactivity arising from activities using nuclear materials, the important elements from the point of view of dose via food consumption are 90Sr, 134Cs, 137Cs and 131I. Most dairy products generally contain 137Cs to levels of 40 Bq/kg while 131I is present to concentrations of 20 Bq/litre (Mathur et al., 1999).

4. Veterinary Drug Residues

Modern veterinary drugs are used in livestock production to reduce mortality and morbidity from infections, to increase growth rates, to relieve stress or to tranquillise, to improve feed conversion, to increase milk secretion and to increase productivity in general. The veterinary drugs and active compounds associated with milk are chemotherapeutics (antibiotics and sulfonamides), endoparasiticides (fasciolicides and antihelminthics), ectoparasiticides, hormones, and teat and skin disinfectants (iodophores, chlorphenidine, quats, linear alkyl benzyl sulfate). Antimicrobial residues in milk should be taken into account for health aspects such as possible impact on the emergence of antimicrobial resistance against antimicrobials administered in human therapy, disorders of the intestinal flora and possible occurrence of allergic symptoms.

Antibiotic residues in milk may lead to severe allergic reactions in sensitive consumers. In the dairy industry, antimicrobial residues cause economic losses. Apart from the concern to public health, the presence of residues has created other problems in the dairy industry, including inadequate curdling of milk during manufacture of curds and yoghurt, inadequate ripening of cheese, etc. In fermented milk products, the effect of antimicrobials slows or inhibits acid formation and results in inadequate aroma formation.

Even low levels of residue below the MRL can cause defects in cheese, including off-flavours, uneven texture, uneven eye development and a tendency for butyric acid fermentation. As a preventive measure in most countries, milk is withheld for fixed times and/ or milk containing antimicrobials is discarded so that drug residues are not present or drop below the permissible level. There are several different groups of antibiotics available to treat infected dairy cattle.

The most common groups include the β-lactams (e.g. penicillin, cephapirin), sulphonamides (e.g., sulfamethazine), aminoglycosides (e.g. streptomycin, gentamycin), tetracycline (oxytetracycline), macrolides (e.g. erythromycin), quinolones (e.g. fluroquinolone), etc. These antibiotics may be used singly or, at times, in combination when treating dairy cattle for certain diseases like mastitis and others. Dairy cattle that have been treated with antibiotics produce milk containing antibiotic residues for a period of time after treatment.

Milk from treated cows is therefore required to be excluded from the milk supply for a specific time period to ensure that antibiotic residues no longer persist in their milk. Antibiotic residues enter the milk supply when treated cows are returned to the milking herd early or when a cow retains antibiotic residues in her system for a long time. The excretion of a residue in the mammary gland depends on the degree of its ionisation, its solubility in fat and water and on the difference between the pH of plasma and milk.

Lipid-soluble antibiotics which ionise as bases move readily from blood to milk. Organic bases ionise at the pH of milk and cannot diffuse back towards the blood. The level of drug residues in milk is also affected by their route of administration. In a study to determine whether extra-label use, the route of administration or the type of drug is responsible for prolonged shedding of residues in milk, it was found that extra-label use of antibiotics was significantly associated with increased risk of antibiotic residues in milk beyond the label withholding time.

No significant differences in risk were observed among the various antibiotic products used in the study, but the route of administration had an effect, as the proportion of antibiotic residue positive samples was significantly higher when given subcutaneously rather than by other routes, such as intramuscular, intravenous, intramammary, etc. (McEwen et al., 1992).

5. Mycotoxins

The word mycotoxin is derived from myco, meaning mould, and toxin, a poison. Mycotoxins or fungal toxins are low molecular weight secondary metabolites (i.e., metabolites not essential to the normal growth and reproduction of the fungus) formed by a consecutive series of enzyme-catalysed reactions from a few biochemically simple intermediates of primary metabolism in a wide range of fungi, particularly moulds.

There are about 400 secondary metabolites of moulds but only a few of them are toxins causing pathological, physiological or biochemical alterations in other species, including humans and animals. Such poisoning is referred to as mycotoxicosis. There are over 100 known

species of moulds that produce mycotoxins, but most of the mycotoxins of importance are produced mainly by the species of three genera, viz. Aspergillus, Penicillium and Fusarium. The most studied mycotoxins are aflatoxins (AFB1, AFB2, AFG1, AFG2, AFM1 and AFM2), ochratoxins, fumonisin, zearalenone and trichothecenes, particularly deoxy- nivalenon (DON), diacetoxyscirpenole (DAS), nivalenone (NIV), T-2 and HT-2 toxins.

Mycotoxin contamination of animal fodder occurs in the field owing to infestation by particular pathogenic fungi or with symbiotic endophytes and also during processing and storage of harvested products and feed in appropriate environmental conditions. In general, mould growth and mycotoxin production require a moisture content of greater than 14%, a temperature of at least 25 °C and some degree of aeration. When these conditions are met, mould infestation and mycotoxin production in target crops are likely to occur.

During the process of host colonisation, fungi may produce toxins that are harmful for humans and animals. This occurs mainly under specific conditions of plant stress, such as those produced by adverse weather conditions. For instance, drought has been shown to prompt aflatoxin production in groundnuts. Differences in mycotoxin production are also related to plant genotype. Clinical manifestation of myco- toxicoses in animals depends on exposure, concentration and type of mycotoxin.

Differences in tolerances to mycotoxins exist among poultry, pigs and cows. The most important mycotoxin is aflatoxin (bisfurano coumarins) produced by Aspergillus flavus, A. parasiticus and A. nomius. The aflatoxins are potent carcinogenic, mutagenic, teratogenic and immunosuppressive agents and cause liver damage, decreased milk yield and egg production, and suppression of immunity in animals. Aflatoxin-producing strains of Aspergillus are widely distributed in soil and air. Under favourable environmental conditions, colonisation and mould growth can easily occur in substrates like feed or seed.

The nature and concentration of aflatoxin will vary greatly according to prevailing weather conditions, substrate and the type of mould. A. flavus produces primarily aflatoxin B1 and B2 on corn, while A. parasiticus on corn produces all four major aflatoxins B1, B2, G1 and G2. Three major feedstuffs, viz. corn, cottonseed and peanuts, have high potential for invasion by Aspergillus spp. during growth, harvest, transportation or storage.

In the case of soybean and small grains, colonisation generally occurs during storage. Poor storage conditions, high moisture and temperature, lack of aeration, kernel damage and spore dissemination caused by storage insects, etc., aid in colonisation and promote aflatoxin production. Chronic poisoning or aflatoxicosis can result when low

levels of toxins are ingested over a prolonged period. The toxic effects are not nearly as specific or clinically evident as in acute intoxication. In general, affected livestock exhibit decreased growth rate, lower productivity and immunosuppression.

Among the biological effects, aflatoxins have an effect on protein synthesis as they interact with a number of different cell components involved in the process. Aflatoxins have a high affinity for nucleic acids and polynucleotides (RNA and DNA), particularly at nucleophilic sites on the guanine base. Formation of these aflatoxin-nucleic acid covalently bonded adducts can lead to a number of different biological effects, including cancer.

Aflatoxin is carcinogenic in several species including rats, ducks, mice, trout and primates. AFB1 is considered the most potent carcinogen known, with AFG1, AFB2 and AFG2 in order of decreasing potency. Aflatoxins are the major mycotoxins known to enter milk and many milk- based foods. The naturally occurring aflatoxin B1 in feed is biotransformed by mammals into various metabolites including aflatoxin M1 (a hydroxylated derivative of AFB1, also known as milk toxin) excreted in milk, which is hepatotoxic and hepatocarcinogenic.

The acute toxicity of AFM1 is similar to or slightly less than that of AFB1 but its carcinogenic potential is probably one or even two orders of magnitude lower than that of AFB1 (Henry et al., 2001). AFB1 has been categorised by the IARC (International Agency for Research on Cancer) as a class I human carcinogen and AFM1 as a class IIB human carcinogen. The presence of AFM1 in cattle and buffalo milk and milk products was reported from various parts of India (Paul et al., 1976; Patel et al., 1981; Shemakayani and Patel, 1989; Tiwari and Chauhan, 1991; Rajan et al., 1995, Choudhary et al., 1997; Dhand et al., 1998; Vaidya et al., 1998).

The extent of contamination varied from very few to 94.17%. The average level of AFM1 in buffalo milk was lower than that in cow milk, probably due to the lower conversion of AFB1 to AFM1 through hydroxylation in buffaloes.

6. Nitrates and Nitrites

Nitrates are a naturally occurring form of nitrogen and an integral part of the nitrogen cycle. Manures, nitrogenous fertilisers, decaying plants and animals and other organic residues generate nitrates, which are also produced in the human and animal body. Due to increased use of fertilisers and manures in intensive crop production systems, the level of nitrates is increasing in cereals, other field crops and water. Forage crops may also be a source of high levels of nitrates.

Sudan grass, oats, rape, wheat, barley and maize accumulate nitrate. Nitrate is relatively non-toxic, but approximately 5% of all ingested nitrate is converted in saliva and the GI tract to the toxic nitrite by bacterial enzymes. Nitrite and N-nitroso compounds formed by binding nitrite to other substances like amines, amides, etc., are toxic and can lead to severe pathologies in humans. N-nitroso compounds have also been reported to be carcinogenic in more than 40 animal species, including mammals, birds, reptiles and fish. The best known effect of nitrite is its ability to react with haemoglobin (Oxyhb) to form methaemoglobin (metHb) and nitrate.

MetHb cannot bind oxygen and as a result oxygen delivery to tissues is impaired. Once the proportion of metHb reaches 10% of normal Hb levels, clinical symptoms like cyanosis and asphyxia occur. Nitrate reduction in the rumen competes with other metabolic reactions such as methanogenesis and results in the formation of other end products, leading to abortion, infertility and goitre. Only good agricultural practices can negate this problem.

Although milk and milk products are, either secretory or post-secretory, contaminated with nitrates, nitrites or nitrosamines, the level does not pose any health risk. The nitrate level in milk is naturally relatively constant in the range of 1-12 mg/kg raw milk, while nitrite is practically absent. Traces of volatile nitrosamines can be detected in milk. The nitrate and nitrite residues in milk were studied after application of KNO_3 to dairy cows at two-week intervals in single peroral doses of 75, 37.5, 18.75 and 9.5 g two hours before evening milking.

A marked increase in nitrate content appeared in dependence on applied KNO_3 up to 38 hours, and in samples taken at 50 hours the level was at par with the control. The average value of residual nitrate in milk at two hours after administration of 150 g of KNO_3 was 34.60 mg NO_3 /litre. The nitrate concentration, however, did not exceed 0.05 mg/litre in any single sample (Baranova et al., 1993).

7. Detergents and Disinfectants

Detergents and disinfectants are used in milk production for cleaning and sanitation at various stages. Cleaning of udders of the cows is done with disinfectants. Detergents are necessary to clean milking and ancillary equipment effectively before disinfection. Detergents contain inorganic alkalis (e.g. sodium carbonate and silicates and trisodium phosphate), chlorine and chlorine compounds, iodine and iodine compounds, alcohols, phenolic compounds, nitrogen compounds, surface-active agents (or wetting agents), sequestering (water-softening) agents (e.g. polyphosphates) and acids for de-scaling. Many proprietary, purpose-made detergents are usually available, but otherwise an inexpensive mixture can be made to give a concentration in solution of 0.25% sodium carbonate (washing soda) and 0.05% polyphosphate. The detergents and disinfectants can contaminate milk and its products if proper care is not taken, such as adequate rinsing and drainage of the installations. Impairment of organoleptic quality in milk occurs due to the presence of residues of detergent.

Management of Contaminants

Feed and fodder are the main sources of contamination of animals with pesticide residues and many other contaminants. Therefore, care needs to be taken at all points of production, right from the sowing of the crops, their management, harvesting, processing and storage, so that there is ultimately no contamination of pesticides in the fodder or feed to be fed to the animals.

Comprehensive action is required from all stakeholders to keep the contamination at minimum possible levels. The government agencies should be proactive in taking various measures like proper treatment of municipal sewage, sludge and irrigation water, control of pesticide markets, sale and use of pesticides, veterinary drugs, etc. The use of good agricultural practices (GAP), good manufacturing practices (GMP) and good hygienic practices (GHP) should be invoked in all possible areas. A few points in this regard may be mentioned here.

1. Seed/planting material, soil and irrigation water used for growing crops, which are supposed to be offered to animals either as main crop (fodder) or as their by-product (feed), should be free from any contamination.
2. Synthetic pesticides should be used only when absolutely necessary. Integrated pest management (IPM), integrated nutrient management (INM) and organic agriculture should be introduced as much as possible, keeping in view that there is no decrease in productivity or total yield.
3. Proper pesticides should be used under the proper conditions as per the manufacturer's recommendations.
4. Pesticides should be applied at the exact dose.
5. Farmers/users should not be guided by dealers/retailers in choosing chemicals.
6. Application equipment should be in perfect working order and there should be minimum possible drift.
7. Application should be avoided immediately before harvest.
8. A safe waiting period or pre-harvest interval (the prescribed minimum time gap between the last application of a pesticide

to a crop/commodity/animal and harvesting or grazing/feeding the livestock or milking/slaughtering the animal for human consumption) should be strictly maintained.

9. In case of feed/fodder, the maintenance of a safe waiting period becomes all the more important and should be strictly adhered to, because the fodder is normally fed straightaway to animals and thus there is little chance for decontamination. Freshly treated material should never be fed to animals, particularly to those that are being milked.
10. When pesticides are applied on animals by drenching their skin to control ectoparasites, milking should be avoided for a certain period depending on the type of chemical.

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